ME 262 –Unit 2
Mechanism Synthesis and Analysis



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KNUST

Mechanism Synthesis and Analysis

Mechanism: is a mechanical device that has the purpose of transferring motion from a source to an output.

- Analysis – determination of position, velocity, acceleration, etc. for a given mechanism
- Synthesis – design of mechanism to do a specific task.

Logical synthesis of mechanisms

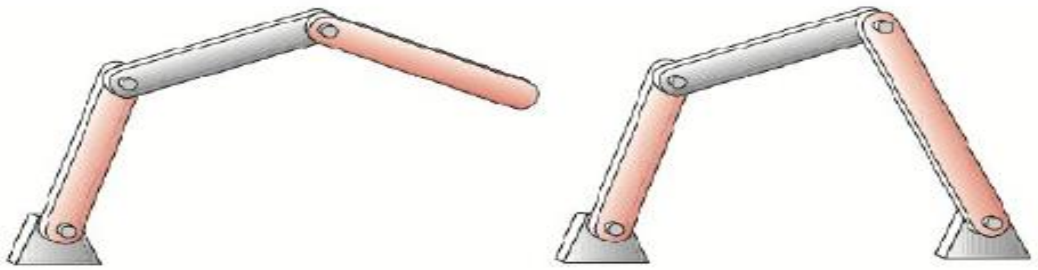
- Type, number and dimensional synthesis
  - Type synthesis: Kind of mechanism selected
  - Number synthesis: Number of joints or links
  - Dimensional synthesis: The determination of the proportions (lengths) of the links necessary to accomplish the desired motions

Terminology and Definitions

- Kinematics: the study of the motion of bodies without reference to mass or force
- Links: one of the rigid bodies or members joined together to form a kinematic chain considered as rigid bodies
- Kinematic pair/ mechanical joint : a connection between two bodies that imposes constraints on their relative movement.
- Ground/frame: a fixed or stationary link in a mechanism.
- Lower pair: the two lower pair have surface contact with one on another.
- Higher pair: the two lower pair have point or line contact with one another.
- Degree of freedom (DOF)/Mobility: of a mechanical system is the number of independent parameters that define its configuration.
- Frame: the fixed link in a mechanism.

Terminology and Definitions

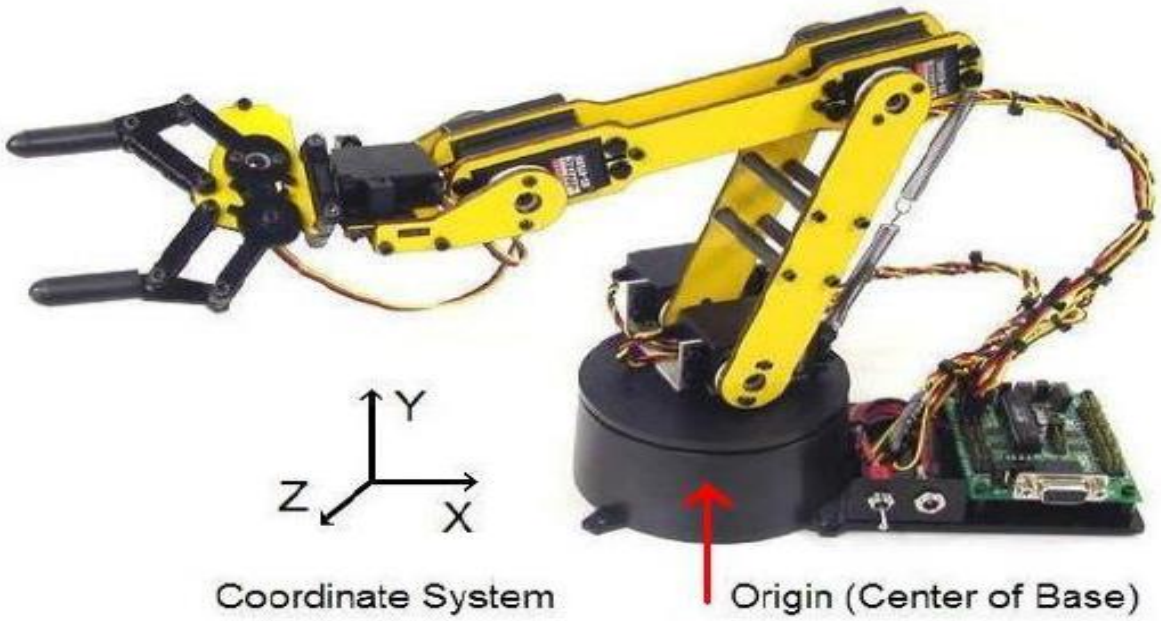
- Closed-Loop Kinematic Chains: a kinematic chain is an assembly of links and joints. Each link in a closed-loop kinematic chain is connected to two or more other links.
- Open-Loop Kinematic Chains: at least one link is connected to only on other link.



(a) Open mechanism chain (b) Closed mechanism chain

Terminology and Definitions

- Manipulators: manipulators designed to simulate human arm and hand motion are an example of open-loop kinematic chains.
- Robots: programmable manipulators

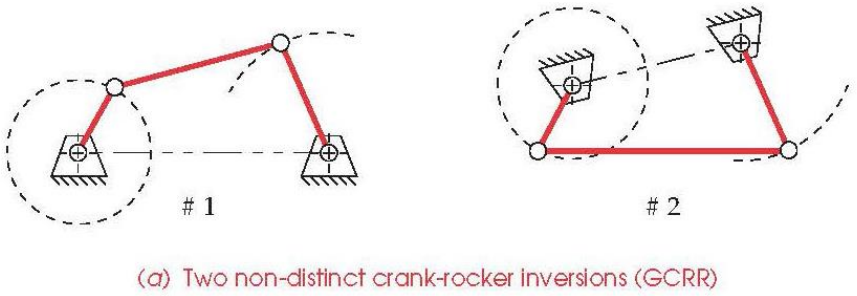


Terminology and Definitions

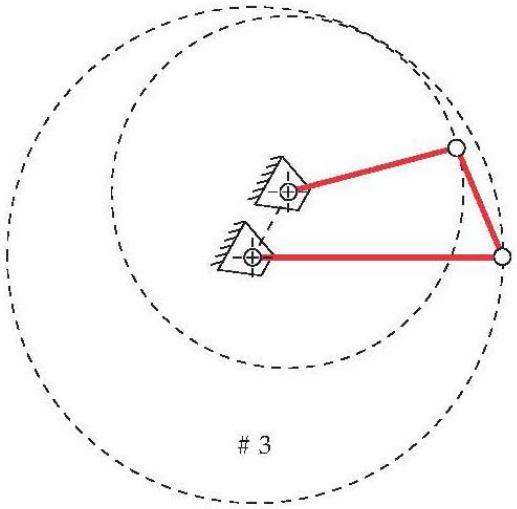
- Linkage/mechanism/machine/engine: an assemblage of rigid bodies connected by kinematic pairs.
- Planar motion: motion of all linkages is 2D (projected onto a common plane).
- Spacial motion: at least the motion of one link is 3D.

Terminology and Definitions

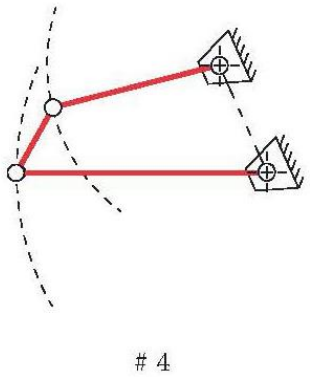
Inversion: if two otherwise identical linkages have different fixed links, then each is an inversion of the other.



(*a*) Two non-distinct crank-rocker inversions (GCRR)



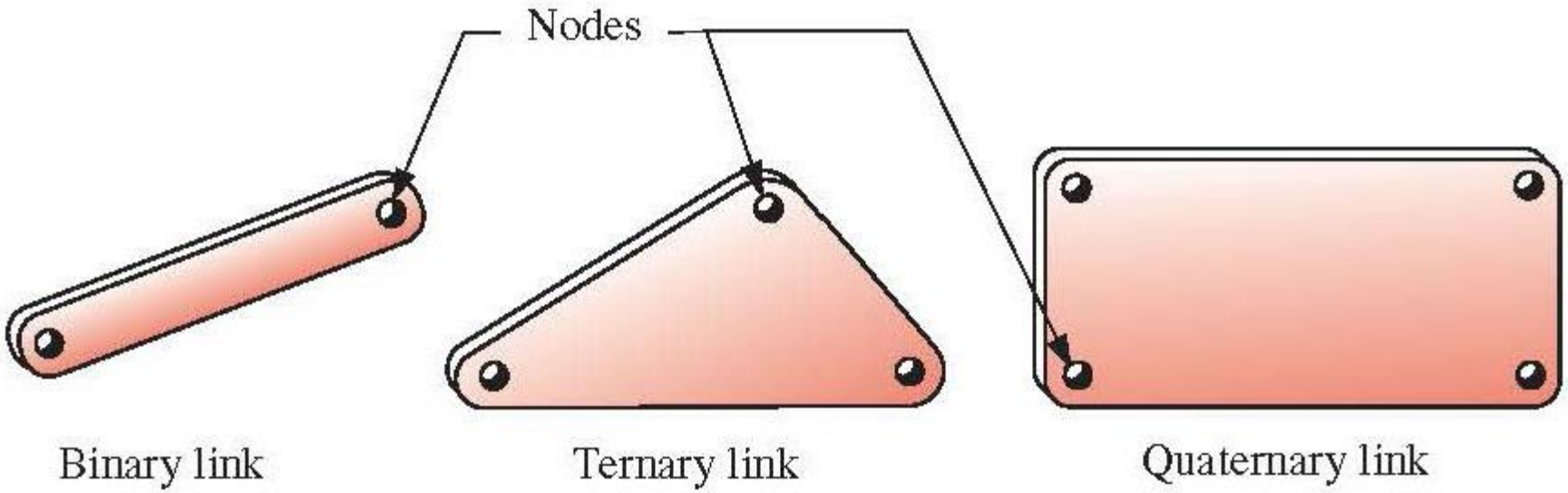
(*b*) Double-crank inversion (GCCC)
(drag link mechanism)



(*c*) Double-rocker inversion (GRCR)
(coupler rotates)

Link Types

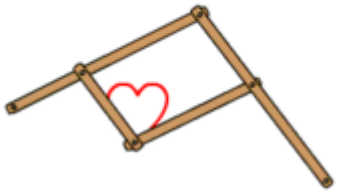
A link is a rigid connection between two or more elements of different kinematic pairs.

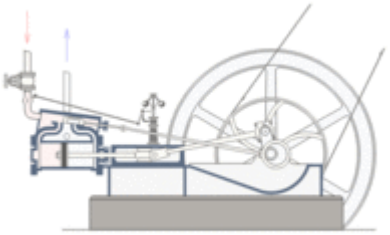


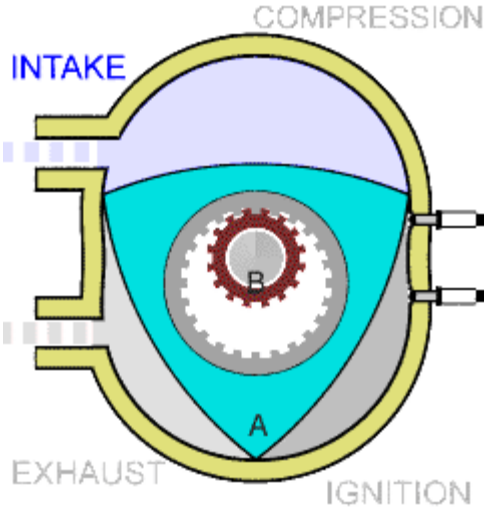
Mechanisms for specific applications

- Drafting Instrument
- Pantograph Linkages
- Slider-Crank Mechanism
- Rotating Combustion(Wankel) Engine







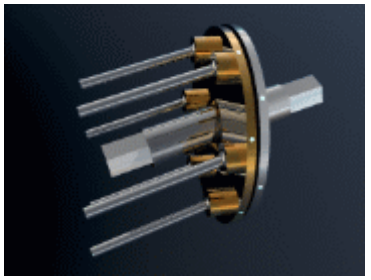


Mechanisms for specific applications

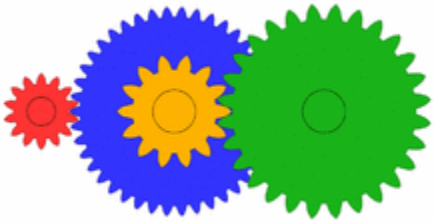
Fluid Links



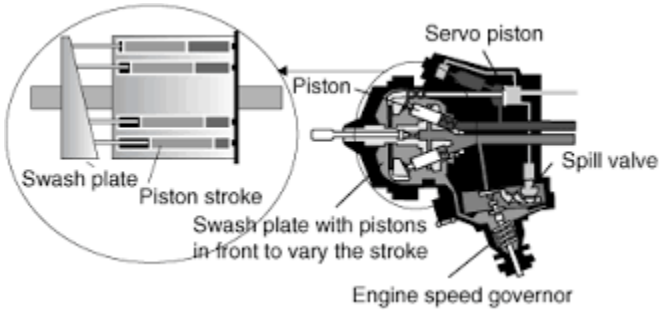
Swash Plate



Gear trains

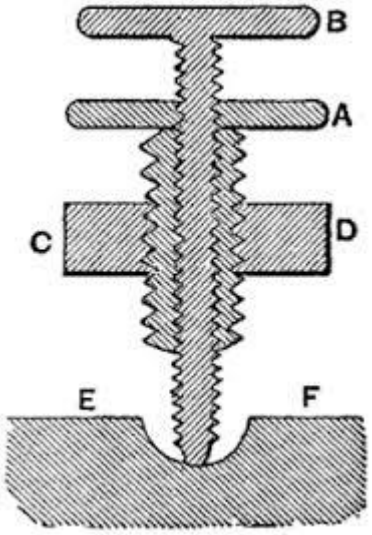


Lamination-Type Impulse Drive



Mechanisms for specific applications





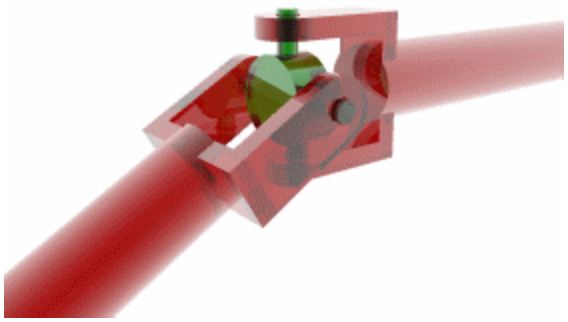


Power Screws

- Differential Screws
- Ball Screws

Mechanisms for specific applications

Universal Joints



Special-Use Clutches
- Sprag-Type Reverse-Locking Clutches
- One-Way Clutches

Universal Joints

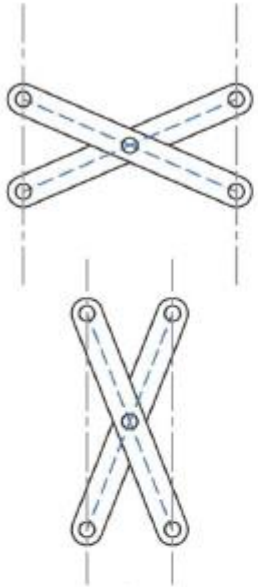
Automotive Steering Linkage

Computer Controlled Industrial Robots

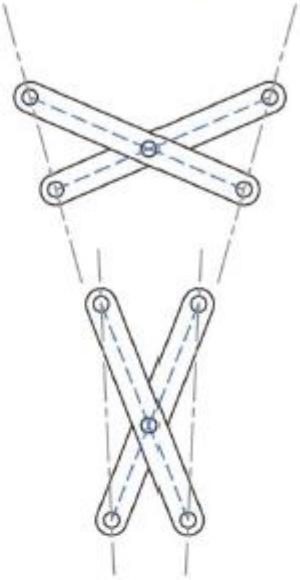
Scissor mechanisms



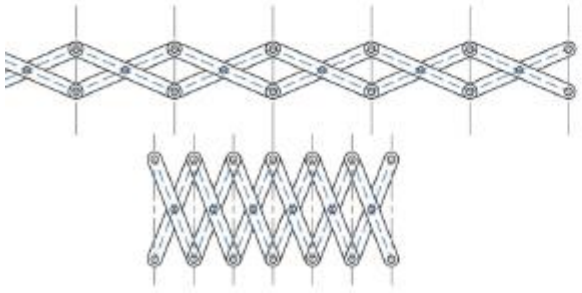
Parallel / symmetric



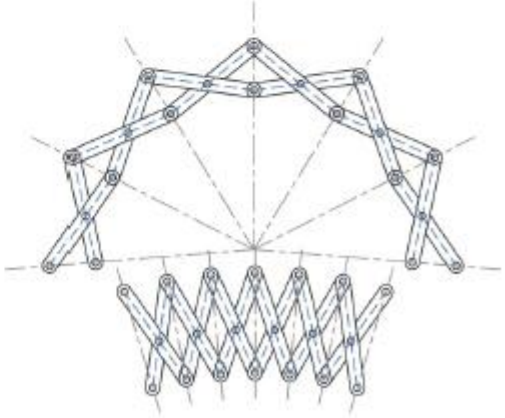
Parallel / asymmetric



No curvature

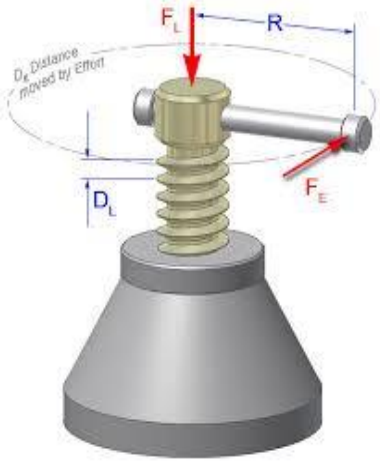


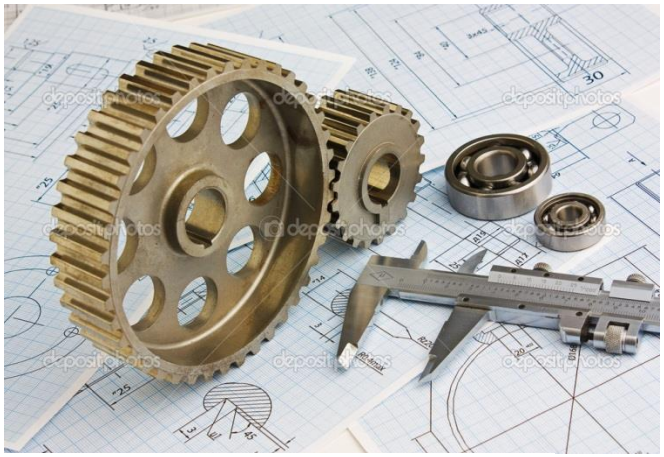
Variable curvature

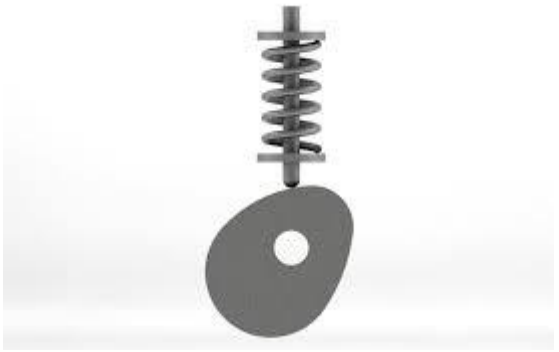


Classification of mechanisms

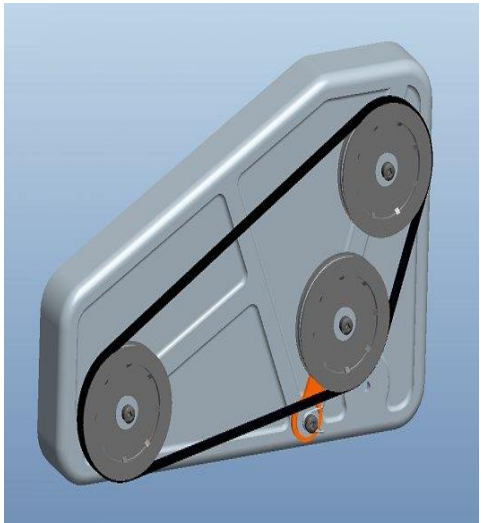
No	Description
1	Screw
2	Roller
3	Wheel
4	Cam
5	Linkage
6	Ratchet

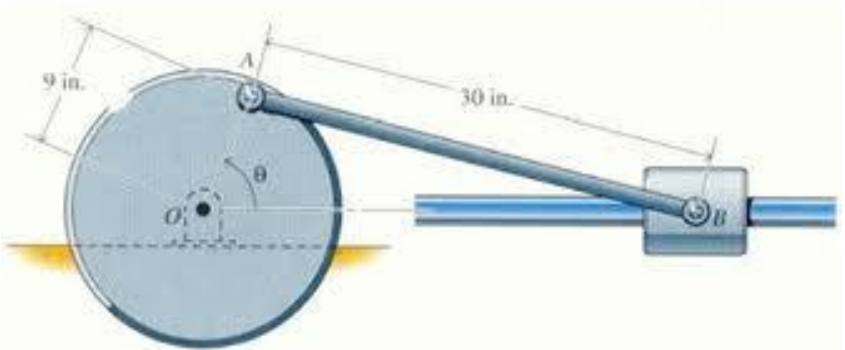










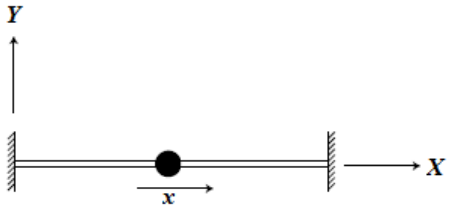


Kinematic Pair(Joints)

- Lower pair(six types)
- Higher pair(cam and its follower have line or point contact between the elements)
- Wrapping(chains and belts)

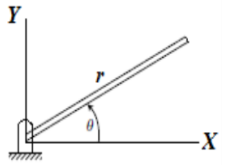
Degree of freedom

For example the bead shown in A has only one coordinate needed to specify its location.



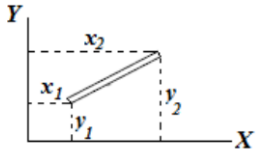
A

The bar shown in B needs two coordinates(complete, independent and with admissible variation) to locate the end point. (x, y) or (r, θ)



B

The bar shown in D needs three coordinates(complete, independent and with admissible variation) to locate the end point. (x, y, θ)

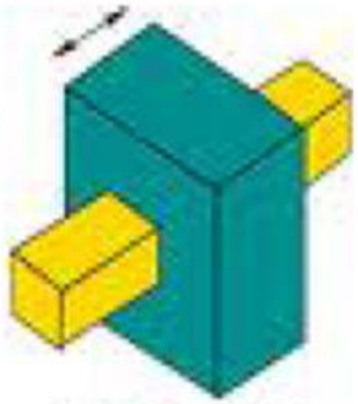


C

Lower Pairs(surface contacts)



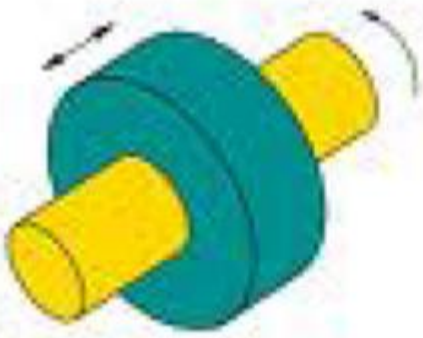
Revolute
1 Degree of Freedom



Prismatic
1 Degree of Freedom



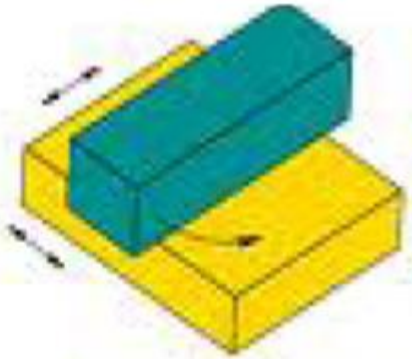
Screw
1 Degree of Freedom



Cylindrical
2 Degrees of Freedom

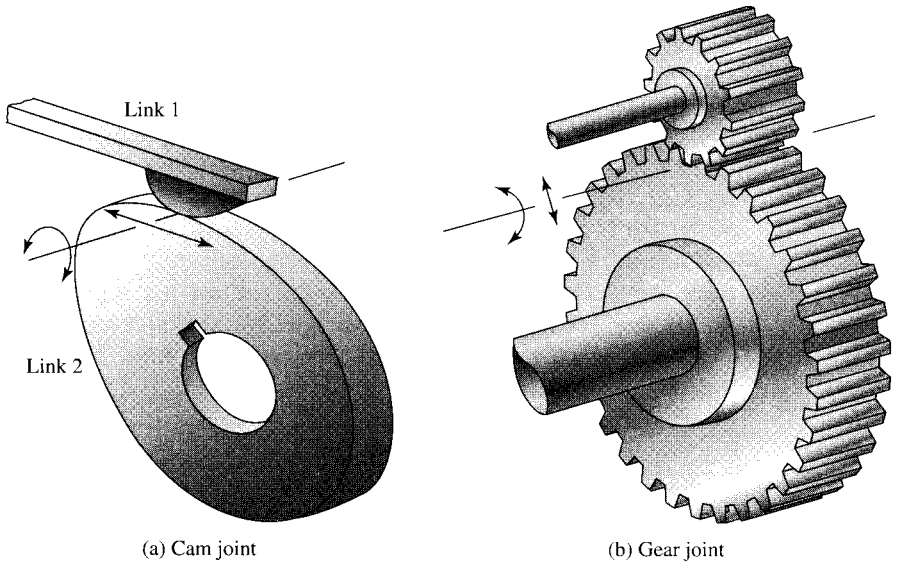


Spherical
3 Degrees of Freedom



Planar
3 Degrees of Freedom

Higher Order Joints



(a) Cam joint

(b) Gear joint

Degree of freedom(dof)/Mobility

- Any unconstrained rigid body has six dof
- A body moving in a plane has three dof
- A cluster of n bodies not connected have dof=3n
- A fixed link(frame) has no degree of freedom
- Each lower pair reduces total dof by 2
- Each higher pair reduces total dof by 1
- Grubbler's Criterion: the total dof of a planar mechanism is

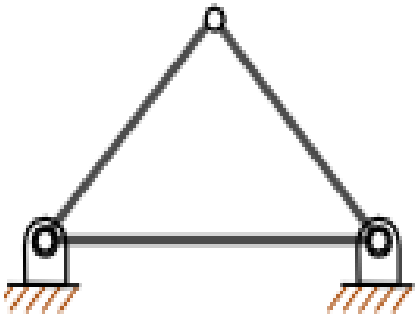
dof=3(n-1)-2j-h

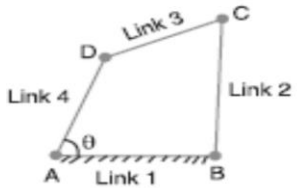
Where n=number of links

j=number of lower pairs

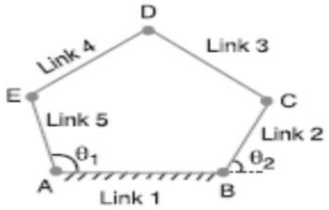
h=number of higher pairs

Examples

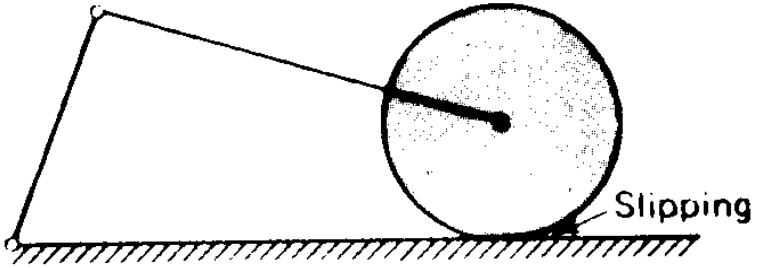


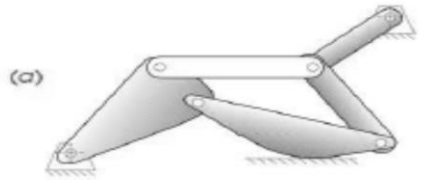


(a) Four bar chain.



(b) Five bar chain.



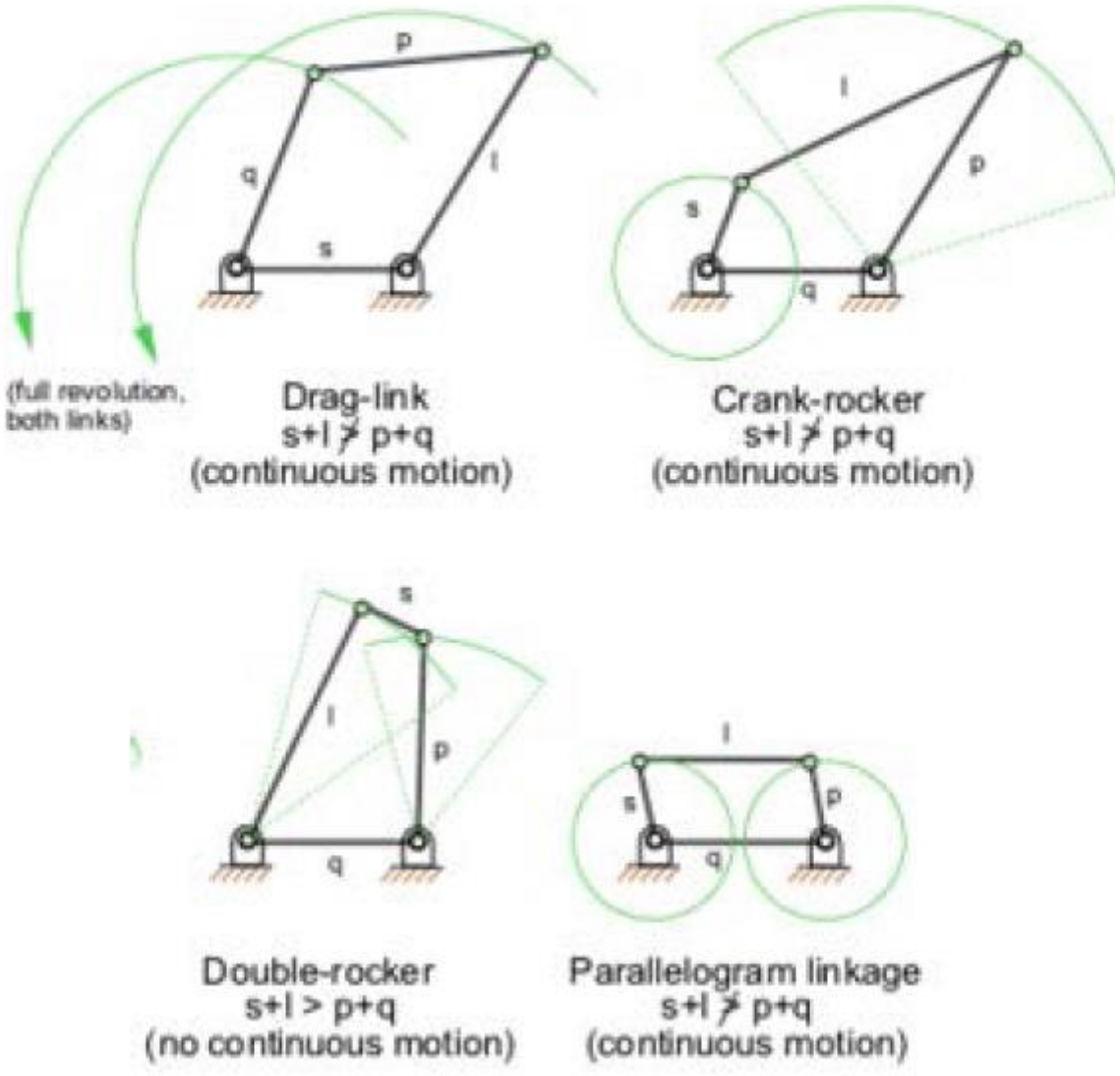




Classification of Closed Planar Four-Bar Linkages-The Grashof Criterion

- Closed planar linkages of four pin-connected rigid links are usually identified as **four-bar linkages.**
- If one of the links can perform a full rotation relative to the other three links, the linkage is called **Grashof mechanism**

Four-Bar Linkages



Grashof condition

A simple relationship which predicts the behavior of a four-link mechanism inversion based only on link lengths.

S=length of shortest link

L=length of longest link

P=length of one link

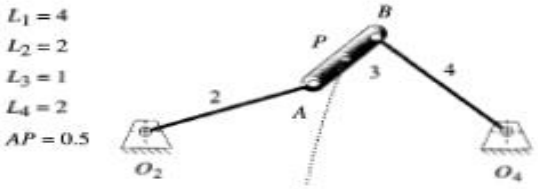
Q=length of other link

If $S+L \leq P+Q$ then at least one link will crank, and the mechanism is called a Grashof mechanism.

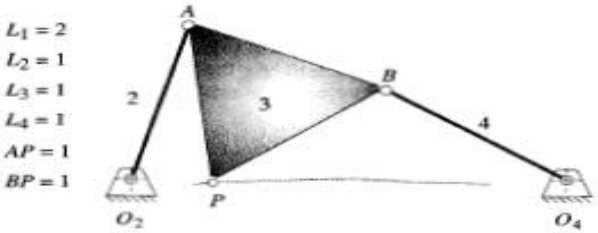
Criteria of motion for each class of Four-Bar Linkages

Type of Mechanism	Shortest link	Relationship between link lengths
Grashof	Any	$S+L \leq P+Q$
crank rocker	Driver crank	$S+L < P+Q$
drag link	Fixed link	$S+L < P+Q$
double rocker	coupler	$S+L < P+Q$
crossover position(change point)		$S+L = P+Q$
Non-Grashof		$S+L < P+Q$
double rocker of the second kind(triple rocker)		$S+L > P+Q$

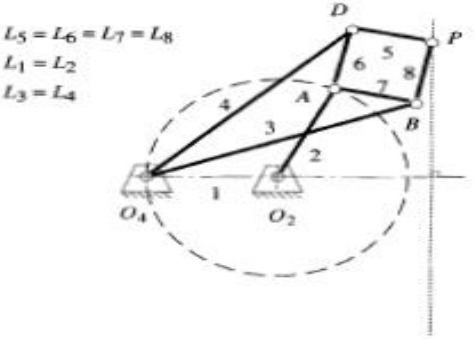
Straight line mechanisms



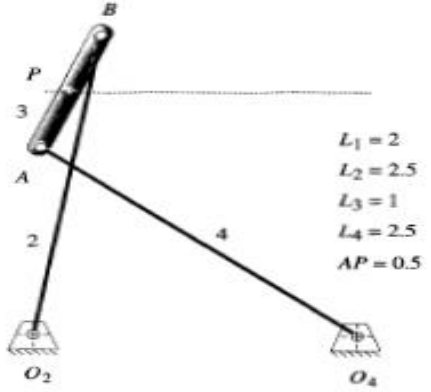
(a) Watt straight-line linkage



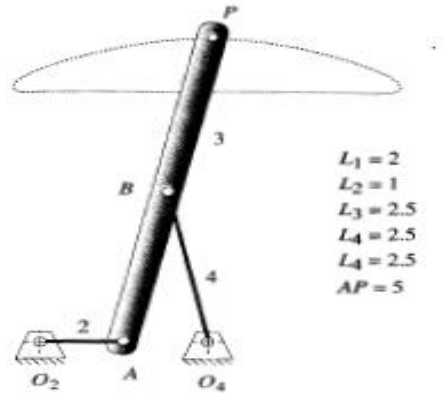
(b) Roberts straight-line linkage



(e) Peaucellier exact straight-line linkage

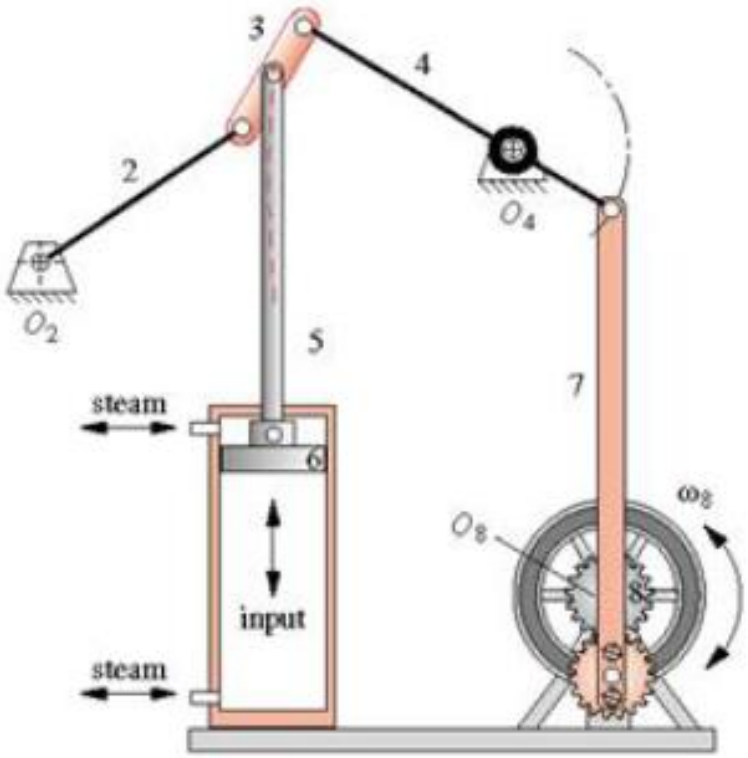


(c) Chebyschev straight-line linkage

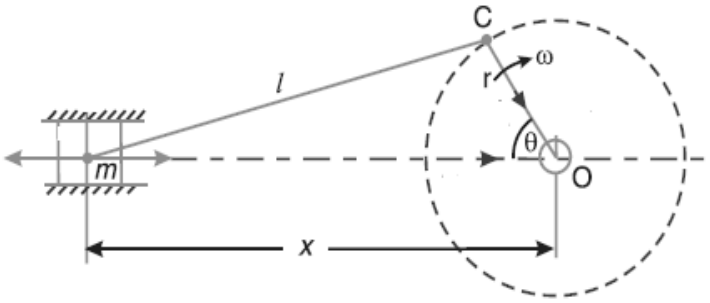


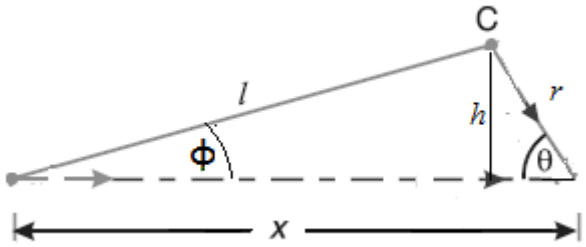
(d) Hoeken straight-line linkage

Straight line mechanisms
watt's linkage as used in his steam engine



Slider-crank mechanism





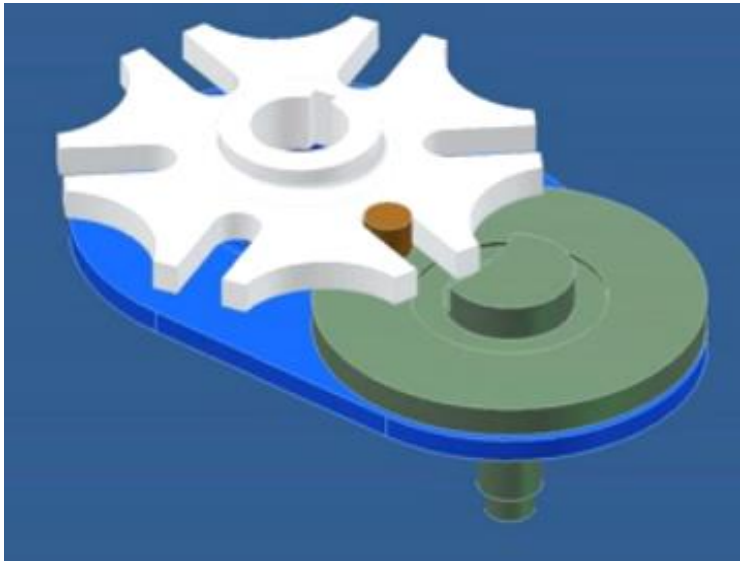
$$h = r\sin\theta = l\sin\phi$$

$$x = r\cos\theta + l\cos\phi$$

$$\cos^2\phi + \sin^2\phi = 1 \Rightarrow \cos\phi = \sqrt{1-\left(\frac{r}{l}\sin\theta\right)^2}$$

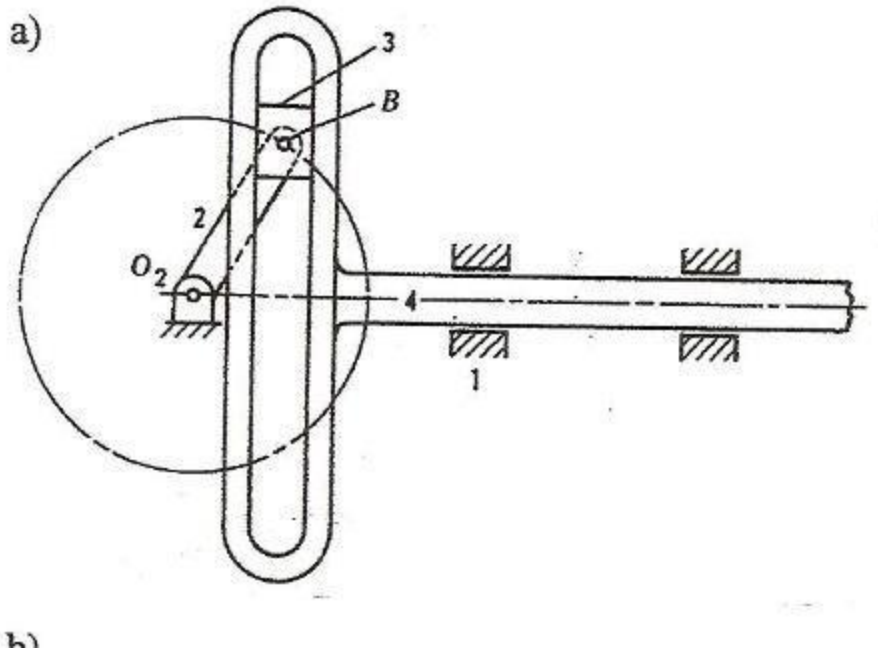
$$\therefore x = r\cos\theta + l\sqrt{1-\left(\frac{r}{l}\sin\theta\right)^2}$$

Geneva Wheel

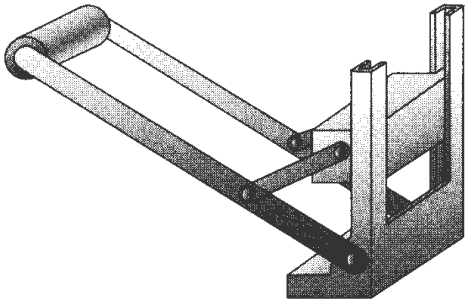




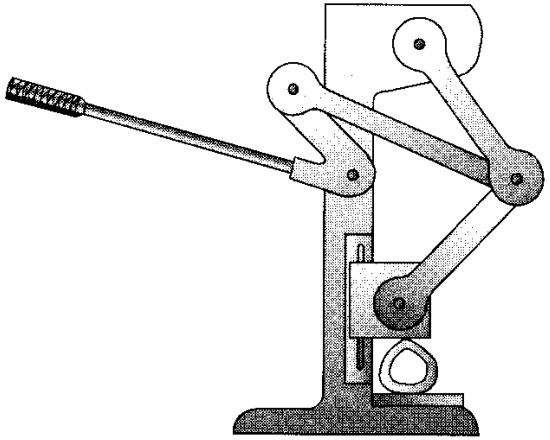
Scotch yoke



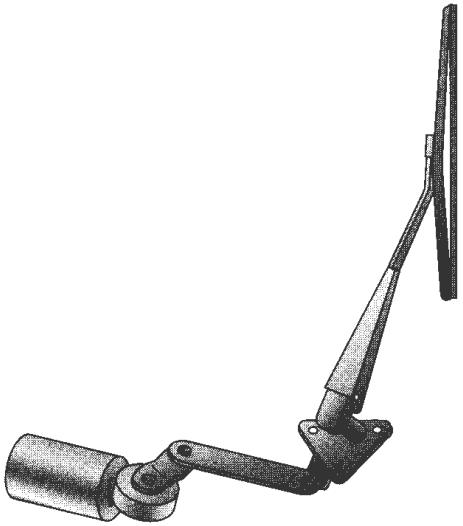
EXAMPLE OF MECHANISM



Can crusher

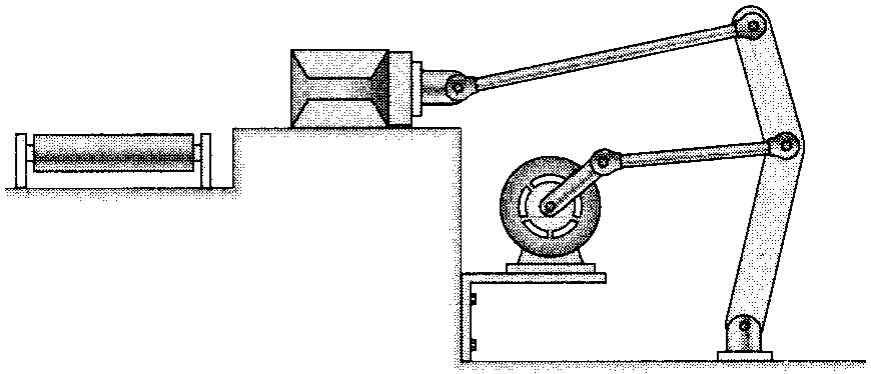


Simple press

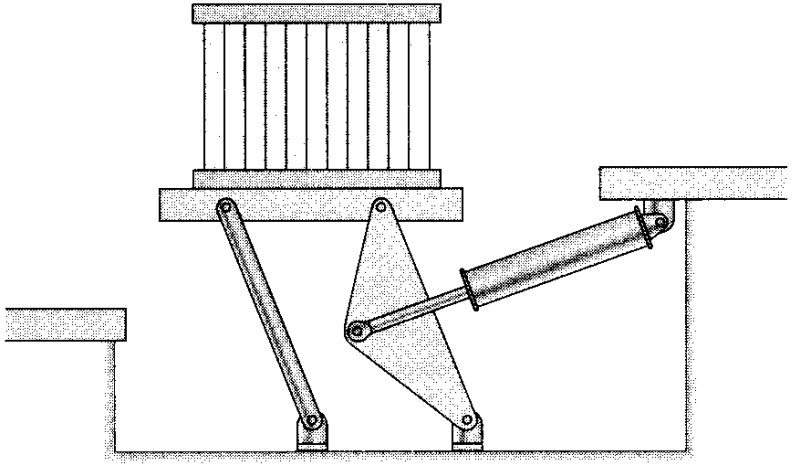


Rear-window wiper

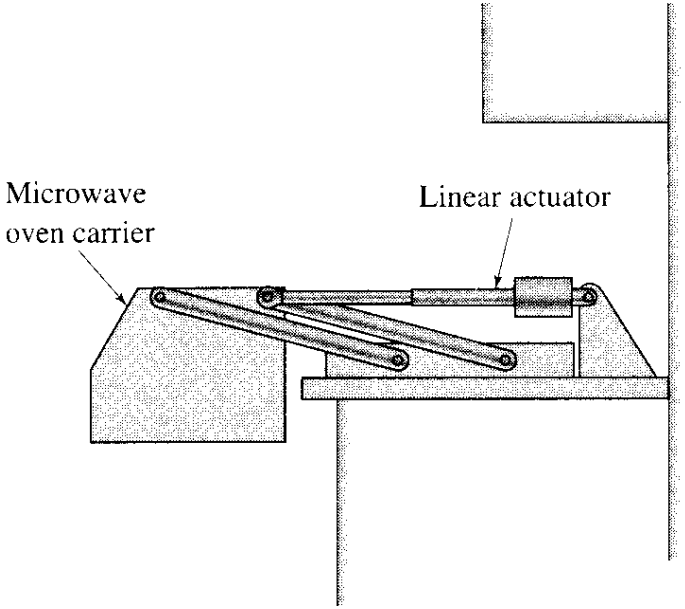
EXAMPLE OF MECHANISMS



Moves packages from an assembly bench to a conveyor

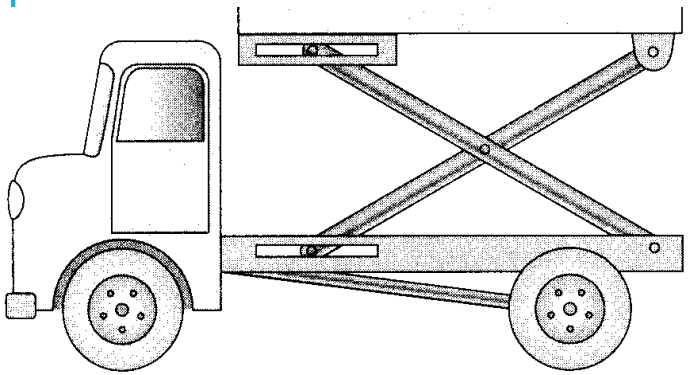


Lift platform

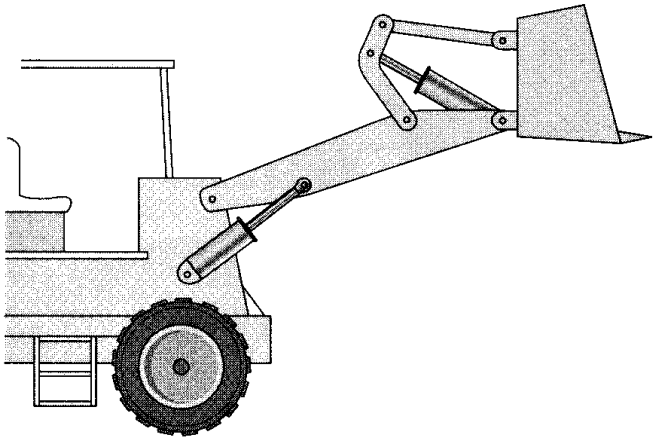


Microwave carrier to assist people on wheelchair

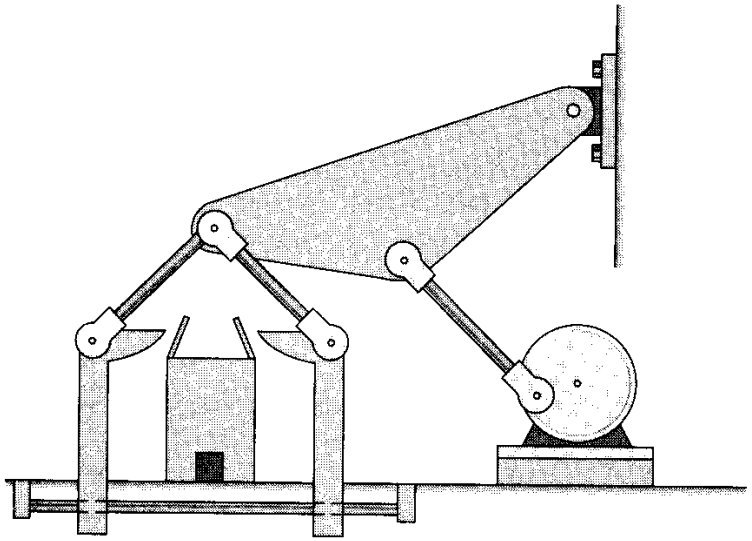
EXAMPLE OF MECHANISMS



Lift platform



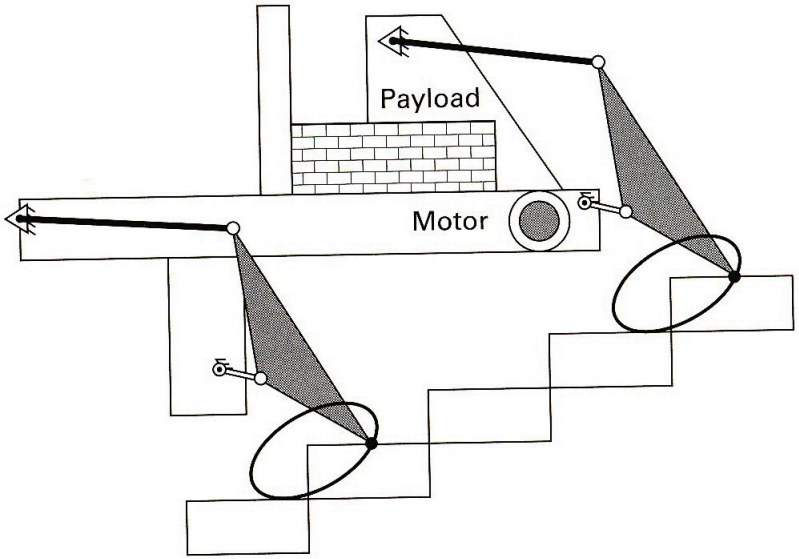
Front loader

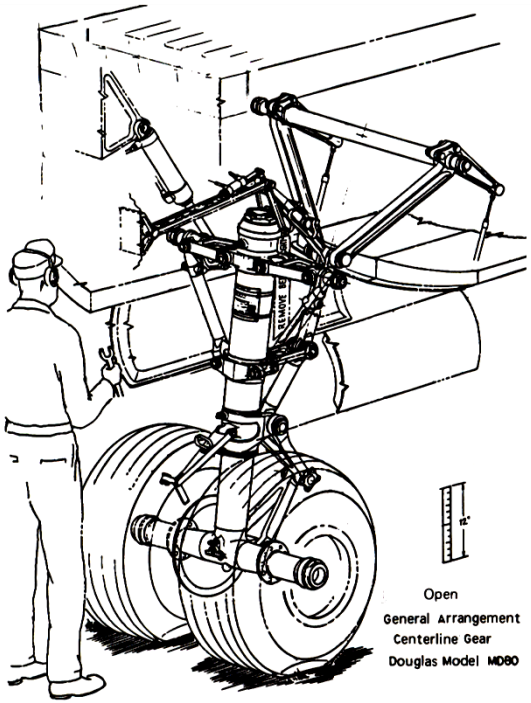


Device to close the
top flap of boxes

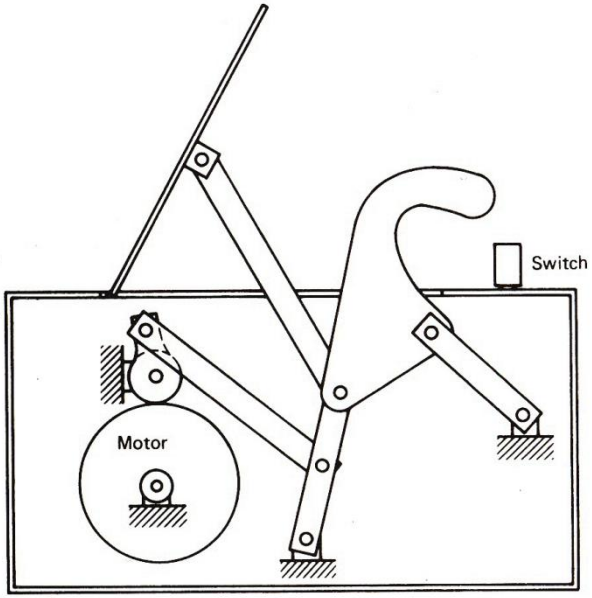
EXAMPLE OF MECHANISMS

Stair climbing mechanism

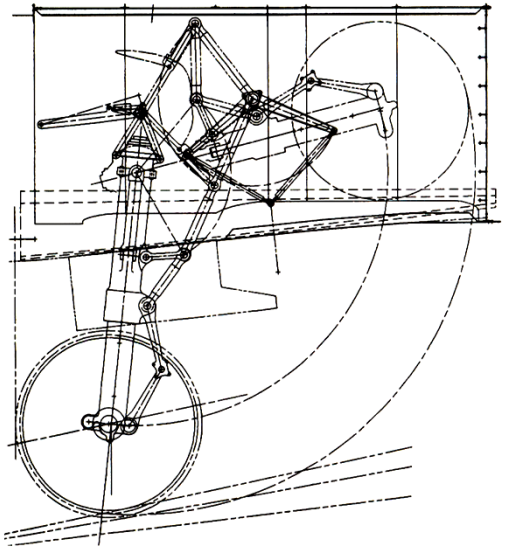




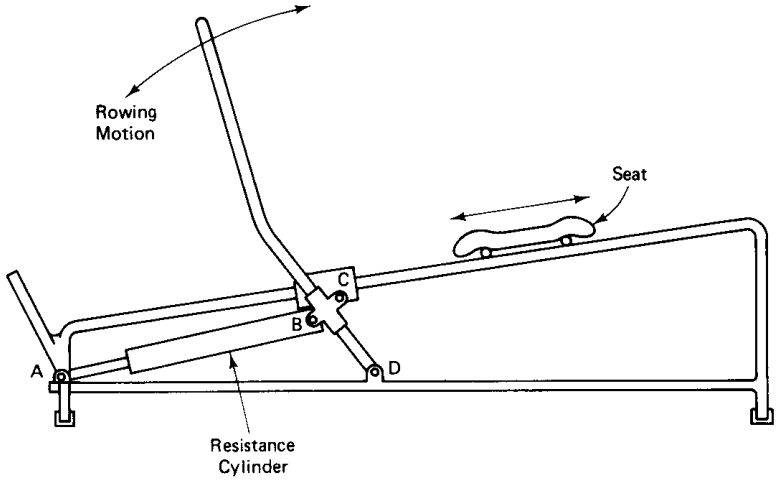
A box that
turns itself off



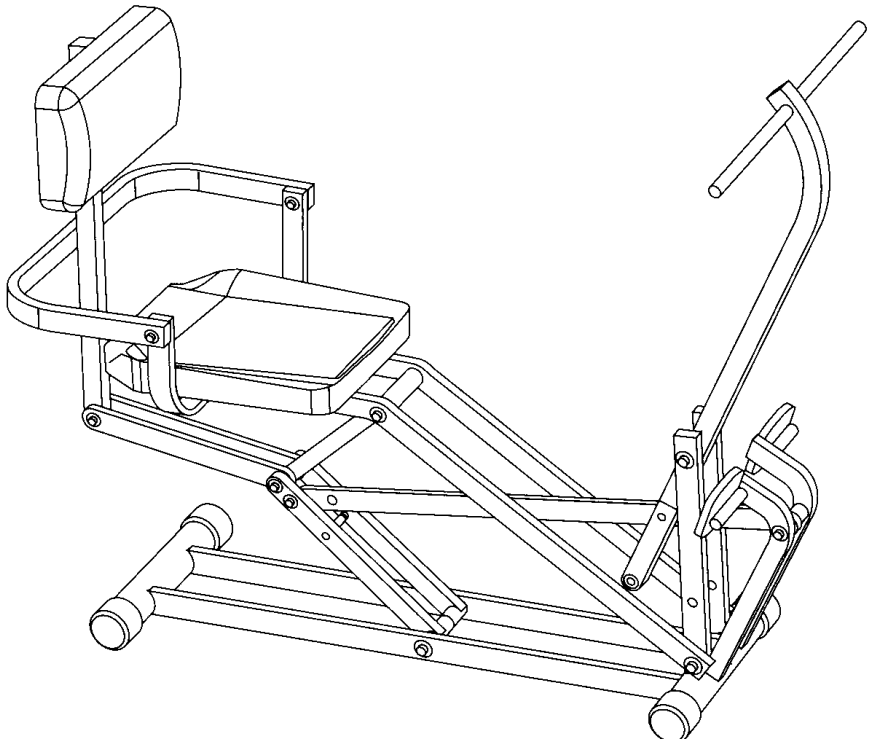
Airplane landing
gear mechanism



EXAMPLE OF MECHANISMS

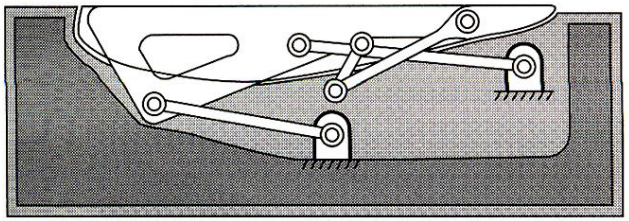


Rowing type exercise machine

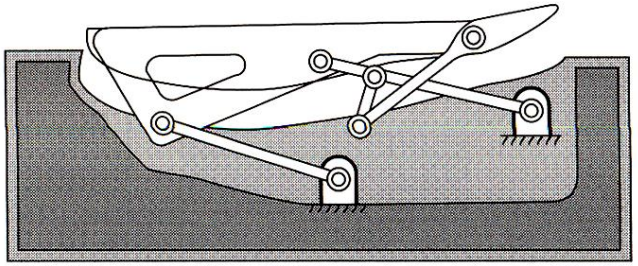


Conceptual design for an exercise machine

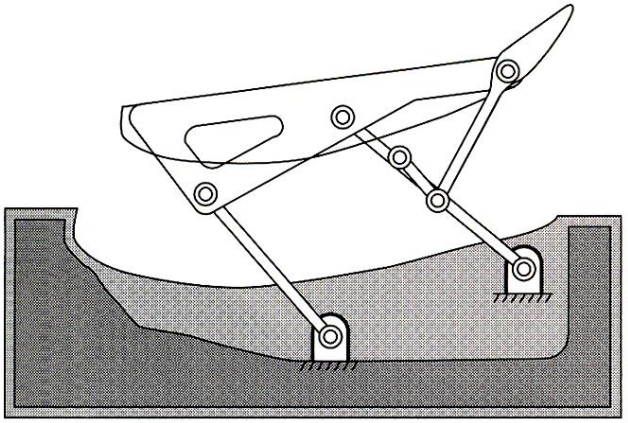
EXAMPLE OF MECHANISMS



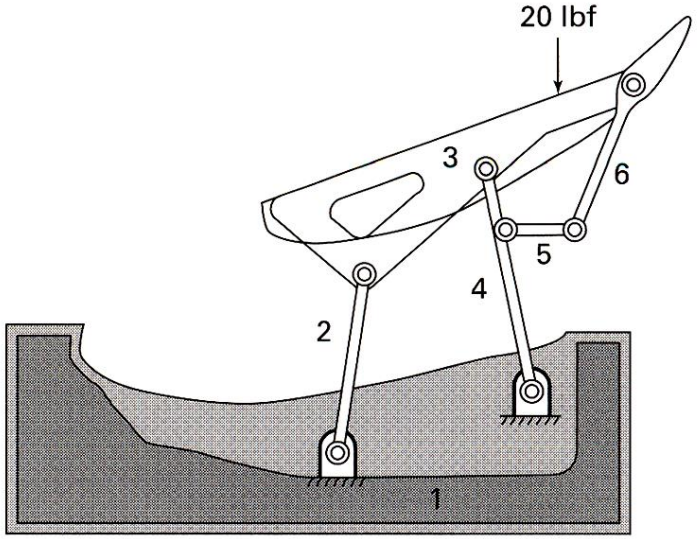
(a) Precision Position 1
(Fully Retracted)



(b) Precision Position 2



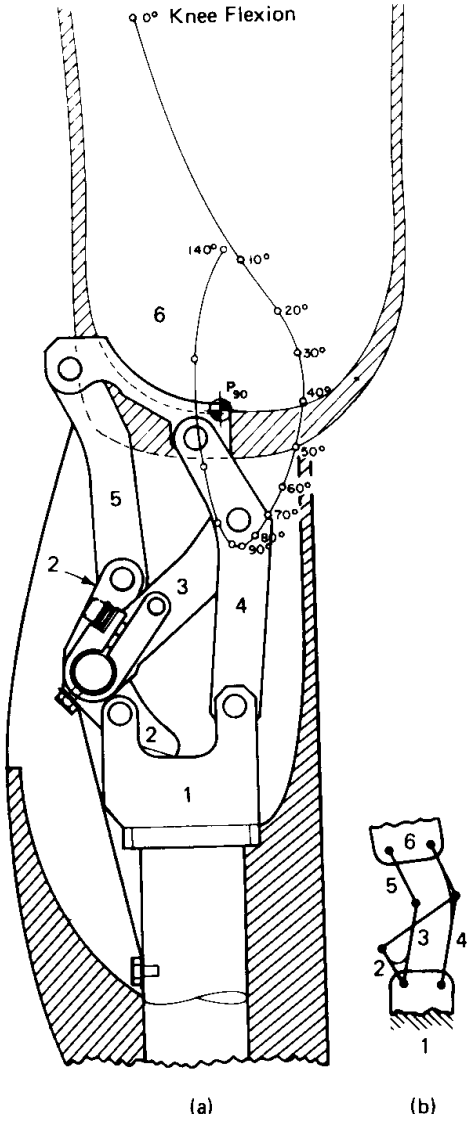
(c) 50% Elevation

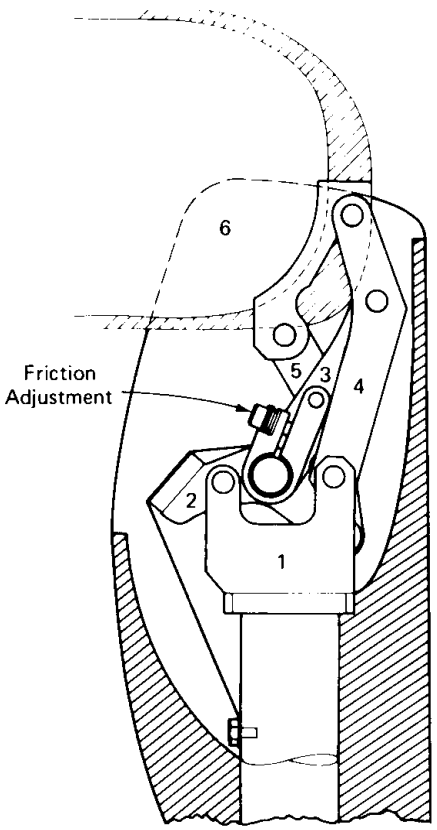


(d) Precision Position 3
(Fully Elevated)

EXAMPLE OF MECHANISMS

Extension position





Flexed position

Six-bar linkage prosthetic knee mechanism

Dimensional Synthesis

- The determination of the proportions (length) of the links necessary to accomplish the desired motions.
- Function generation, path generation and Motion Generation (body guidance).

Function generation, path generation and Motion Generation (body guidance).

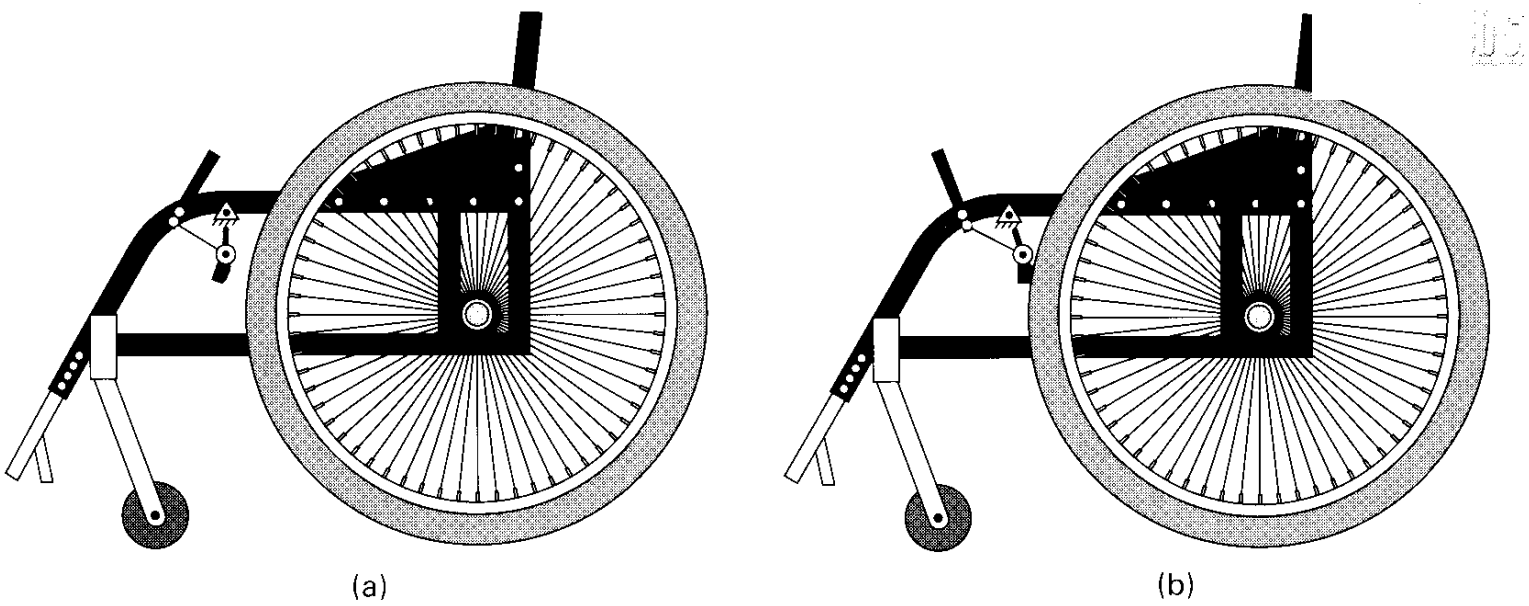
- Function Generation: the correlation of an input motion with an output motion in a mechanism

- Path Generation: Control of a point in the plane such that it follows some prescribed path

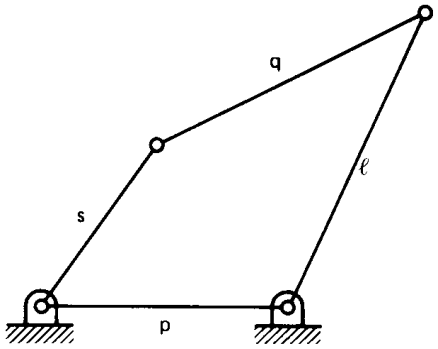
- Motion Generation: Control of a line in the plane such that is follows some prescribed set of sequential positions

Mechanism Categories

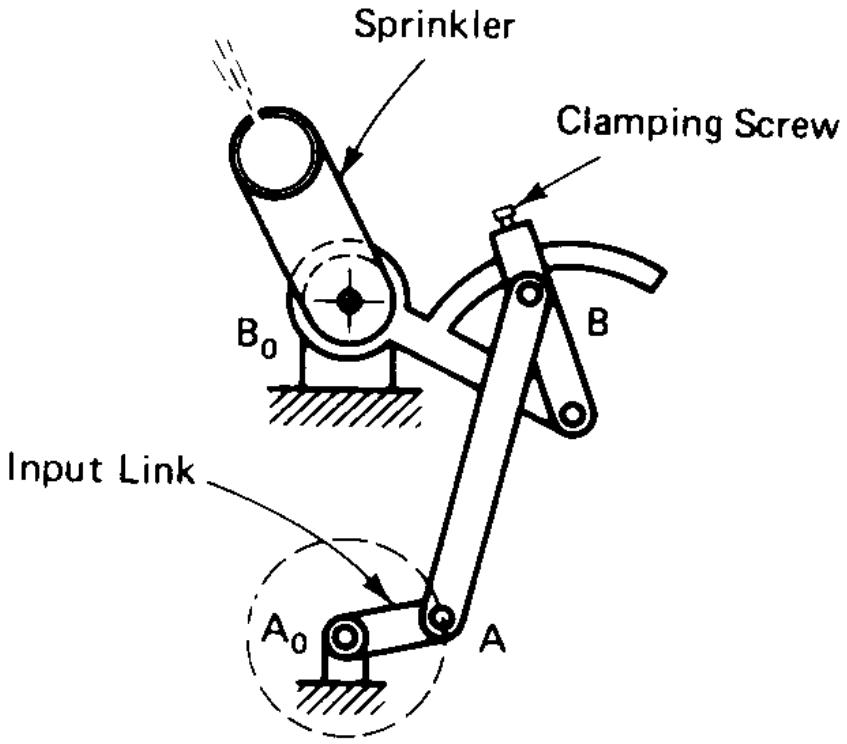
A function generator is a linkage in which the relative motion between links connected to the ground is of interest.



A four-bar hand actuated wheelchair brake mechanism

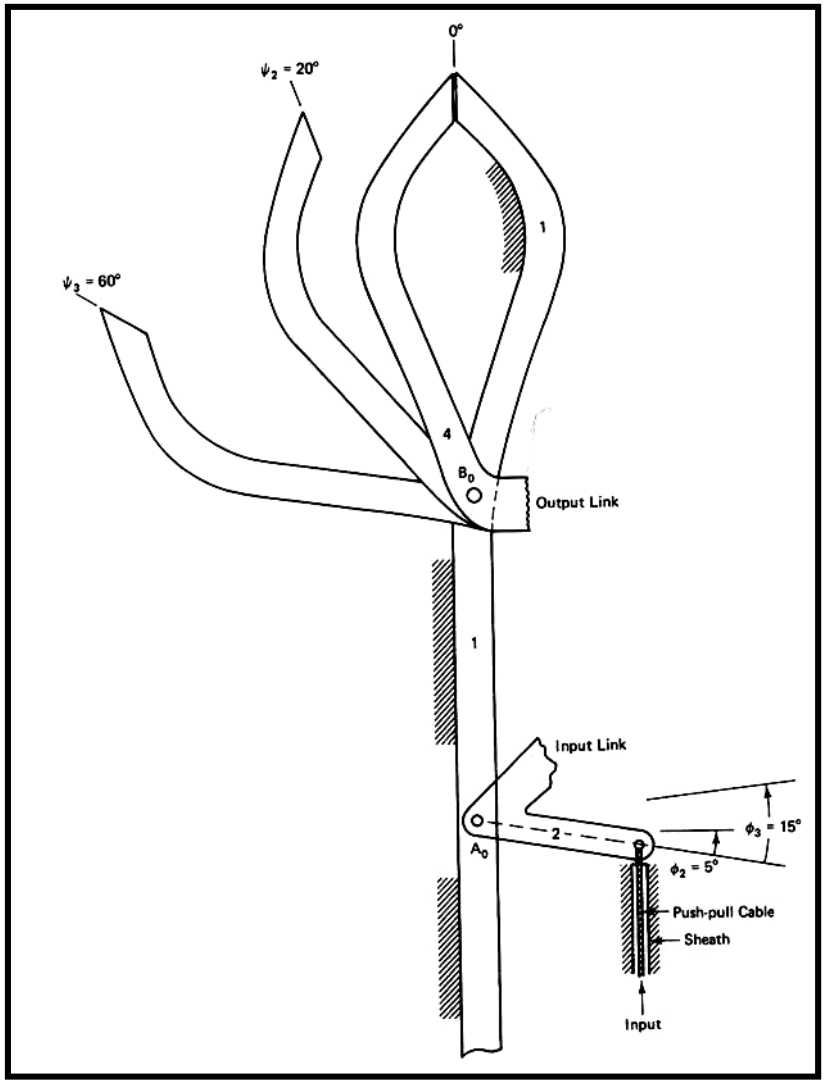


Mechanism Categories

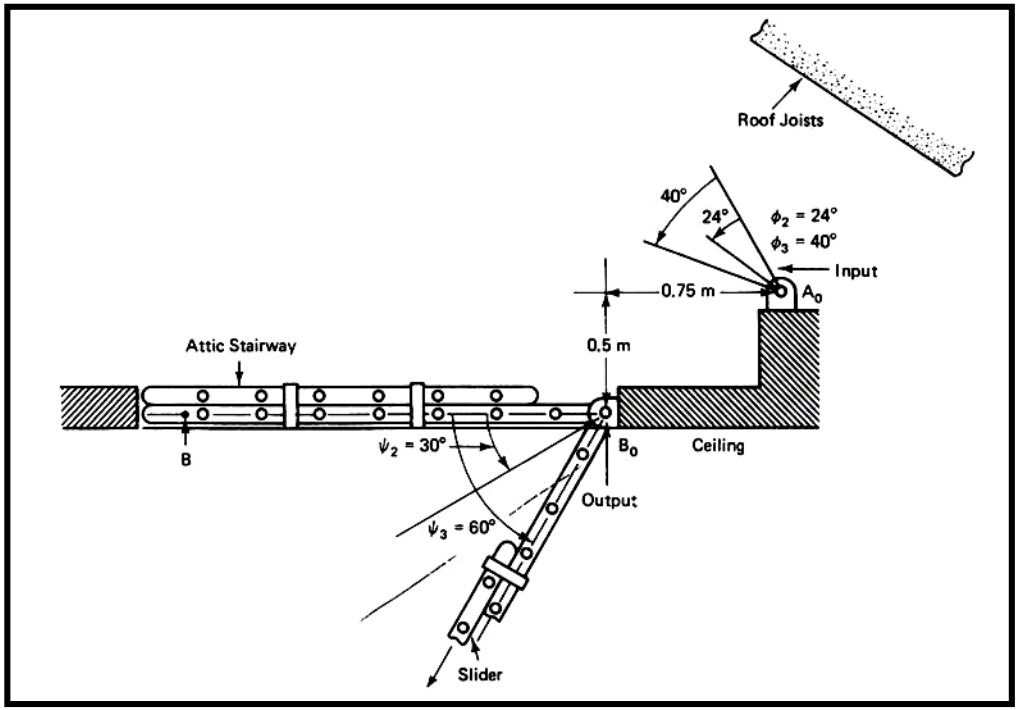


A four-bar drive linkage for a lawn sprinkler

Mechanism Categories



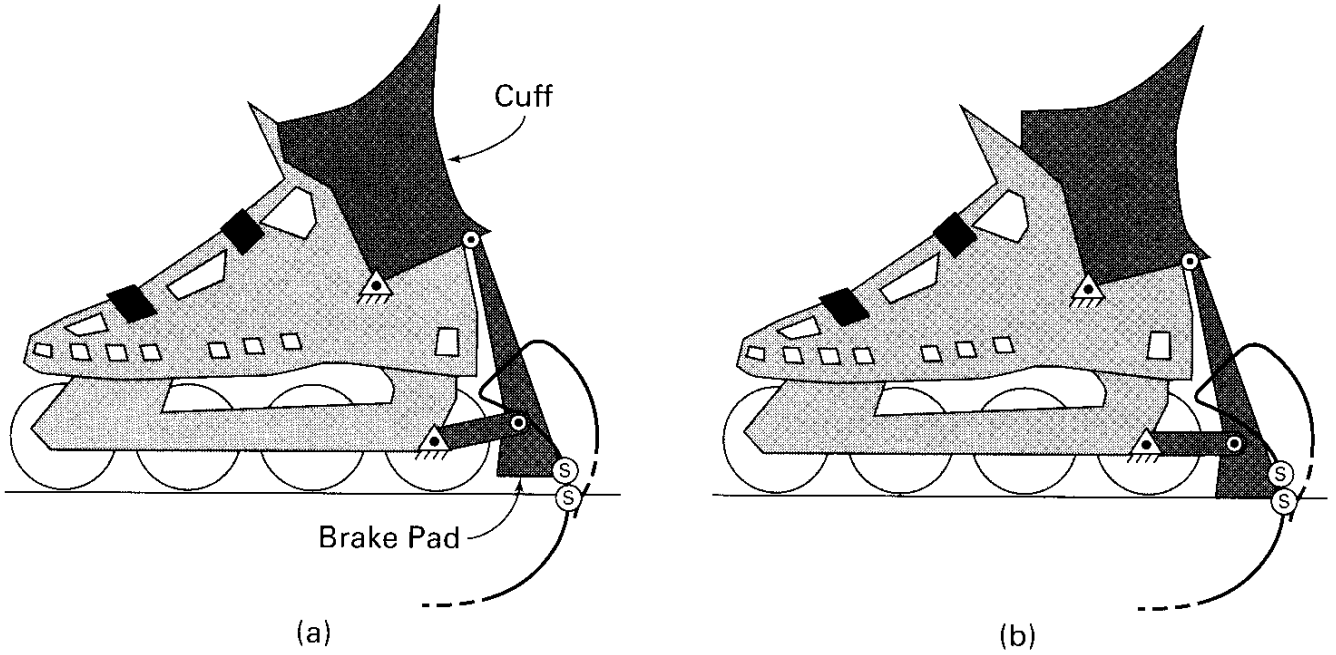
A four-bar function generation mechanism to lower an attic stairway.



A four-bar function generation mechanism to operate an artificial hand used for gripping.

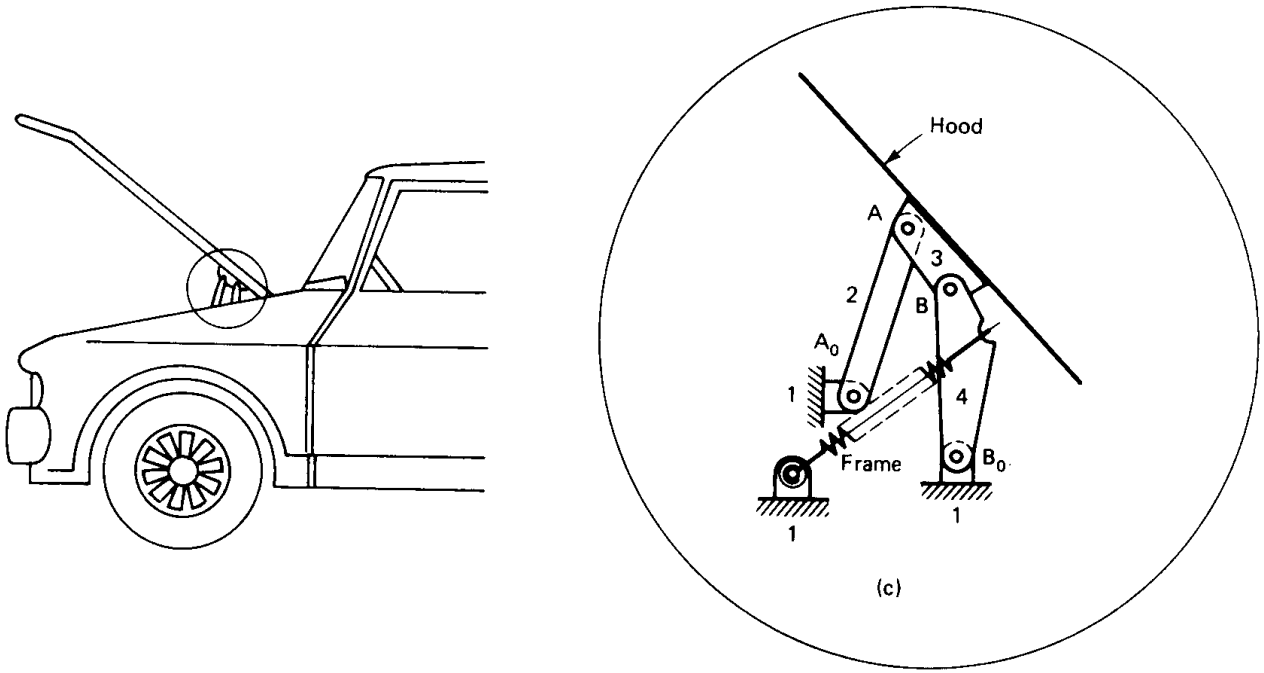
Mechanism Categories

In **motion generation**, the entire motion of the coupler link is of interest (*rigid body guidance*).



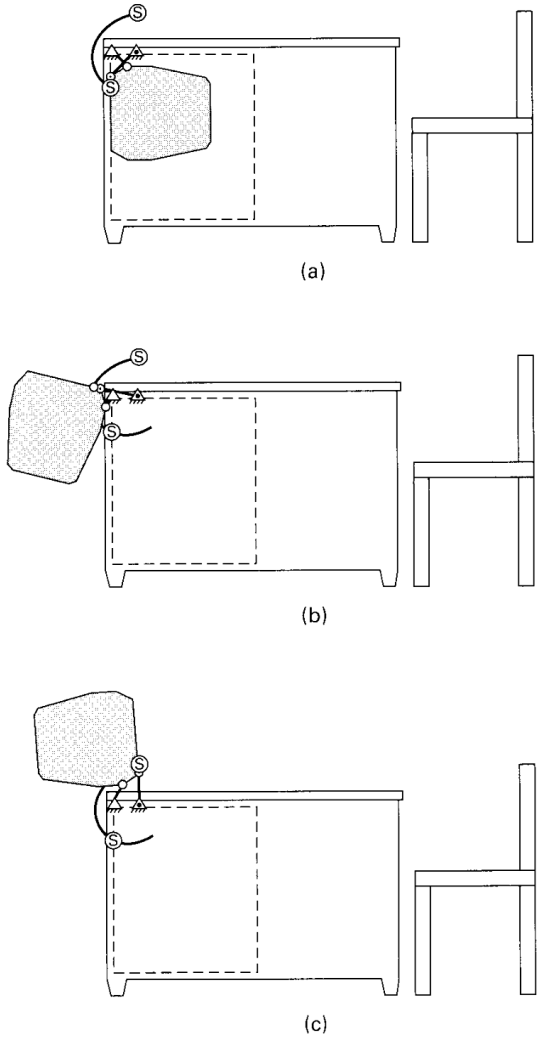
New Rollerblade brake system

Mechanism Categories

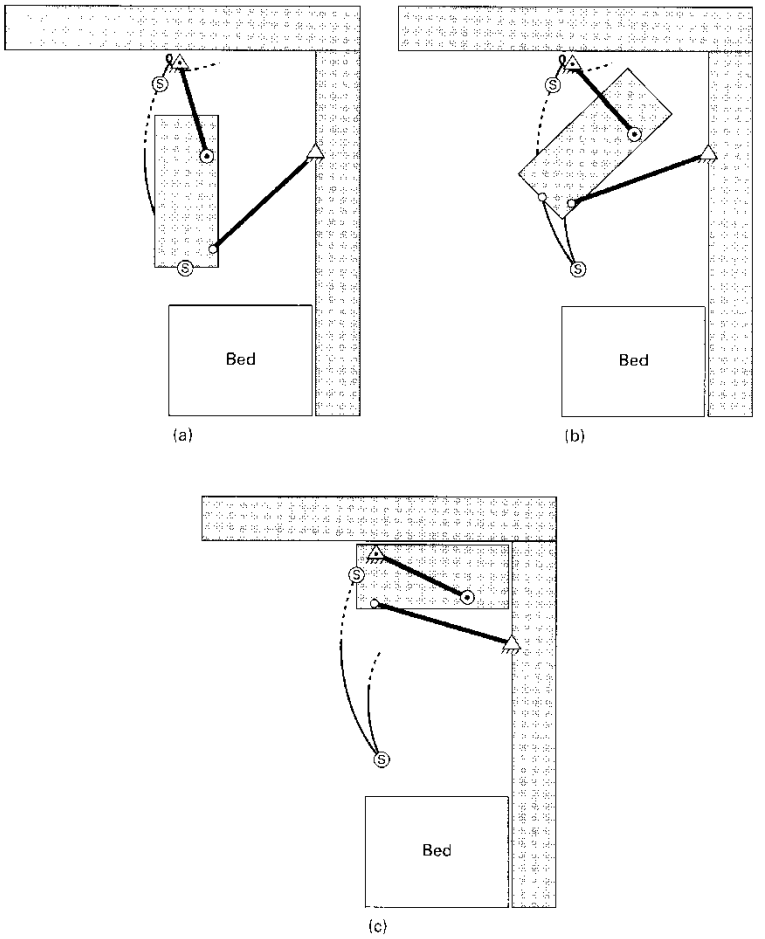


Four-bar automobile hood linkage design

Motion Generation Mechanisms

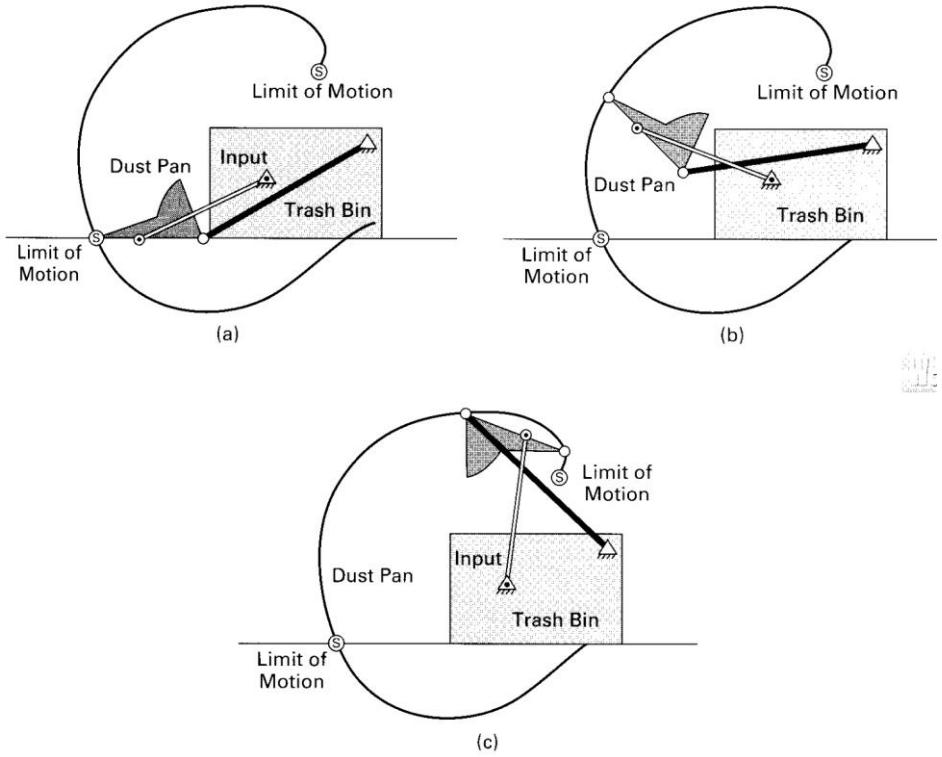


Rotating a monitor into a storage position

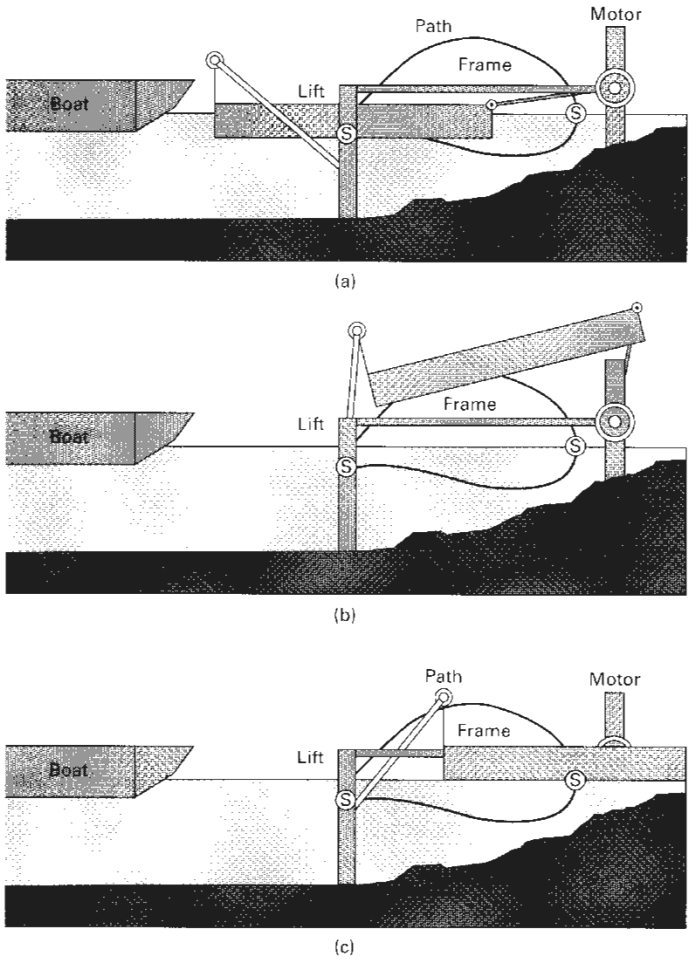


Moving a storage bin from an accessible position to a stored position

Motion Generation Mechanisms



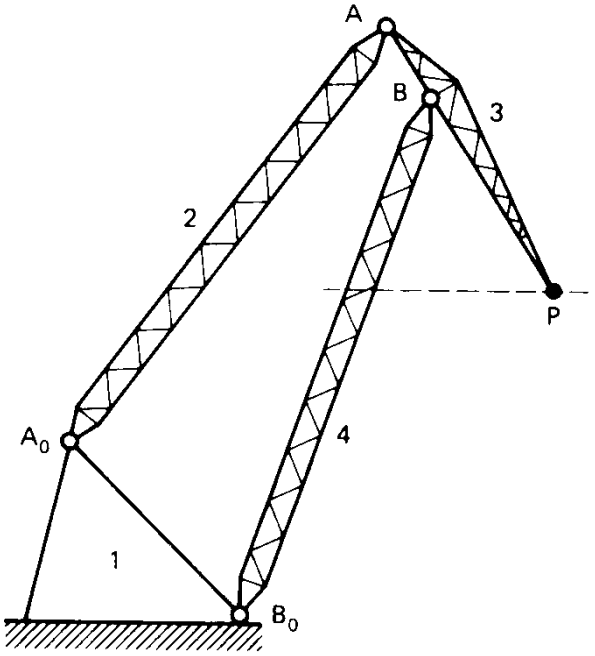
Moving a trash pan from the floor up over a trash bin and into a dump position



Lifting a boat out of water

Mechanism Categories

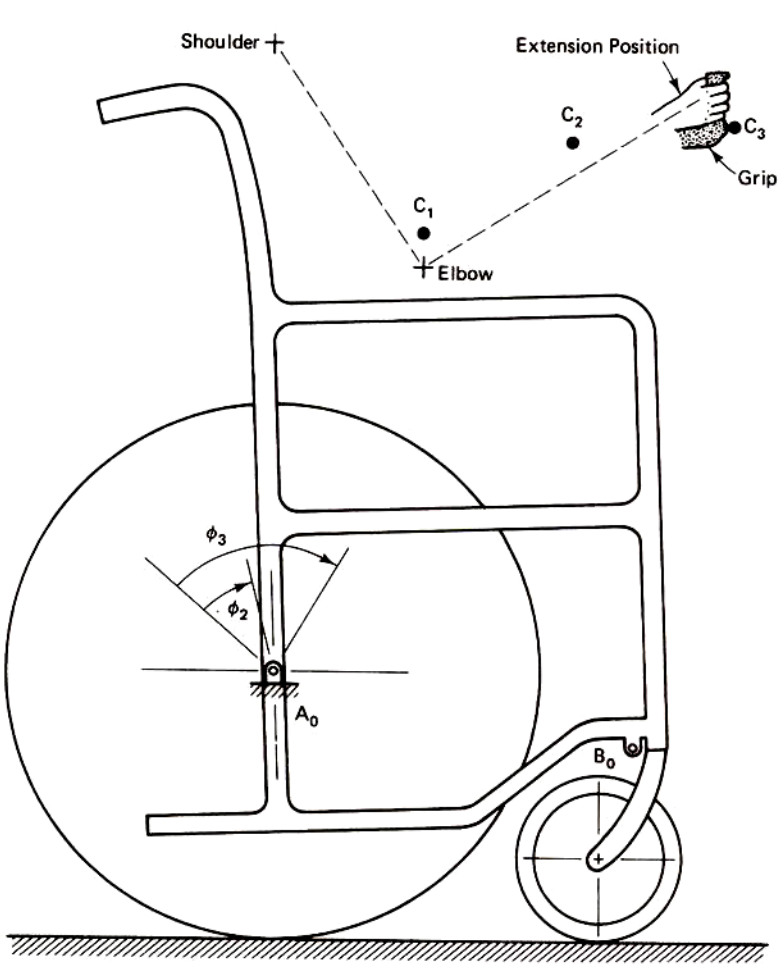
In **path generation**, we are concerned only with the path of a tracer point and not with the motion (rotation) of the coupler link.

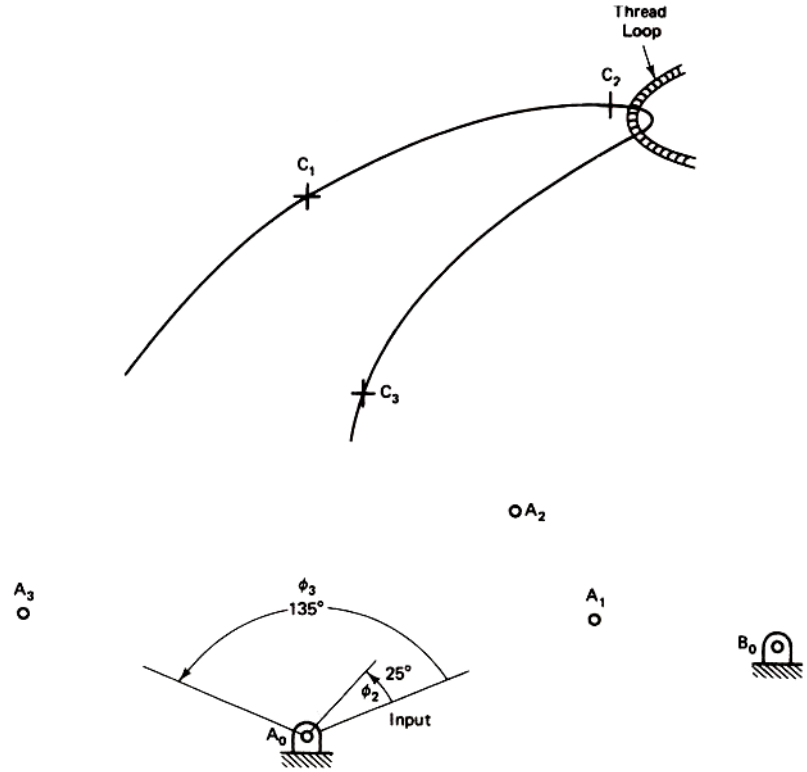


Crane – straight line motion

Mechanism Categories

A four-bar path generation mechanism to guide a thread in an automatic sewing machine

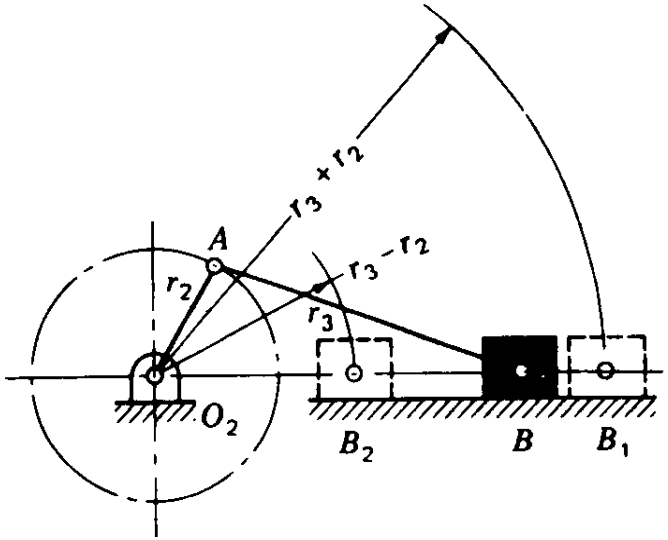




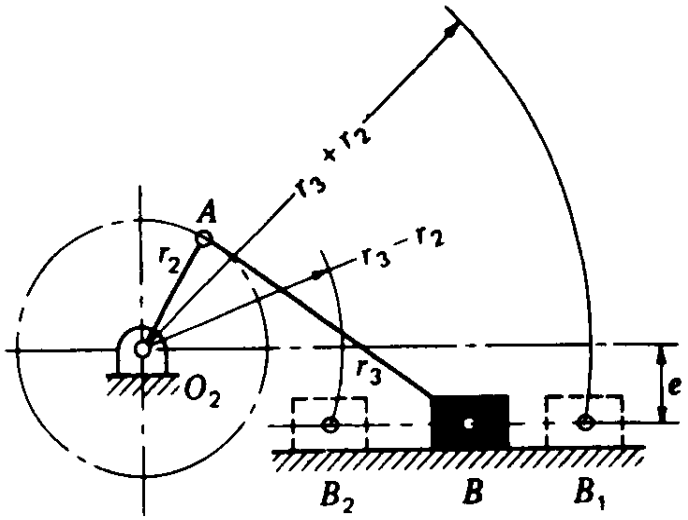
A four-bar path generation mechanism as part of an arm-actuated propulsion system for a wheelchair

Slider-Crank Mechanism

- The slider-crank mechanism below has stroke $B_1B_2 = 2r_2$
- The extreme positions $B_1\ and\ \ B_2$ are found by constructing circle arcs through O_2 of length $r_3 - r_2$ and $r_3 + r_2$ respectively.



Centered slider-crank mechanism



General/ offset slider-crank mechanism

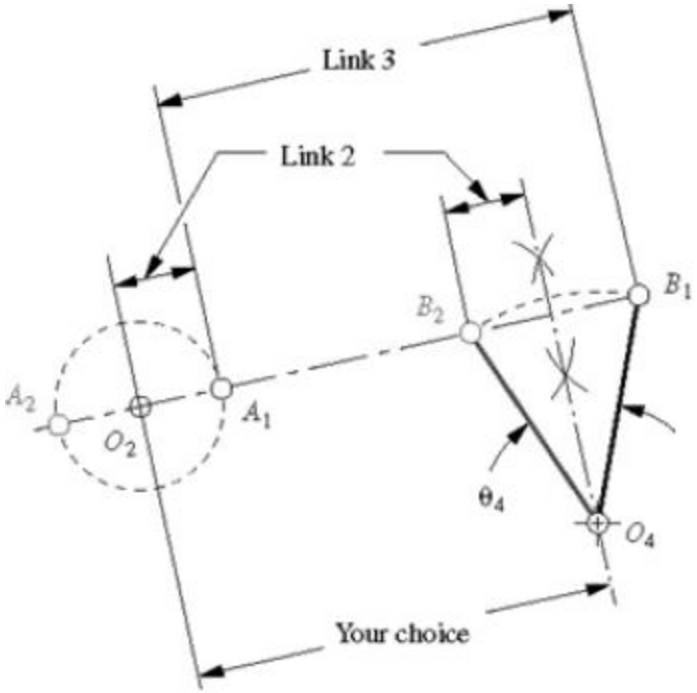
Rocker Output- Two Position with Angular Displacement

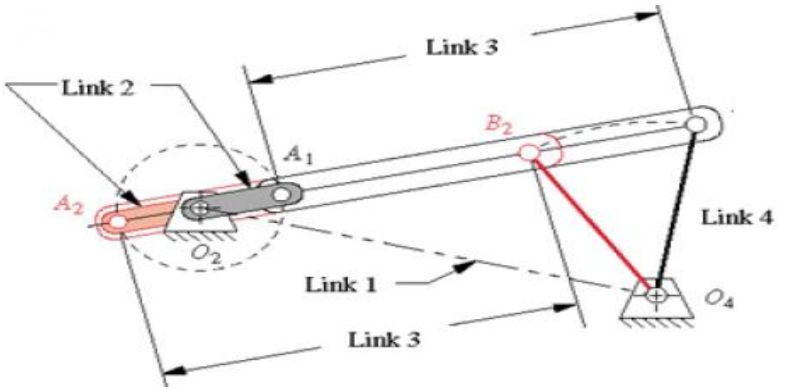
Example 1

Design a four bar Grashof crank-rocker mechanism to give 45° of rocker motion with equal time forward and back, from a constant speed motor input.

Example 1

- Draw the output link O_4B in both extreme positions, B_1 and B_2 in any convenient location, such that the desired angle of motion θ_4 is subtended.
- Draw the chord B_1B_2 and extend it in either direction.
- Select a convenient point O_2 on the line B_1B_2 extended.
- Bisect line segment B_1B_2, draw a circle of that radius about O_2.
- Label the two intersection of the circle and B_1B_2 extended, A_1 and A_2.
- Measure the length of the coupler as A_1 to B_1 or A_2 to B_2.
- Measure ground length 1, crank length 2, and rocker length 4.
- Find the Grashof condition. If non-Grashof, redo steps 3 to 8 with O_2 further from O_4.
- Make model of the linkage and check its function and transmission angles.

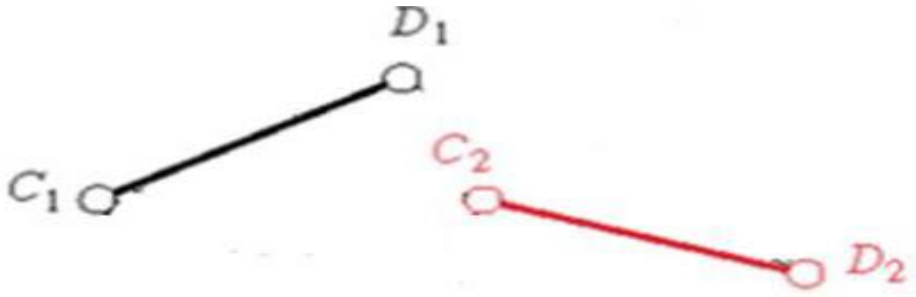




Example 2

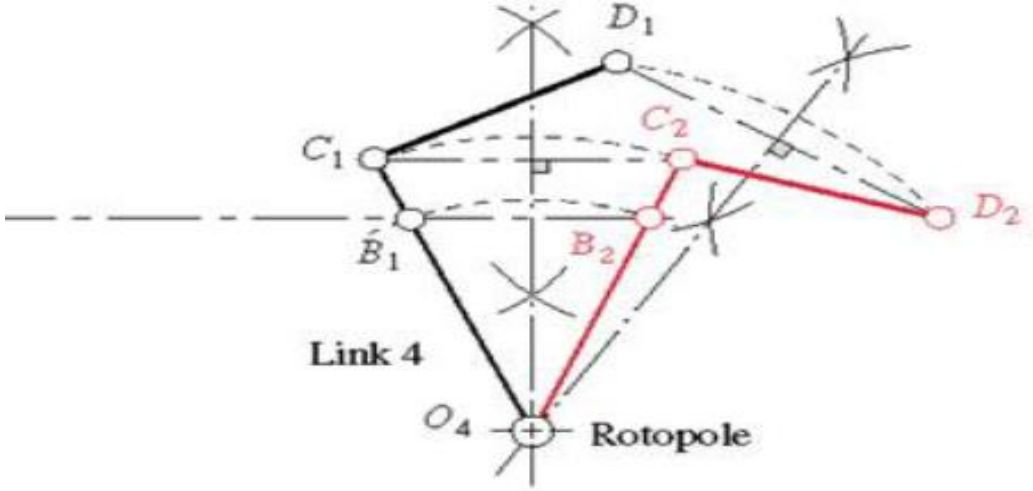
- Rocker Output- Two Position with Complex Displacement (Motion)

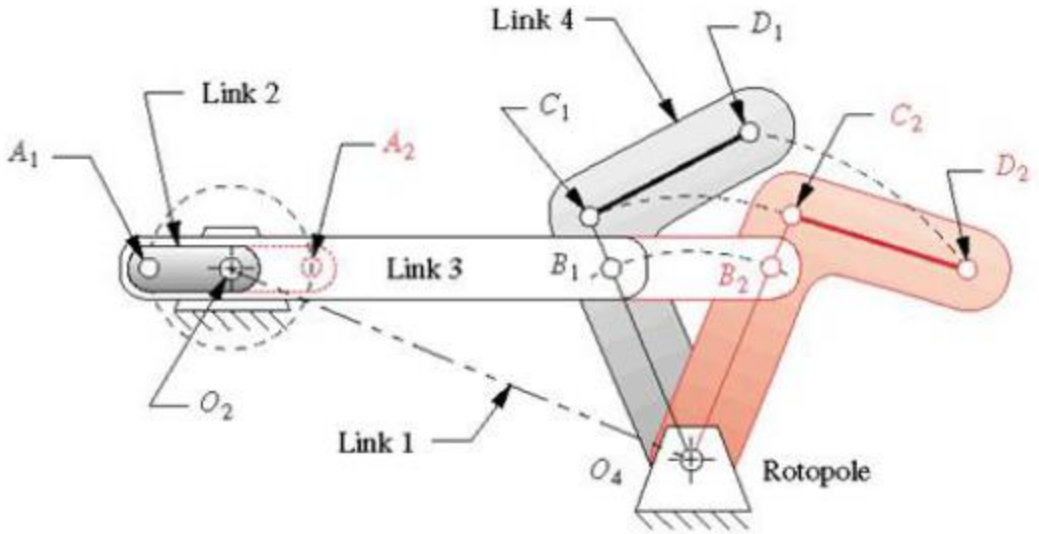
– Design a fourbar linkage to move link CD from C_1D_1 to C_2D_2.



Example 2

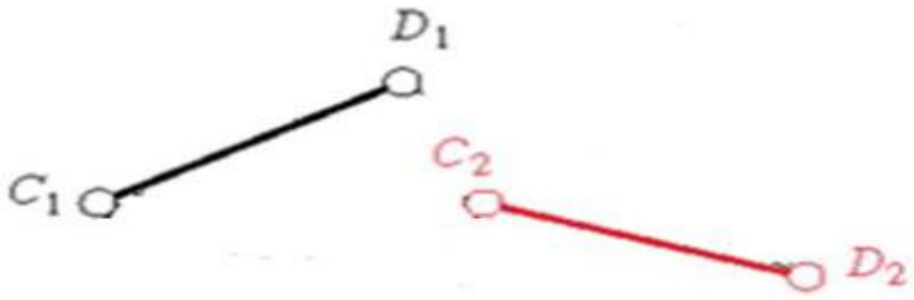
- Draw the link CD in its two desired positions, C_1D_1 and C_2D_2 in plane as shown.
- Draw construction line from point C_1 to C_2 and from point D_1 to D_2.
- Bisect line C_1C_2 and line D_1D_2 and extend their perpendicular bisectors to intersect at O_4. Their intersection is the **rotopole**.
- Select a convenient radius and draw an arc about the rotopole to intersect both lines O_4C_1 and O_4C_2. Label the intersection B_1 and B_2.
- Do steps 2 to 8 of example 1 to complete the linkage.
- Make a model of the linkage and articulate it to check its function and its transmission angles.





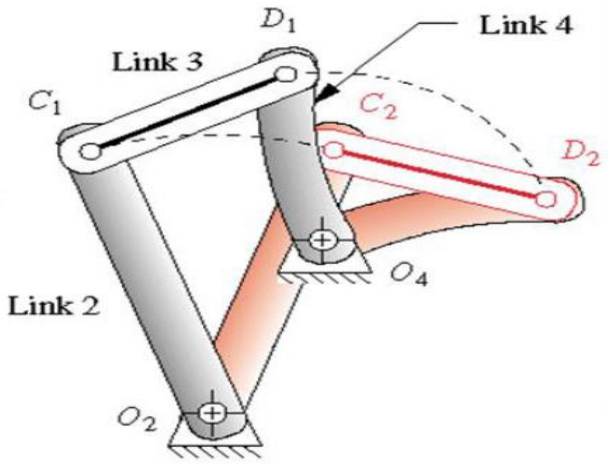
Coupler Output- Two Position with Complex Displacement (Motion)

Example 3: Design a fourbar linkage to move link CD from C_1D_1 to C_2D_2 (with moving pivots at C and D).



Example 3

- Draw the link CD in its two desired positions, C_1D_1 and C_2D_2 in plane as shown.
- Draw construction line from point C_1 to C_2 and from point D_1 to $^{D}2$.
- Bisect line C_1C_2 and line D_1D_2 and extend their perpendicular bisectors in convenient directions. The rotopole will not be used in this solution.
- Select any convenient point on each bisector as the fixed pivots O_2 and O_4, respectively.
- Connect O_2 with C_1 and call it link 2. Connect O_4 with D_1 and call it link 4.
- Line C_1D_1 is link 3. Line O_2O_4 is link 1.
- Check the Grashof condition, and repeat steps 4 to 7 if unsatisfied. Note that any Grashof condition is potentially acceptable in this case.
- Construct a model and check its function to be sure it can get from the initial to final position without encountering ant limit (toggle) positions.
- Check the transmission angles.

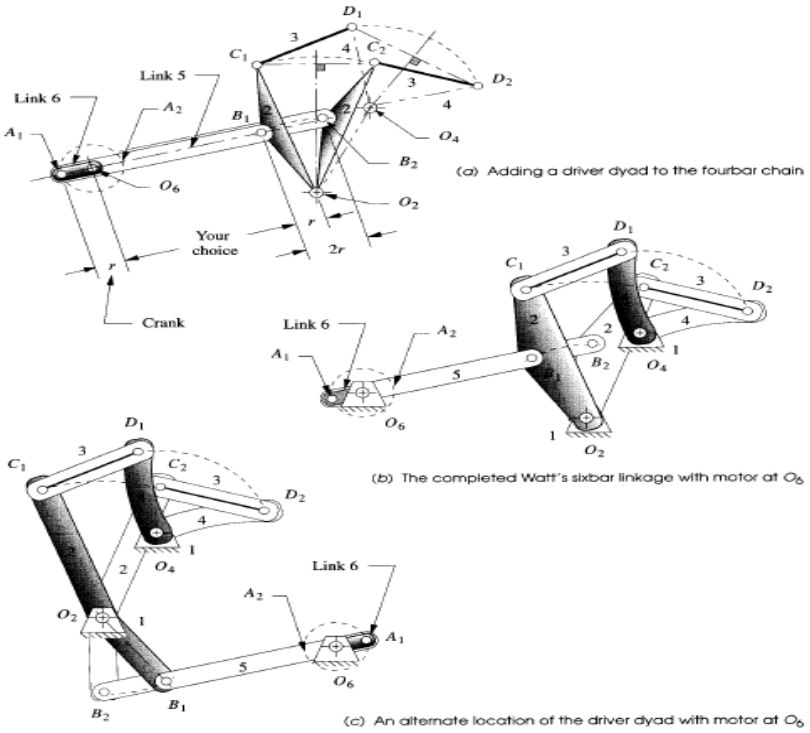


Review Example 3

Design a dyad to control and limits the extremes of motion of the linkages in the previous example to its two design positions

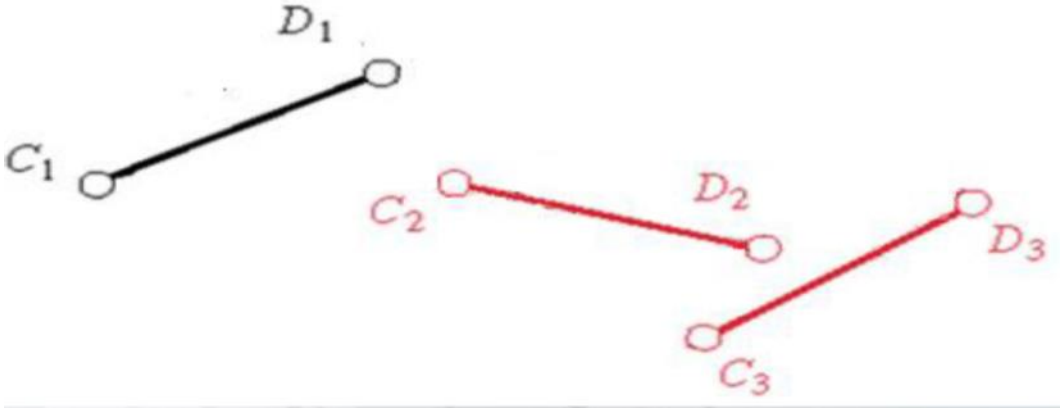
Review Example 3

- Select a convenient point on link 2 of the linkage designed in Example 3. Note that it need not be on the line O_2C_1. Label this point B_1.
- Draw an arc about center O_2 through B_1 to intersect the corresponding line O_2B_2 in the second position of link 2. Label this point B_2. The chord B_1B_2 provides us with the same problem in Example 1.
- Do steps 2-9 of Example 1 to complete the linkage, except add links 5 and 6 and center O_6 rather than links 2 and 3 and center O_2 . Link 6 will be the driver crank. The fourbar subchain of the links O_6, A_1, B_1, O_2 must be a Grashof crank-rocker.



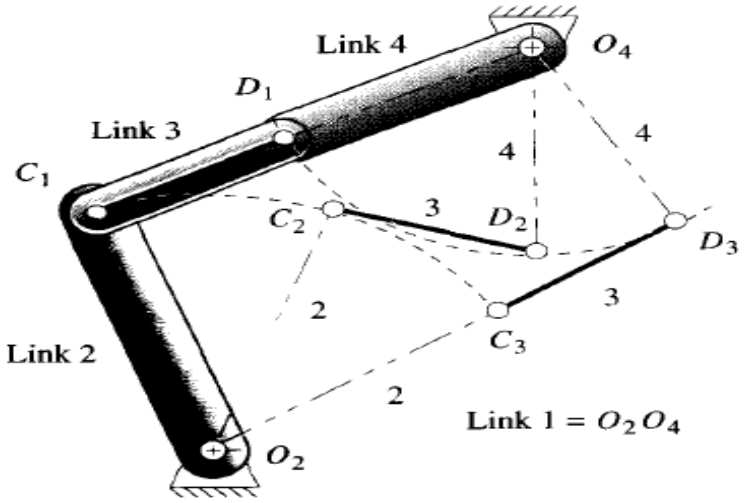
Three-Position Synthesis

- **Example 4** – Coupler Output – 3 position with Complex Displacement
- Design a fourbar linkage to move the link CD shown from position C_1D_1 to C_2D_2 and then to position C_3D_3. Moving pivots are C and D. Find the fixed pivot locations.



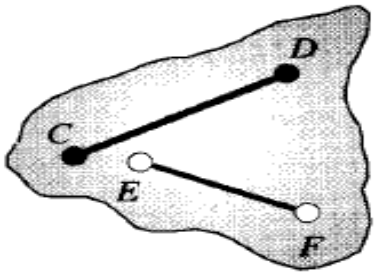
Example 4

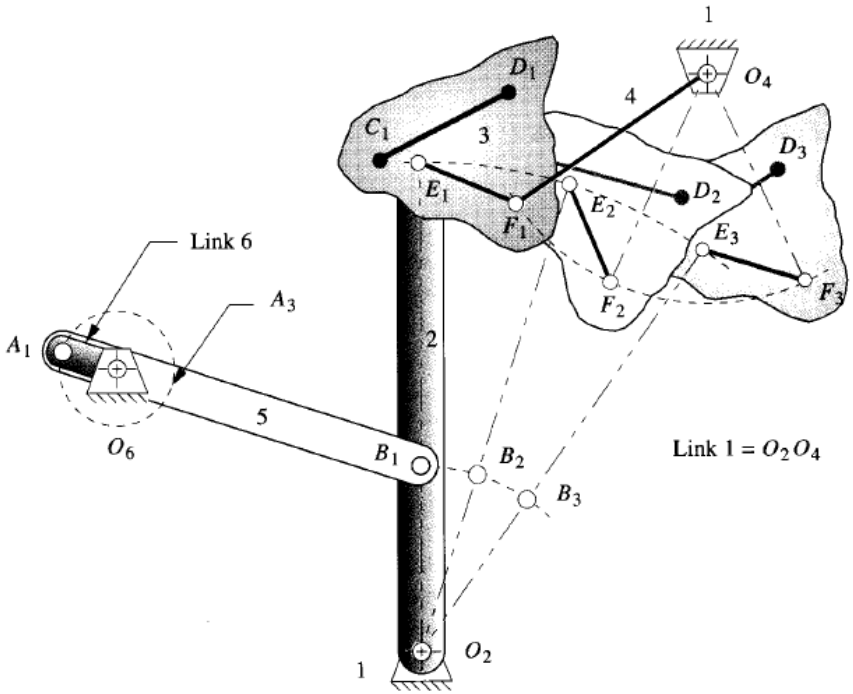
- Draw link CD in its three position
- C_1D_1, C_2D_2 , C_3D_3 in the plane as shown.
- Draw construction lines from point C_1 to C_2 and from C_2 to C_3.
- Bisect line C_1C_2 and line C_2C_3 and extend their perpendicular bisector until they intersect. Label their intersection O_2.
- Repeat steps 2 and 3 for lines D_1D_2 and D_2D_3. Label the intersection O_4.
- Connect O_2 with C_1 and call link 2. Connect O_4 with D_1 and call link 4.
- Line C_1D_1 is link 3. Line O_2O_4 is link 1.
- Check the Grashof condition. Note that any Grashof condition is potentially acceptable in this case.
- 8.Construct a model and check its function to be sure it can get from initial to final position without encountering any limits positions.



Three-Position Synthesis – Example 5

- Coupler Output – 3 position with Complex Displacement (Alternate Attachment Points for Moving Pivots)
- Design a fourbar linkage to move the link CD shown from position C_1D_1 to C_2D_2 and then to position C_3D_3. Use different moving pivot than CD. Find the fixed pivot locations.

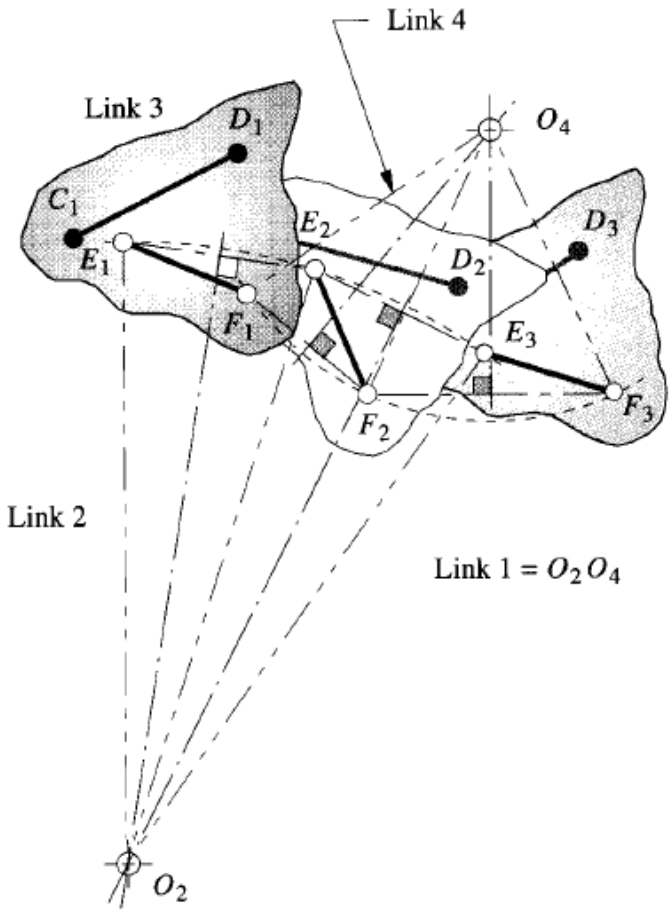




(c) Completed Watt's sixbar linkage with motor at O_6

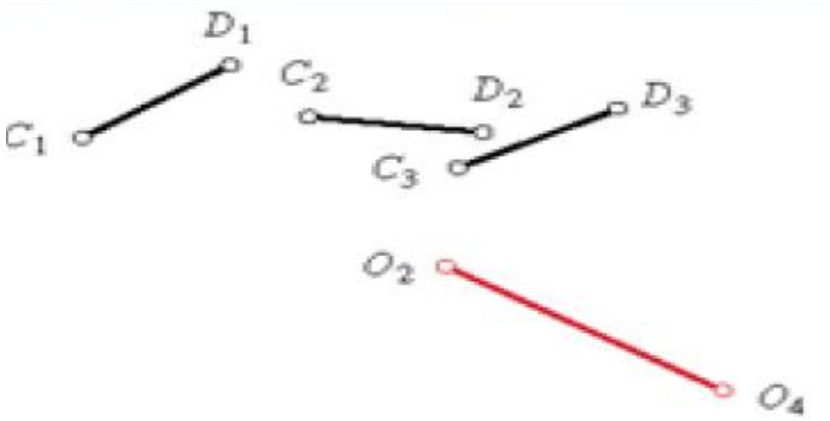
Example 5

- Draw link CD in its three position C_1D_1, C_2D_2 , C_3D_3 in the plane as shown.
- Define new attachment points E_1 and F_1 that have a fixed relationship between C_1D_1 and E_1F_1 within the link. Now use E_1F_1 to define the three position of the link.
- Draw construction lines from point E_1 to E_2 and from E_2 to E_3.
- Bisect line E_1E_2 and line E_2E_3 and extend their perpendicular bisector until they intersect. Label their intersection O_2.
- Repeat steps 2 and 3 for lines F_1F_2 and F_2F_3. Label the intersection O_4.
- Connect O_2 with E_1 and call link 2. Connect O_4 with F_1 and call link 4.
- Line E_1F_1 is link 3. Line O_2O_4 is link 1
- Check the Grashof condition. Note that any Grashof condition is potentially acceptable in this case.
- Construct a model and check its function to be sure it can get from initial to final position without encountering any limits positions. If not, change locations of point E and F and repeat steps 3 to 9.



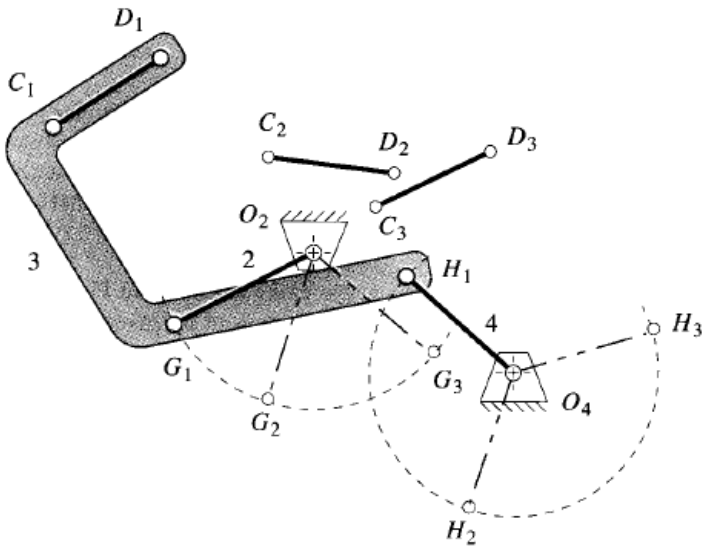
Example 6

- Example 6 – Three –Position Synthesis with Specified Fixed Pivots - Inverting the 3-position problem
- Invert a fourbar linkage which move the link CD shown from position C_1D_1 to C_2D_2 and then to position C_3D_3. Use specified fixed pivots O_2 and O_4.

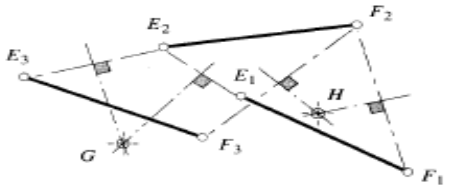


Example 6

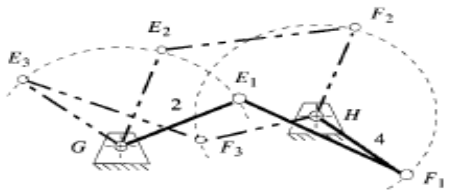
- Draw link CD in its three position C_1D_1, C_2D_2 , C_3D_3 in the plane as shown.
- Draw the ground link O_2O_4 in its desired position in the plane with respect to the first coupler position C_1D_1
- Draw construction arc from point C_2 to O_2 and from D_2 to O_2 whose radii define the side of triangle $C_2O_2D_2$ This define the relationship of the fixed pivot O_2 to the coupler line CD in the second coupler position.
- Draw construction arc from point C_2 to O_4 and from D_2 to O_4 whose radii define the side of triangle $C_2O_4D_2$ This define the relationship of the fixed pivot O_4 to the coupler line CD in the second coupler position.
- Now transfer this relationship back to the first coupler position C_1D_1 so that the ground plane position $O_2'O_4'$ bears the same relationship to C_1D_1 as O_2O_4 bore to the second coupler position C_2D_2. In effect, you are sliding C_2 along the dotted line C_2C_1 and D_2 along the dotted D_2D_1. By doing this, we have pretended that ground plane moved from O_2O_4 to $O_2'O_4'$ instead of the coupler moving from C_1D_1 to C_2D_2. We have inverted the problem.
- Repeat the process for the third coupler position as shown in the figure and transfer the third relative ground link position to the first, or reference, position.
- The three inverted position of the ground plane that correspond to the three desired coupler positions are labeled O_2O_4,$O_2'O_4'$, and $O_2''O_4''$ and have also been renamed
- E_1F_1, E_2F_2 and E_3F_3 as shown in the figure.



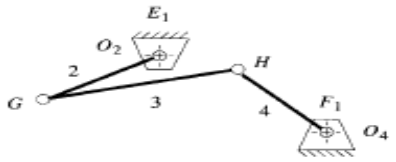
Example 6



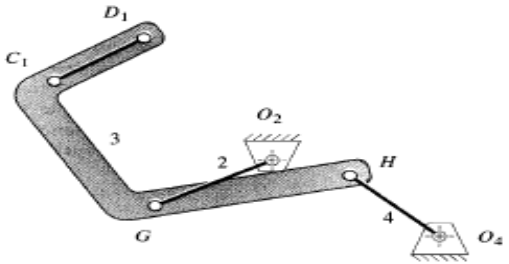
(a) Construction to find rotopoles G and H



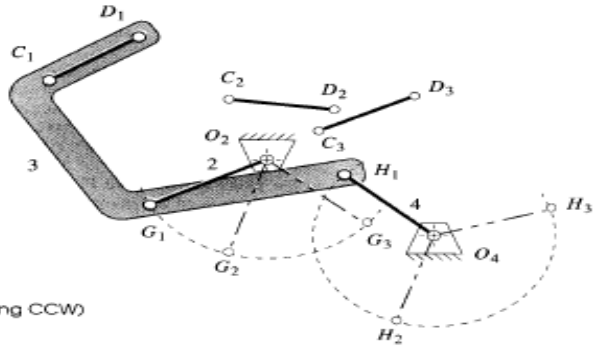
(b) The correct inversion of desired linkage



(c) Reinvert to obtain the result



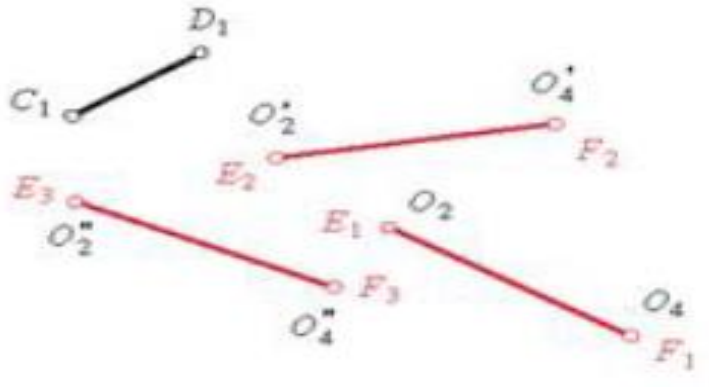
(d) Re-place line CD on link 3



(e) The three positions (link 4 driving CCW)

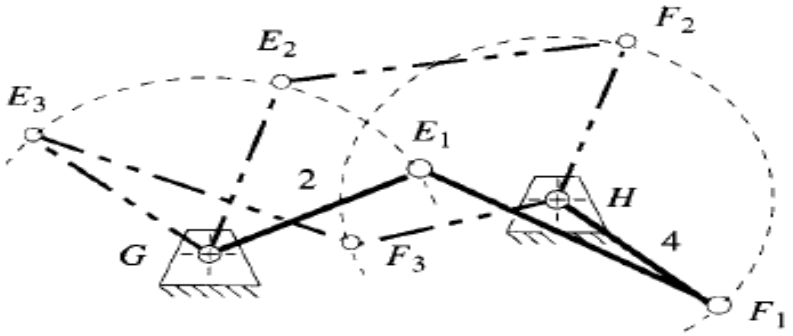
Example 7

- Example 7 – Finding the Moving Pivots for Three Positions and Specified Fixed Pivots
- – Design a fourbar linkage to move the link CD shown from position C_1D_1 to C_2D_2 and then to position C_3D_3. Use specified fixed pivots O_2 and O_4. Find the required moving pivot location on the coupler by inversion.



Example 7

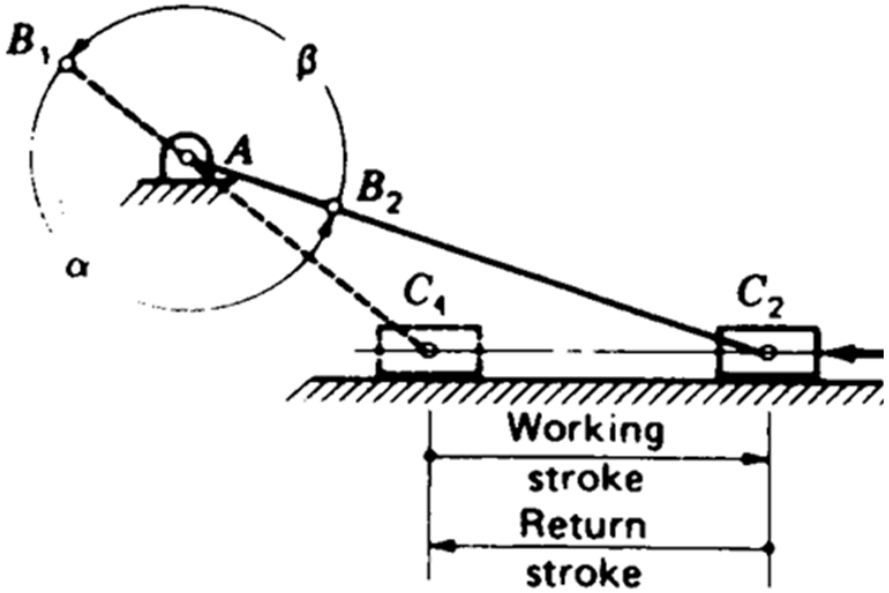
- Start with inverted three positions plane as shown in the figures. Lines E_1F_1, E_2F_2 and E_3F_3 define the three positions of the inverted link to be moved.
- Draw construction lines from point E_1 to E_2 and from point E_2 to E_3.
- Bisect line E_1E_2 and line E_2E_3 extend the perpendicular bisector until they intersect. Label the intersection G.
- Repeat steps 2 and 3 for lines F_1F_2 and line F_2F_3. Label the intersection H.
- Connect G with E_1 and call it link 2. Connect H with F_1 and call it link 4.
- In this inverted linkage, line E_1F_1 is the coupler, link 3. Line GH is the "ground" link1.
- We must now reinvert the linkage to return to the original arrangement. Line E_1F_1 is really the ground O_2O_4 and GH is really the coupler. The figure shows the reinversion of the linkage in which points G and H are now the moving pivots on the coupler and E_1F_1 has resumed its real identity as ground link O_2O_4.
- The figure reintroduces the original line C_1D_1 in its correct relationship to line O_2O_4 at the initial position as shown in the original example 3. This form the required coupler plane and defines a minimal shape of link 3.
-

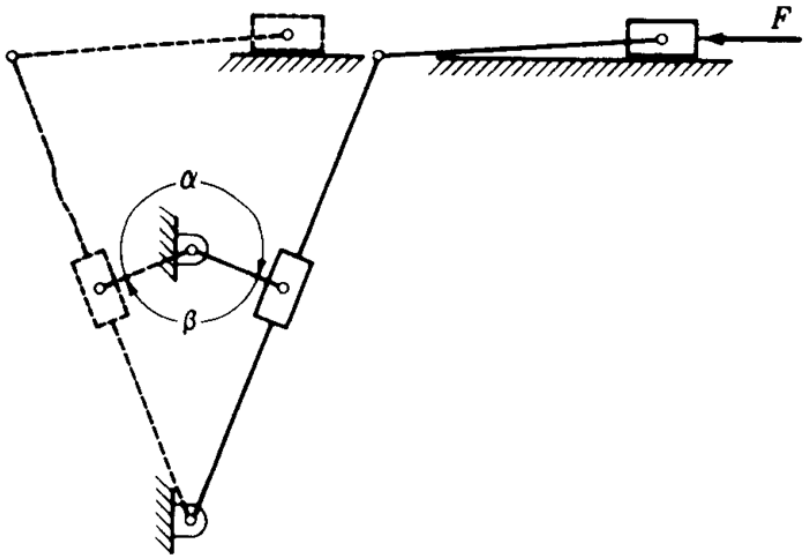


Example 7

- The angular motions required to reach the second and third position of line CD shown in the figure are the same as those defined in figure b for the linkage inversion. The angle F_1HF_2 in the figure b is the same as the angle $H_1O_4H_2$ in the figure and F_2HF_3 is the same as angle $H_2O_4H_3$. The angular excursions of link 2 retain the same between figure b and e as well. The angular motions of links 2 and 4 are the same for both inversion as the link excursions are relative to one another.
- Check the Grashof condition. Note that any Grashof condition is potentially acceptable in this case provided that the linkage has mobility among all three position. This solution is a non-Grashof linkage.
- 11. Construct a model and check its function to be sure it can get from initial to final position without encountering any limit (toggle) positions.
- In this case link 3 and 4 reach a toggle position between points H_1 and H_2. This means that this linkage cannot be driven from link 2 as it will hang up at that toggle position. It must driven from link 4.

Quick – Return Mechanism





Quick – Return Mechanism

- Fourbar Quick-Return
- Time ratio (T_R) defines the degree of quick-return of the linkage.

$$T_R = \frac{foward\ stroke\ time/angle}{reverse\ stroke\ time/angle}$$

$$T_R = \frac{\alpha}{\beta}$$

$$\alpha + \beta = 360°$$

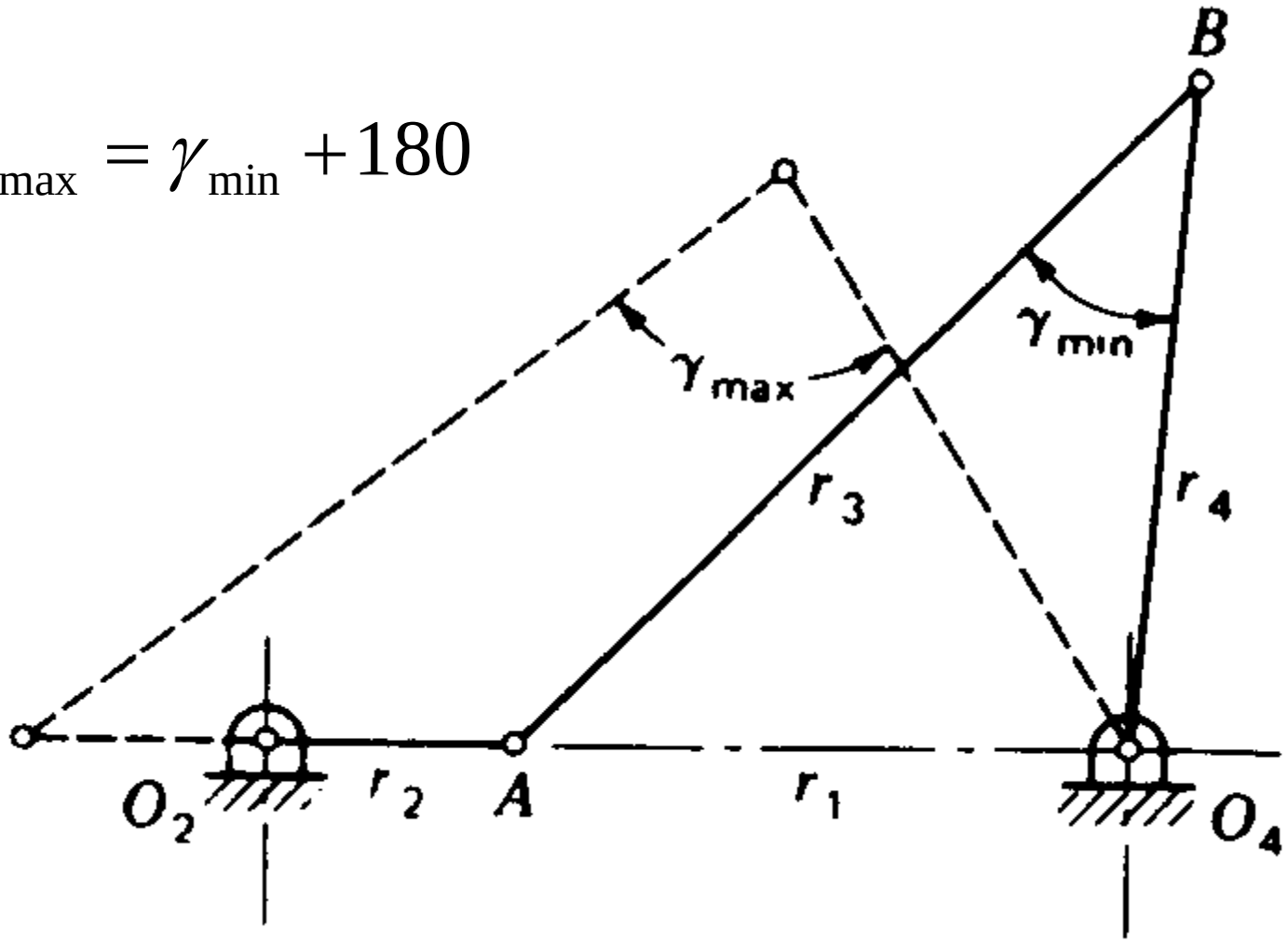
$$\delta = |180 - \alpha| = |180 - \beta|$$

- If T_R >1 the mechanism is a quick return mechanism

- Works well for time ratios down to about 1.5

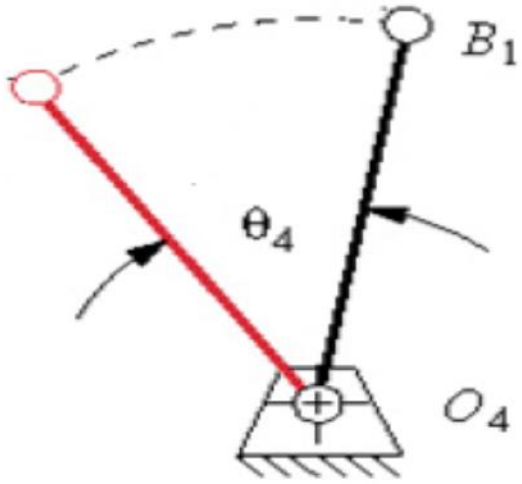
Synthesis of Crank-and-Rocker Mechanism with unit time ratio

$$\gamma_{\max} = \gamma_{\min} + 180$$



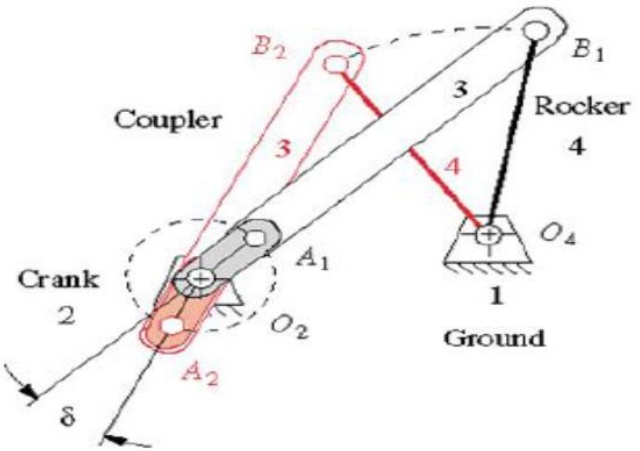
Example 1

- Fourbar Crank-Rocker Quick-Return Linkage for Specified Time Ratio
- Redesign Example 1 (two position) to provide a time 1:1.25 with 45° output rocker motion.



Example 1

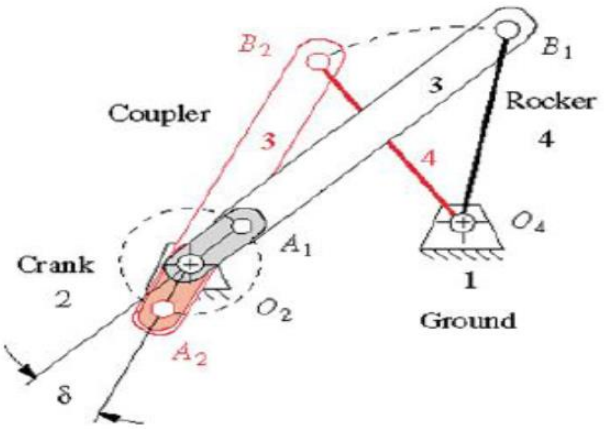
- Draw the output link O_4B in both extreme position, in any convenient location, such that the desired angle of motion, θ_4 , is subtended.
- Calculate α, β, δ using the equations. In this example α=160°, β=200°, δ=20°.
- Draw a construction line through point B_1 at any convenient angle.
- Draw a construction line through point B_2 at an angle δ from the first line.
- Label the intersection of the two construction line O_2.
- The line O_2O_4 define ground.



(b) The finished linkage in its two toggle positions

Example 1

- Calculate the length of crank and coupler by measuring O_2B_1 and O_2B_2 and solve simultaneously;
- coupler + crank=O_2B_1 ; coupler - crank=O_2B_2 or you can construct the crank length by swinging an arc centered at O_2 from B_1 to cut line O_2B_2 extended.
- Label that intersection B_1'. The line B_2B_1' is twice the crank length. Bisect this line segment to measure crank length O_2A_1.
- Calculate the Grashof condition.
- If non-Grashof, repeat steps 3 to 8 with O_2 further O_4.
- Make a model of the linkage and articulate it to check its function.
- Check the transmission angle.



(b) The finished linkage in its two toggle positions

Example 2

- Sixbar Drag Link Quick-Return Linkage for Specified Time Ratio.
- Provide a time ratio of 1:1.4 with 90 degree rocker motion.

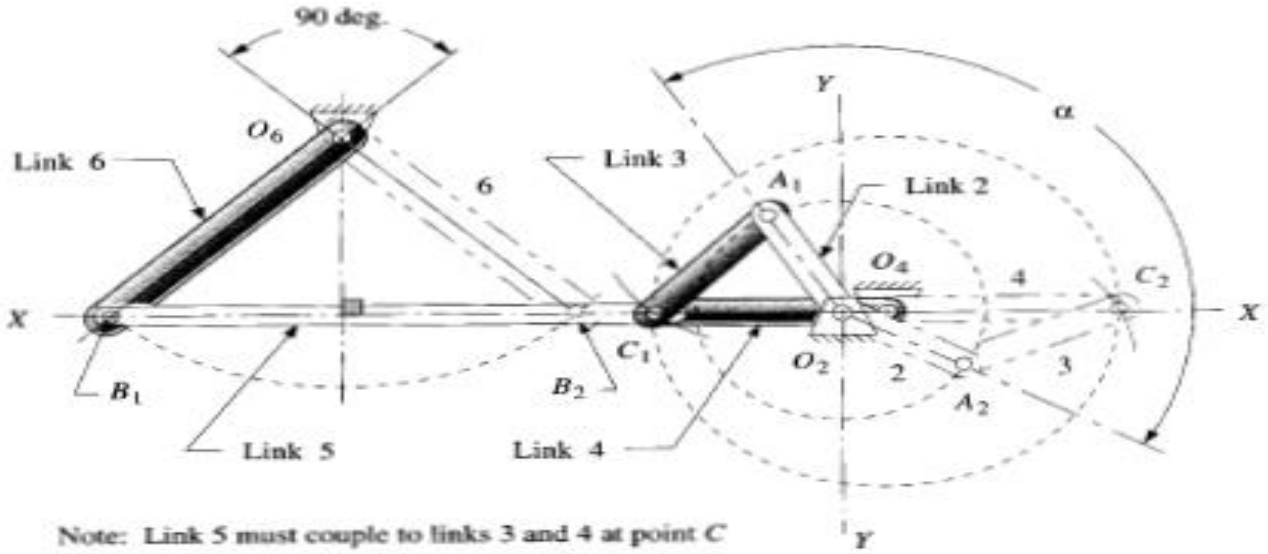
Example 2

- Calculate α, β, δ. In this example $\alpha=150°$, $\beta=210°$.
- Draw a line of centers XX at any convenient location.
- Choose a crank pivot location O_2 on line XX and draw an axis YY perpendicular to XX.
- Draw a line of convenient radius O_2A about center O_2
- Lay out angle α with vertex at O_2, symmetrical about quadrant one
- Label points A_1 and A_2 at the intersection of the lines subtending angle α and the circle of radius O_2A.

Example 2

- Set the compass to a convenient radius AC long enough to cut XX in two places on either side of O_2 when swung from A_1 and A_2. Label the intersection C_1 and C_2.
- The line O_2A_1 is the driver crank, link 2, and line A_1C_1 is the coupler, link 3.
- The distance C_1C_2 is twice the driven (dragged) crank length. Bisect it to locate the fixed pivot O_4.
- The line O2O4 now defines the ground link. Link O4C1 is the driven crank, link 4.
- Calculate the Grashof condition. If Non-Grashof, repeat steps 7-11 with a shorter radius in step 7.
- invert the method of example 1 (two positions) to create the output dyad using XX as the chord and O_4C_1 as the driving crank. The point B_1 and B_2 will lie on line XX and be spaced apart a distance $2O_4C_1$. The pivot O_6 will lie on the perpendicular bisector of B_1B_2, at a distance from line XX which subtends the specified output rocker angle.
- Check the transmission angle.

Example 2

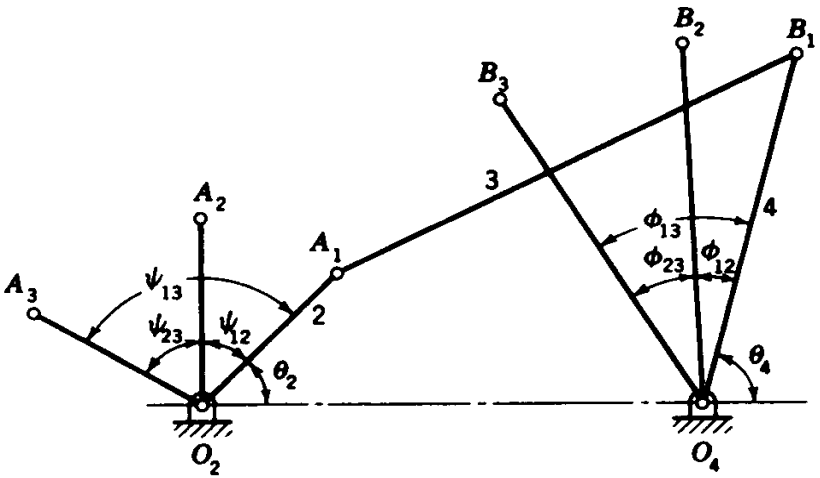


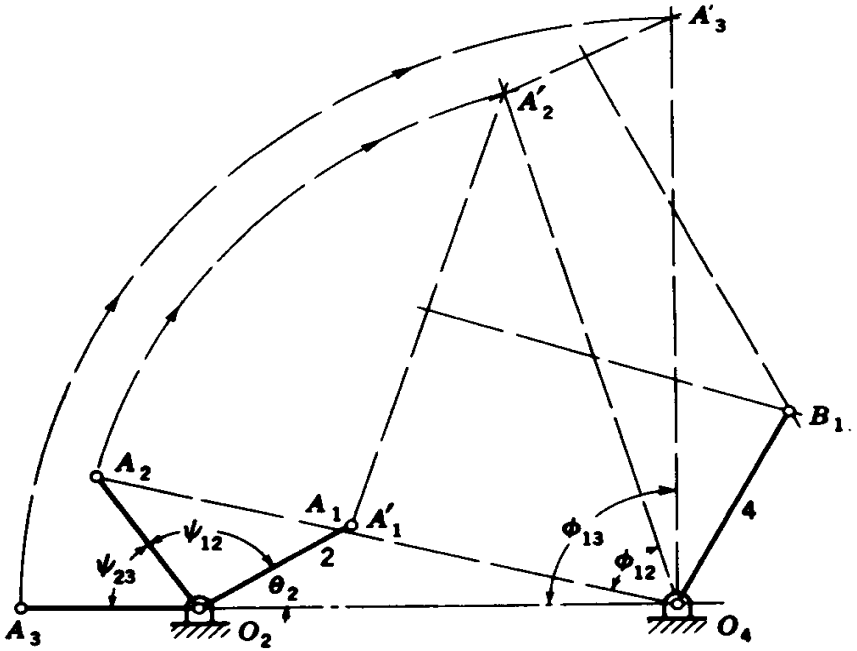
Note: Link 5 must couple to links 3 and 4 at point *C*

Synthesis of Rocker-and-Rocker Mechanism

Three-Position Synthesis

- Given $\psi_{12}, \psi_{23}, \psi_{13}, \emptyset_{12}, \emptyset_{23}, \emptyset_{13}, A, \theta_2, r_1,\ r_2\ and\ (\theta_4, r_4, r_3?)$
- Draw O_2A in its three specified positions and locate O_4
- Draw a ray from $O_4\ to\ A_2$ and rotate it through $\emptyset_{12}$ to locate ${A_2}'$
- Draw a ray from $O_4\ to\ A_3$ and rotate it through $\emptyset_{13}$ to locate ${A_3}'$
- Draw mid-normals to the lines ${A_1}'{A_2}'$ and ${A_2}'{A_3}'$ these intersect at B_1

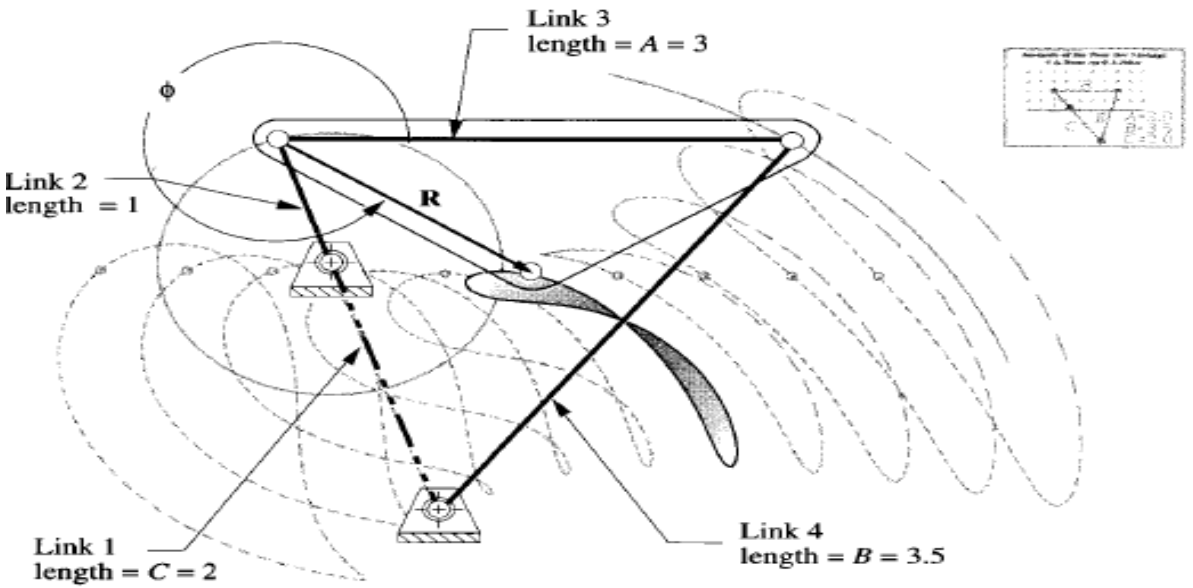




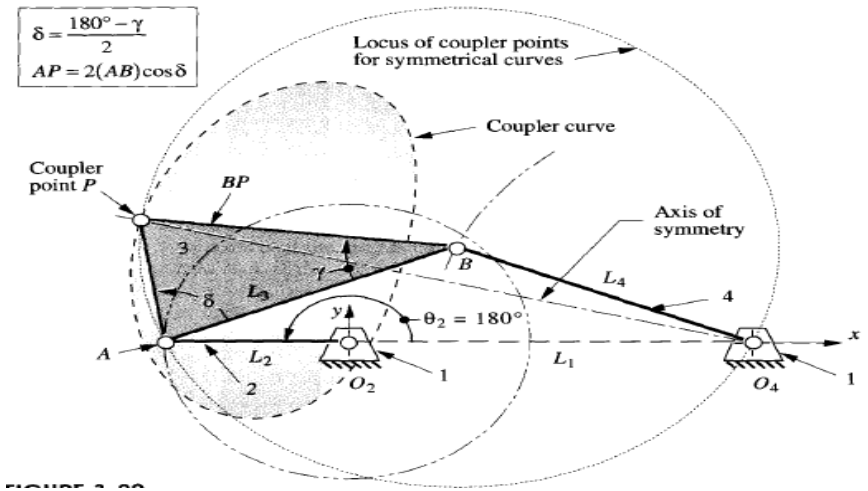
Coupler Curves

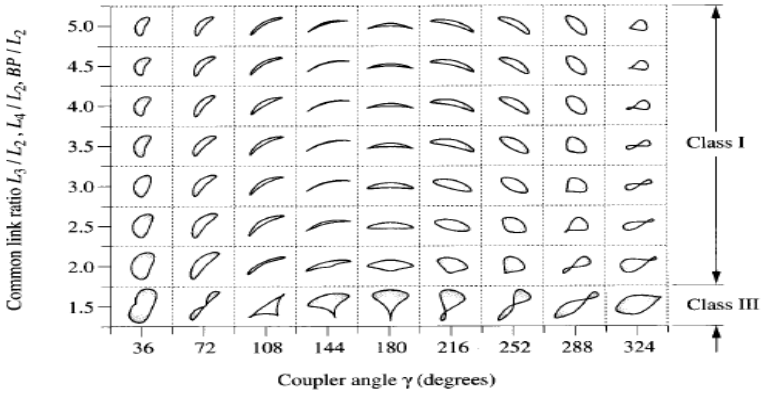
For the mechanisms considered, the displacement of the links joined with the fixed link was the input or output of the simple mechanisms. In great number of applications the output from a simple mechanism is the path traced by one of the points on the coupler link. These paths are generally called “coupler point curves” or “coupler paths”.

Coupler Curves

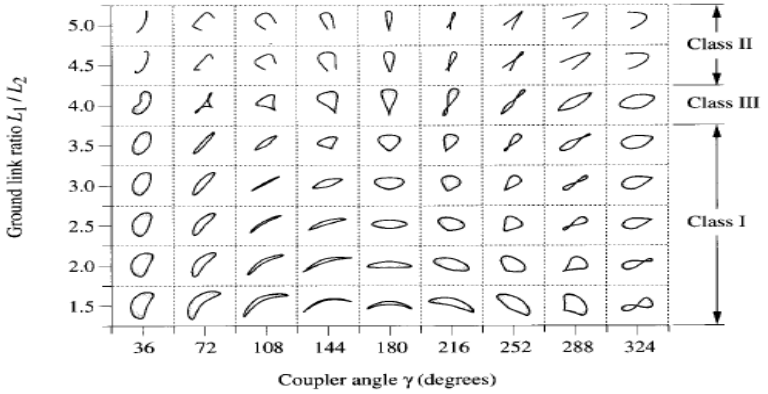


Coupler Curves





(a) Variation of coupler curve shape with common link ratio and coupler angle for a ground link ratio $L_1 / L_2 = 2.0$

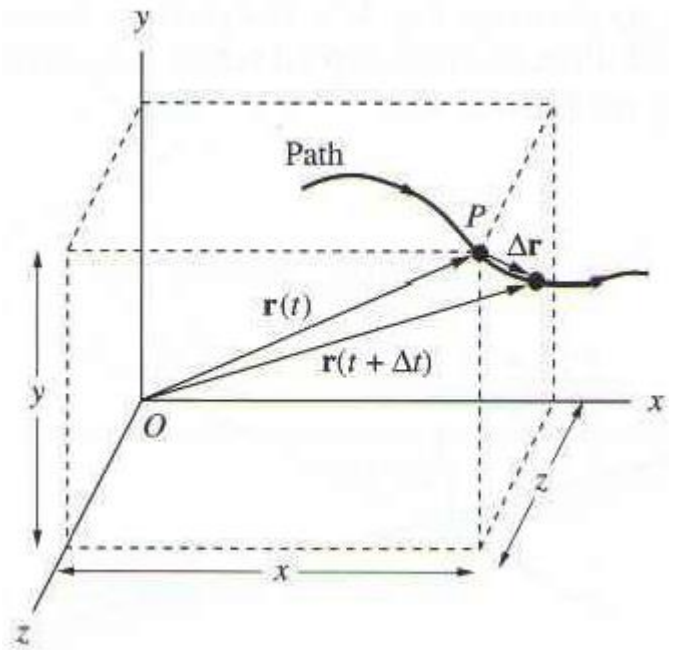


(b) Variation of coupler curve shape with ground link ratio and coupler angle for a common link ratio $L_3 / L_2 = L_4 / L_2 = BP / L_2 = 2.5$

MECHANISM ANALYSIS

Review of Vector Analysis

i. Cartesian/rectangular coordinate system



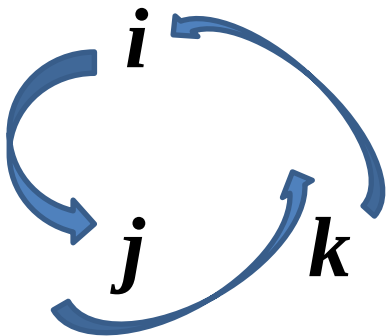
$$\mathbf{r}(t) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

$$\mathbf{v}(t) = \dot{x}\mathbf{i} + \dot{y}\mathbf{j} + \dot{z}\mathbf{k} = v_x\mathbf{i} + v_y\mathbf{j} + v_z\mathbf{k}$$

$$\mathbf{a}(t) = \ddot{\mathbf{r}}(t) = \ddot{x}\mathbf{i} + \ddot{y}\mathbf{j} + \ddot{z}\mathbf{k} = a_x\mathbf{i} + a_y\mathbf{j} + a_z\mathbf{k}$$

$$Also \quad \boldsymbol{i} \times \boldsymbol{k} = -\boldsymbol{j}; \quad \boldsymbol{k} \times \boldsymbol{j} = -\boldsymbol{i}; \boldsymbol{j} \times \boldsymbol{i} = -\boldsymbol{k}$$

$$and \quad \boldsymbol{i} \times \boldsymbol{i} = \boldsymbol{j} \times \boldsymbol{j} = \boldsymbol{k} \times \boldsymbol{k} = \mathbf{0}$$



Example

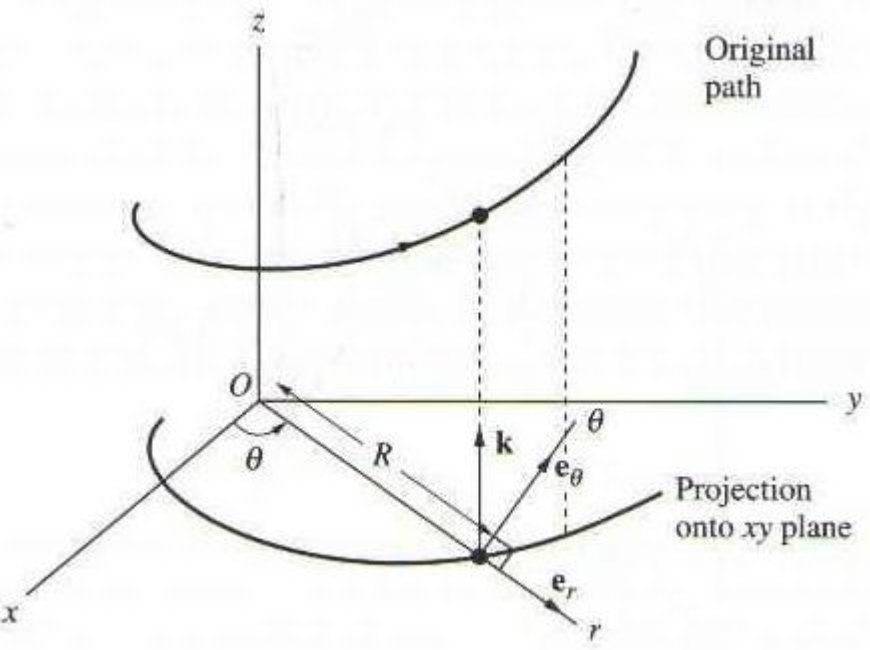
The motion of a particle is described in Cartesian coordinates as

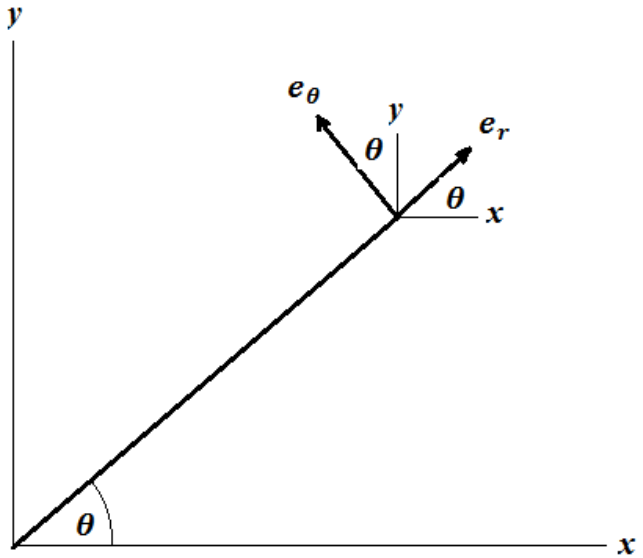
$$x(t) = 2t^2 + 4t$$

$$y(t) = 0.1t^3 + \cos t \qquad z(t) = 3t$$

Find the velocity and acceleration of the particle.

ii. Cylindrical coordinate system





$$\boldsymbol{e_r} = cos\theta\boldsymbol{i} + sin\theta\boldsymbol{j} \;\; and \boldsymbol{e_\theta} = -sin\,\theta\boldsymbol{i} + cos\theta\,\boldsymbol{j}$$

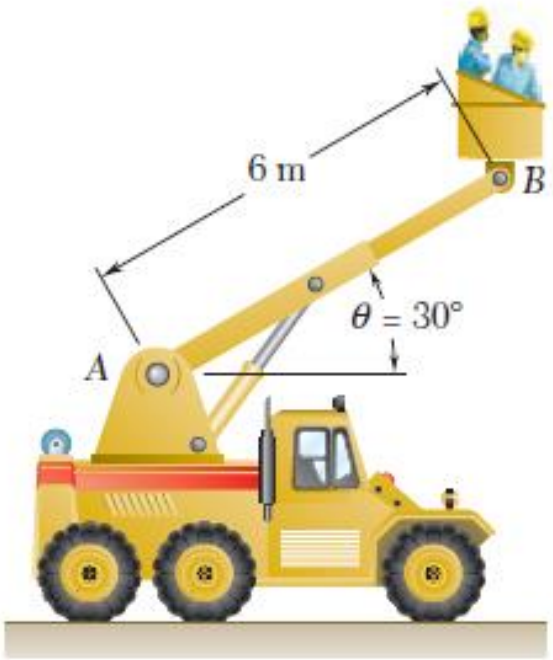
$$\boldsymbol{r}(t) = R\boldsymbol{e_r} + z\boldsymbol{k} = Rcos\,\theta\,\boldsymbol{i} + Rsin\,\theta\boldsymbol{j} + z\boldsymbol{k}$$

$$\boldsymbol{v}(t) = \dot{\boldsymbol{r}}(t) = \dot{R}\boldsymbol{e_r} + R\dot{\theta}\boldsymbol{e_\theta} + \dot{z}\boldsymbol{k}$$

$$\boldsymbol{a}(t) = \left(\ddot{R} - R\dot{\theta}^2\right)\boldsymbol{e_r} + \left(R\ddot{\theta} + 2\dot{R}\dot{\theta}\right)\boldsymbol{e_\theta} + \ddot{z}\boldsymbol{k}$$

Example

At the instant shown the length of the boom *AB* is being *decreased* at the constant rate of 0.2 m/s and the boom is being lowered at the constant rate of 0.08 rad/s. Using cylindrical coordinate system, Determine (*a*) the velocity of point *B*, (*b*) the acceleration of point *B*.



Solution

$$R = 6m; \dot{\theta} = -0.08\, rad/s;$$

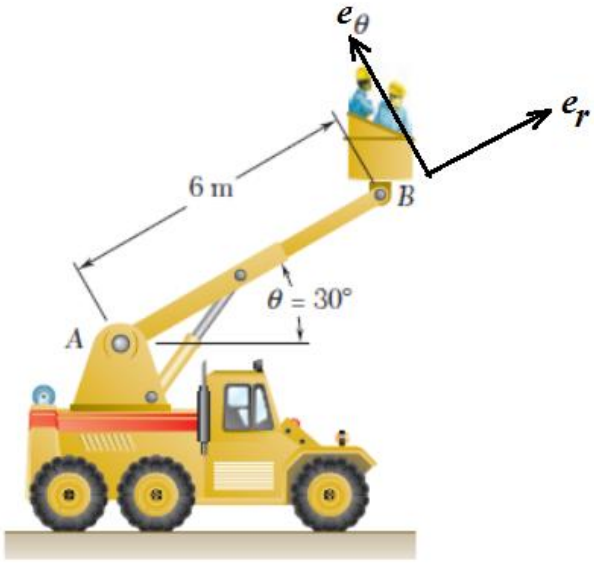
$$\dot{R} = -0.2m/s;$$

$$\dot{z} = \ddot{z} = 0; \ddot{R} = 0; \ddot{\theta} = 0$$

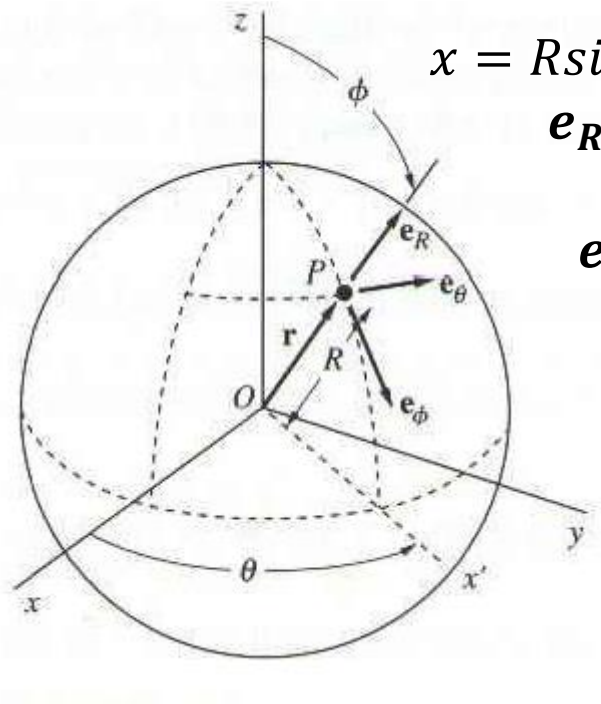
$$\therefore \boldsymbol{v}(t) = (-0.2\boldsymbol{e_r} - 0.48\boldsymbol{e_\theta})\ \mathrm{m/s}$$

$$and\ \ \boldsymbol{a}(t) = -R\dot{\theta}^2\boldsymbol{e_r} + 2\dot{R}\dot{\theta}\boldsymbol{e_\theta}$$

$$\boldsymbol{a}(t) = (-38.4\boldsymbol{e_r} + 32\boldsymbol{e_\theta})mm/s^2$$



iii. Spherical coordinate system



$$\boldsymbol{e_R} \times \boldsymbol{e_\phi} = \boldsymbol{e_\theta}$$

$$\boldsymbol{r}(t) = R\boldsymbol{e_r}$$

$$x = Rsin\phi cos\,\theta \qquad y = Rsin\phi sin\,\theta \qquad z = Rcos\phi$$

$$\boldsymbol{e_R} = sin\phi cos\,\theta\,\boldsymbol{i} + sin\phi sin\,\theta\boldsymbol{j} + cos\phi\boldsymbol{k}$$

$$\boldsymbol{e_\theta} = -sin\,\theta\,\boldsymbol{i} + cos\,\theta\boldsymbol{j}$$

$$\boldsymbol{e_\phi} = cos\phi(cos\,\theta\,\boldsymbol{i} + sin\,\theta\boldsymbol{j}) - sin\phi\boldsymbol{k}$$

$$\frac{d\boldsymbol{e_R}}{dt} = \dot{\theta}sin\phi\boldsymbol{e_\theta} + \dot{\boldsymbol{\phi}}\boldsymbol{e_\phi}$$

$$\frac{d\boldsymbol{e_\theta}}{dt} = -\dot{\theta}sin\phi\boldsymbol{e_R} - \dot{\theta}\boldsymbol{cos}\phi\boldsymbol{e_\phi}$$

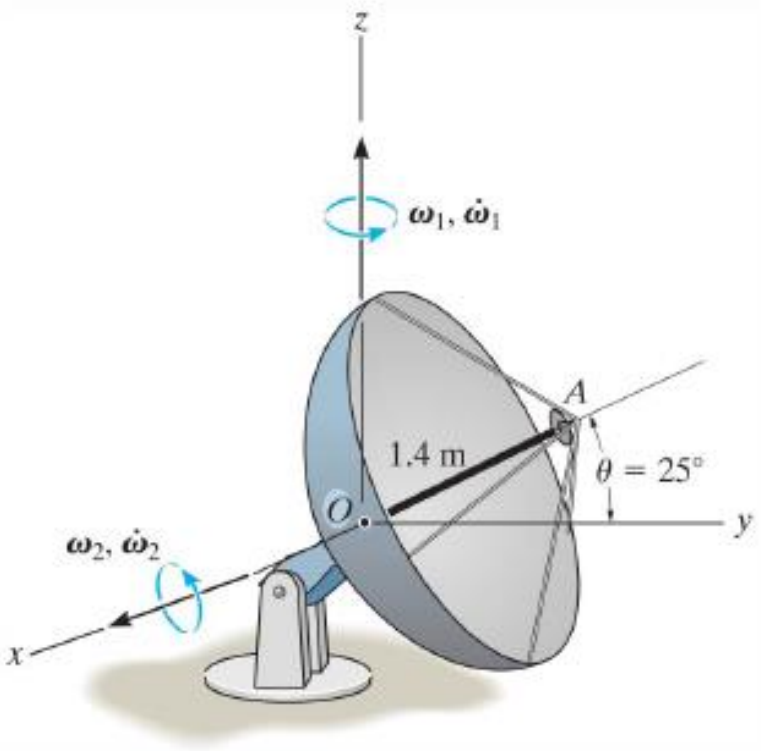
$$\frac{d\boldsymbol{e_\phi}}{dt} = \dot{\theta}cos\phi\boldsymbol{e_\theta} - \dot{\phi}\boldsymbol{e_R}$$

$$\boldsymbol{v}(t) = \dot{R}\boldsymbol{e_R} + R\dot{\theta}sin\phi\boldsymbol{e_\theta} + R\dot{\phi}\boldsymbol{e_\phi}$$

$$\boldsymbol{a}(t) = (\ddot{R} - R\dot{\theta}^2sin^2\phi - R\dot{\phi}^2)\boldsymbol{e_R} + \left(R\ddot{\theta}sin\phi + 2\dot{R}\dot{\theta}sin\phi + 2R\dot{\theta}\dot{\phi}cos\phi\right)\boldsymbol{e_\theta} +$$
$$(R\ddot{\phi} + 2\dot{R}\dot{\phi} - R\dot{\theta}^2sin\phi cos\phi)\boldsymbol{e_\phi}$$

Example

At a given instant, the satellite dish has an angular motion ω_1 = 6 rad/s and $\dot{\omega}_1 = 3\,rad/s^2$ about the z axis. At this same instant $\theta = 25^o$, the angular motion about the x axis is ω_2 = 2 rad/s, and $\dot{\omega}_2 = 1.5\,rad/s^2$. Using spherical coordinate system, determine the velocity and acceleration of the signal horn A at this instant.



Solution

$$R = 1.4\,m; \theta = 90^o;\ \phi = 65^o$$

$$\dot{R} = \ddot{R} = 0;$$

$$\dot{\phi} = \omega_2 = 2\,rad/s; \ddot{\phi} = \dot{\omega}_2 = 1.5\,rad/s^2;$$

$$\dot{\theta} = \omega_1 = 6\,rad/s;\ \ddot{\theta} = \dot{\omega}_1 = 3\,rad/s^2$$

$$\therefore \boldsymbol{v}(t) = (2.8\,\boldsymbol{e_\phi} + 8.16\,\boldsymbol{e_\theta})\ \mathrm{m/s}$$

$$and\ \ \boldsymbol{a}(t) = -R\dot{\theta}^2\boldsymbol{e_r} + 2\dot{R}\dot{\theta}\boldsymbol{e_\theta}$$

$$\boldsymbol{a}(t) = \left(-47\,\boldsymbol{e_r} + 34.25\,\boldsymbol{e_\theta} + 21.4\boldsymbol{e_\phi}\right) m/s^2$$

Transport Theorem

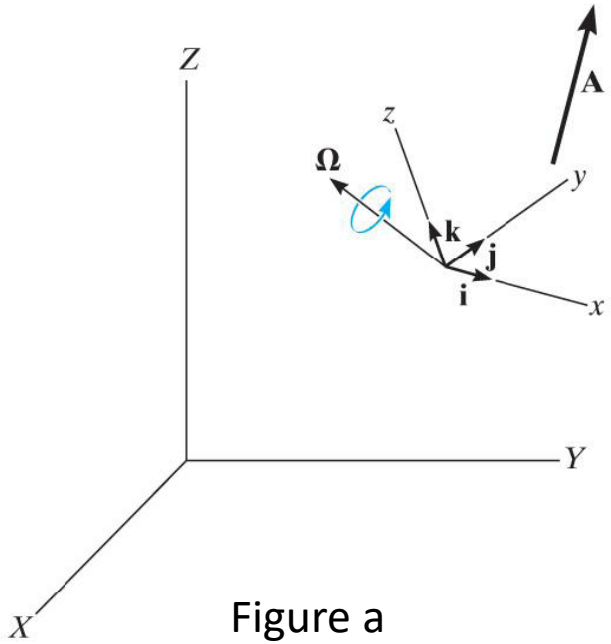


Figure a

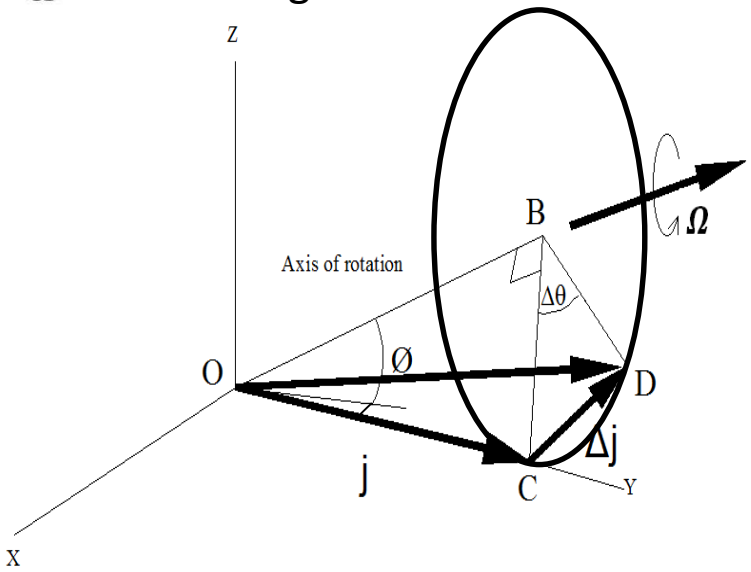


Figure b

- Consider a vector **A** observed from a moving coordinate system *xyz*, which is rotating with angular velocity Ω

$$\vec{A} = A_x \boldsymbol{i} + A_y \boldsymbol{j} + A_z \boldsymbol{k}$$

$$\dot{\vec{A}} = \dot{A}_x \boldsymbol{i} + \dot{A}_y \boldsymbol{j} + \dot{A}_z \boldsymbol{k} + A_x \dot{\boldsymbol{i}} + A_y \dot{\boldsymbol{j}} + A_z \dot{\boldsymbol{k}}$$

If $\dot{A}_{rel} = \dot{A}_x \boldsymbol{i} + \dot{A}_y \boldsymbol{j} + \dot{A}_z \boldsymbol{k}$

Then $\dot{\vec{A}} = \dot{A}_{rel} + A_x \dot{\boldsymbol{i}} + A_y \dot{\boldsymbol{j}} + A_z \dot{\boldsymbol{k}}$

From Figure b,

$$\Delta \boldsymbol{j} = BC\Delta\theta = \boldsymbol{j} sin\phi \, \Delta\theta$$

$$\frac{\Delta \boldsymbol{j}}{\Delta \boldsymbol{t}} = \dot{\boldsymbol{j}} = \left|\frac{\Delta\theta}{\Delta \boldsymbol{t}}\right| |\boldsymbol{j}| sin\phi = \Omega \times \mathbf{j}$$

$$\dot{\boldsymbol{i}} = \Omega \times \boldsymbol{i} \; ; \; \dot{\boldsymbol{j}} = \Omega \times \boldsymbol{j} ; \; \dot{\boldsymbol{k}} = \Omega \times \boldsymbol{k}$$

Therefore $\dot{\vec{A}} = \dot{A}_{rel} + \Omega \times \vec{A}$

General Motion

Position Vector $r_B = r_A + r_{B/A}$

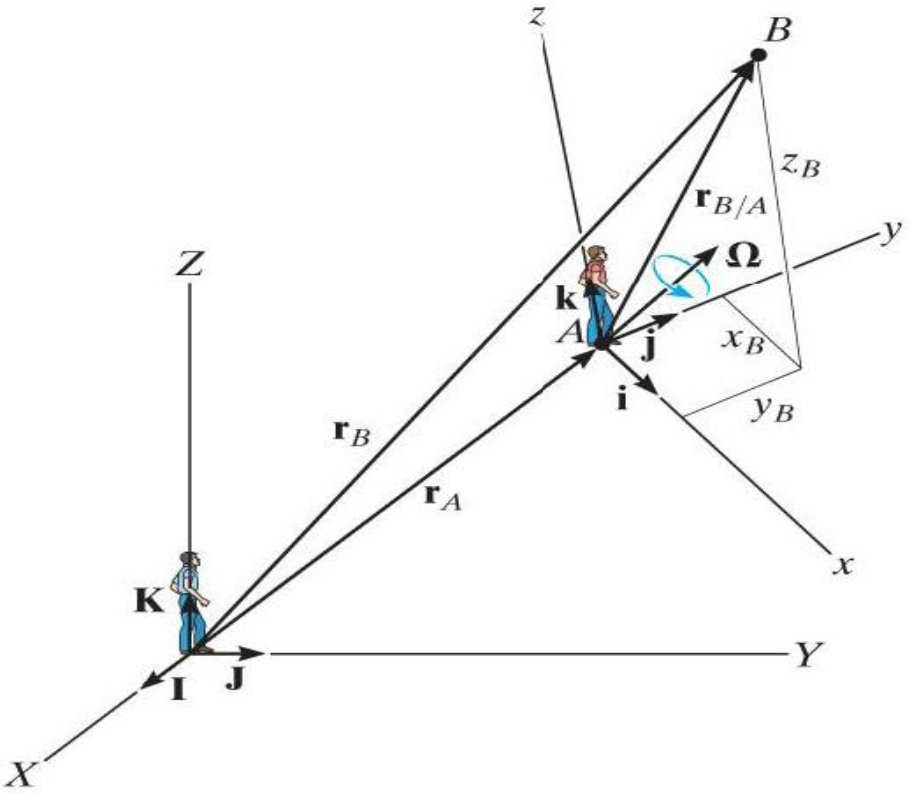
Velocity

$$\frac{d}{dt}(r_B) = \frac{d}{dt}(r_A) + \frac{d}{dt}(r_{B/A})$$

$$v_B = v_A + (v_{B/A})_{xyz} + \Omega \text{ x } \mathrm{r}_{B/A}$$

Acceleration

$$\frac{d}{dt}(v_B) = \frac{d}{dt}(v_A) + \frac{d}{dt}(v_{B/A})_{xyz} + \frac{d}{dt}(\Omega \text{ x } \mathrm{r}_{B/A})$$



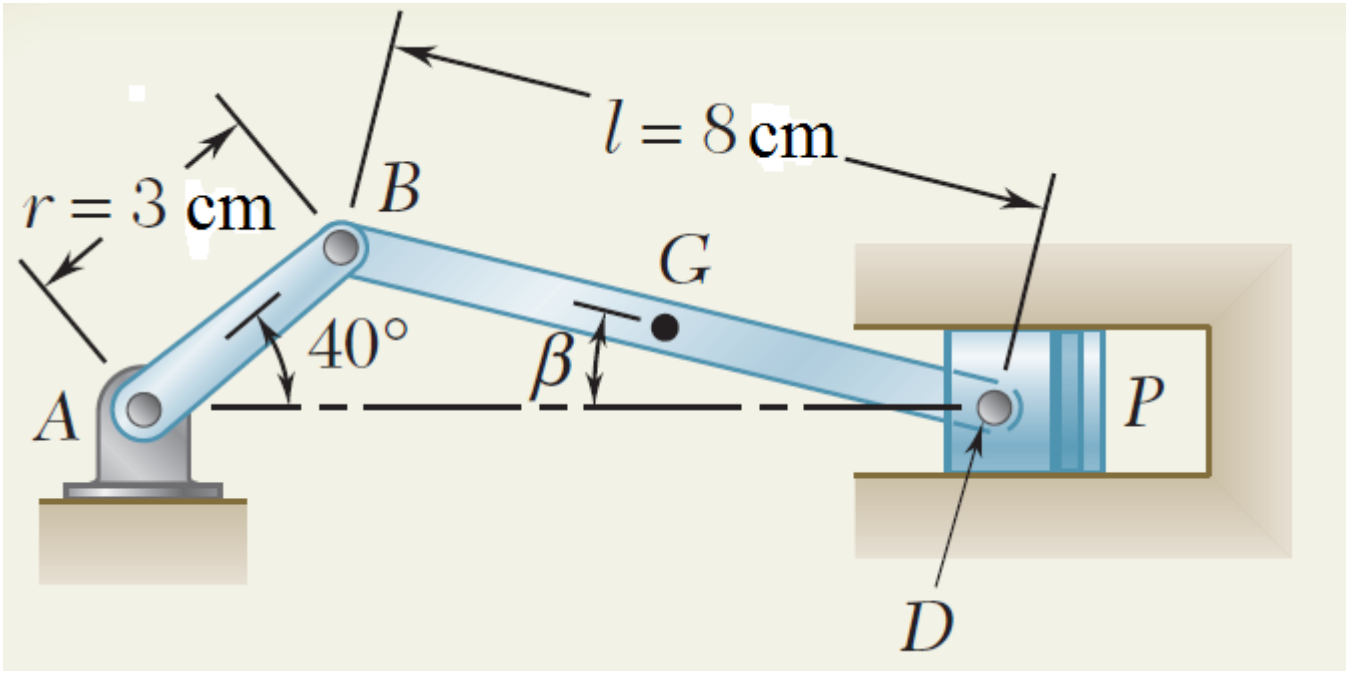
$$a_B = a_A + (\mathrm{a}_{B/A})_{xyz} + \dot{\Omega} \text{ x } \mathrm{r}_{B/A} + 2\Omega \text{ x } (\mathrm{v}_{B/A})_{xyz} + \Omega \text{ x } (\Omega \text{ x } \mathrm{r}_{B/A})$$

For 2D motions, the above equation reduces to

$$a_B = a_A + (\mathrm{a}_{B/A})_{xyz} + \dot{\Omega} \text{ x } \mathrm{r}_{B/A} + 2\Omega \text{ x } (\mathrm{v}_{B/A})_{xyz} - \Omega^2 \ \mathrm{r}_{B/A}$$

Example 1

Crank *AB* of the engine system shown has a constant clockwise angular velocity of 2000 rpm. For the crank position shown, determine the angular velocity as well as the angular acceleration of the connecting rod *BD* and the linear velocity and acceleration of point *D.*



solution

Since the crank rotates about *A* with constant $\omega_{AB} = 2000\, rpm = 209.4\, rad/s$, The velocity of *B can be calculated from*

$$v_B = v_A + v_{B/A} + \Omega_{AB} \text{ x } r_{B/A}$$

$$v_{B/A} = v_A = 0$$

$$v_B = \Omega_{AB} \text{ x } r_{B/A} = 209.4\, \boldsymbol{k} \text{ x } (0.03 \cos 40\, \boldsymbol{i} + 0.03 \sin 40 \boldsymbol{j})$$

$$v_B = -(\, 4.04\, \boldsymbol{i} - 4.81\, \boldsymbol{j}) ms^{-1}$$

$$v_D = v_B + v_{D/B} + \Omega_{DB} \text{ x } r_{D/B}$$

$$v_{D/B} = 0$$

$$v_D \boldsymbol{i} = -4.04\, \boldsymbol{i} + 4.81\, \boldsymbol{j} + \Omega_{DB}\, \boldsymbol{k} \text{ x } (0.03 \cos 40\, \boldsymbol{i} + 0.03 \sin 40 \boldsymbol{j})$$

Expanding and comparing coefficients

$$v_D = -4.04 - 0.019 \Omega_{DB}$$

$$0 = 4.81 + 0.023\, \Omega_{DB}$$

$$\Omega_{DB} = -6.28\, rad/s;\ v_D = -209.13\, ms^{-1}$$

Solution

Since $\alpha_{AB} = 0$. The acceleration of *B can be calculated from*

$$a_B = a_A + a_{B/A} + 2\Omega_{AB} \text{ x } v_{B/A} + \alpha_{AB} \text{ x } r_{AB} - {\Omega_{AB}}^2 r_{B/A}$$

$$v_{B/A} = a_A = a_{B/A} = \alpha_{AB} = 0$$

$$a_B = -{\Omega_{AB}}^2 r_{B/A} = \ -209.4^2(0.03\cos 40\,\boldsymbol{i} + 0.03\sin 40\boldsymbol{j})$$

$$a_B = -(1007.63\,\boldsymbol{i} + 845.56\,\boldsymbol{j})ms^{-2}$$

$$a_D = a_B + a_{D/B} + 2\Omega_{DB} \text{ x } v_{D/B} + \alpha_{DB} \text{ x } r_{D/B} - {\Omega_{DB}}^2 r_{D/B}$$

$$v_{D/B} = a_{D/B} = 0$$

a_D

$= -(1007.63\,\boldsymbol{i} + 845.56\,\boldsymbol{j}) + \alpha_{DB}\,\boldsymbol{k}\,\text{x}(0.03\cos 40\,\boldsymbol{i} + 0.03\sin 40\boldsymbol{j})$

$- 6.28^2(0.03\cos 40\,\boldsymbol{i} + 0.03\sin 40\boldsymbol{j})$

Expanding and comparing coefficients

$$a_D\boldsymbol{i}$$

$$= (-1007.63 - 0.019\alpha_{DB} - 0.91)\,\boldsymbol{i}$$

$$+ (-845.56 + 0.0236\alpha_{DB} - 0.75)\,\boldsymbol{j}$$

$$0 = -846.33 + 0.023\,\alpha_{DB}$$

$$a_D = -1037.84 - 0.019\alpha_{DB}$$

Example 2

Determine the velocity and acceleration of the collar C at this instant

Solution

$$v_D = v_B + v_{D/B} + \Omega_{DB} \text{ x } r_{D/B}$$

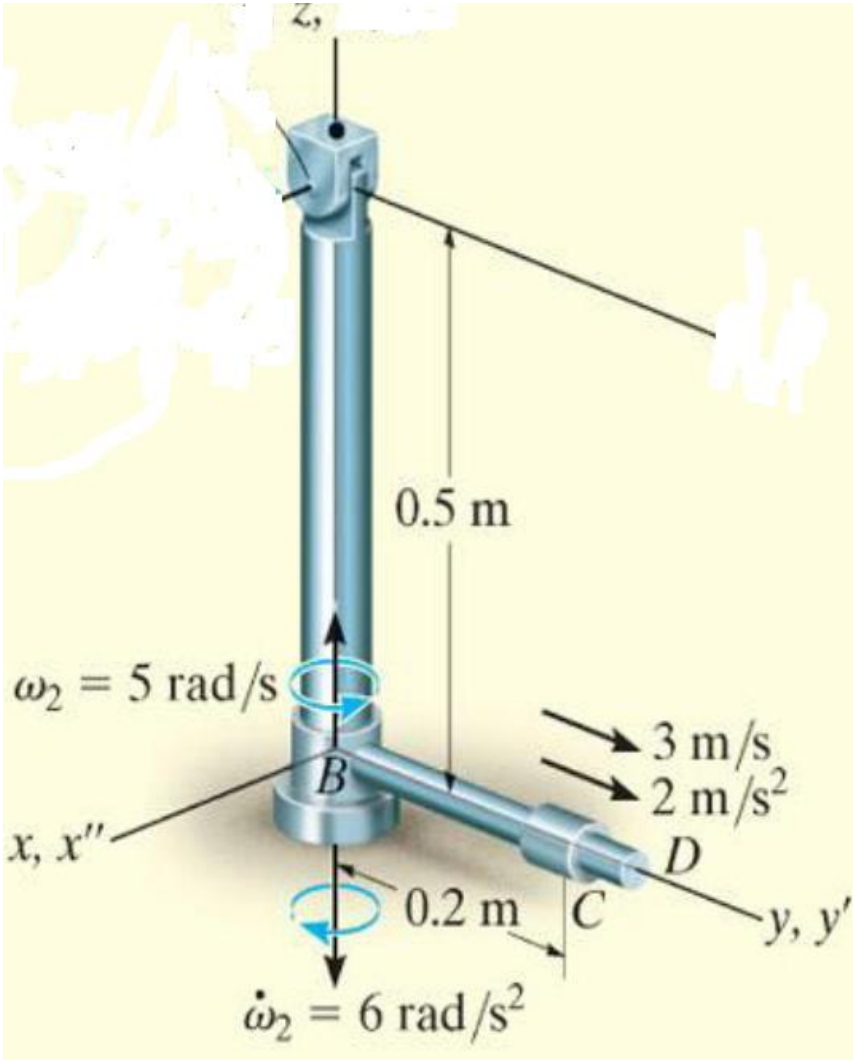
$$v_D = 3\boldsymbol{j} + 5\,\boldsymbol{k} \text{ x } 0.2\,\boldsymbol{j}$$

$$= -\boldsymbol{i} + 3\,\boldsymbol{j}\ m/s$$

$$a_D = a_B + a_{D/B} + 2\Omega_{DB} \text{ x } v_{D/B} + \alpha_{DB} \text{ x } r_{D/B} - \Omega_{DB}{}^2 r_{D/B}$$

$$a_D = 2\boldsymbol{j} + 10\,\boldsymbol{k} \text{ x } 3\boldsymbol{j} - 6\,\boldsymbol{k} \text{ x } 0.2\boldsymbol{j} - 25(0.2\,\boldsymbol{j})$$

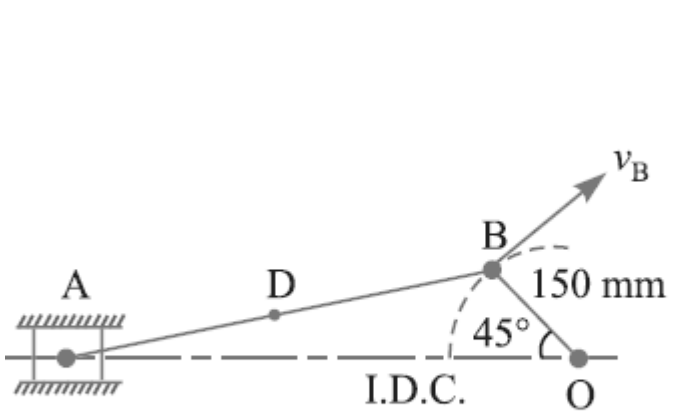
$$a_D = -18.8\,\boldsymbol{i} - 3\boldsymbol{j}\ ms^{-2}$$



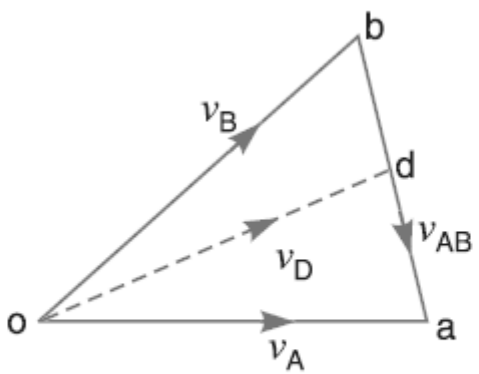
Relative Velocity Method

The crank of a slider crank mechanism rotates clockwise at a constant speed of 300 rpm. The crank is 150 mm and the connecting rod is 600 mm long. Determine:

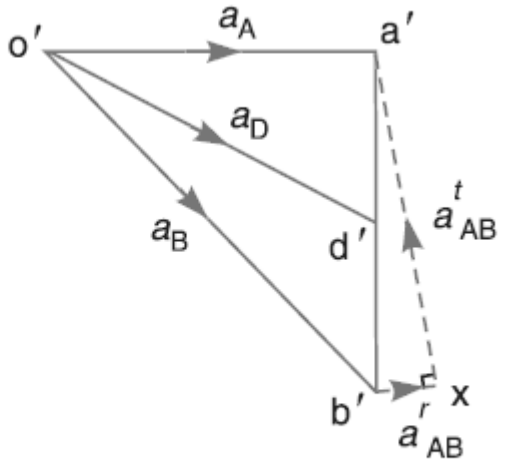
1. Linear velocity and acceleration of the midpoint of the connecting rod, and
2. Angular velocity and angular acceleration of the connecting rod, at a crank angle of 45° from inner dead centre position.



(*a*) Space diagram.



(*b*) Velocity diagram.



(*c*) Acceleration diagram.

solution

Velocity analysis

$$v_B = \omega r_B = 0.15 \times 10\pi = 4.713\ m/s$$

From the velocity diagram

$$v_{AB} = 3.4\ m/s$$

$$v_A = 4\ m/s$$

$$v_D = 4.1\ m/s$$

Acceleration analysis

$${a_B}^r = \frac{{v_B}^2}{OB} = \frac{4.713^2}{0.15} = 148.1\ m/s^2$$

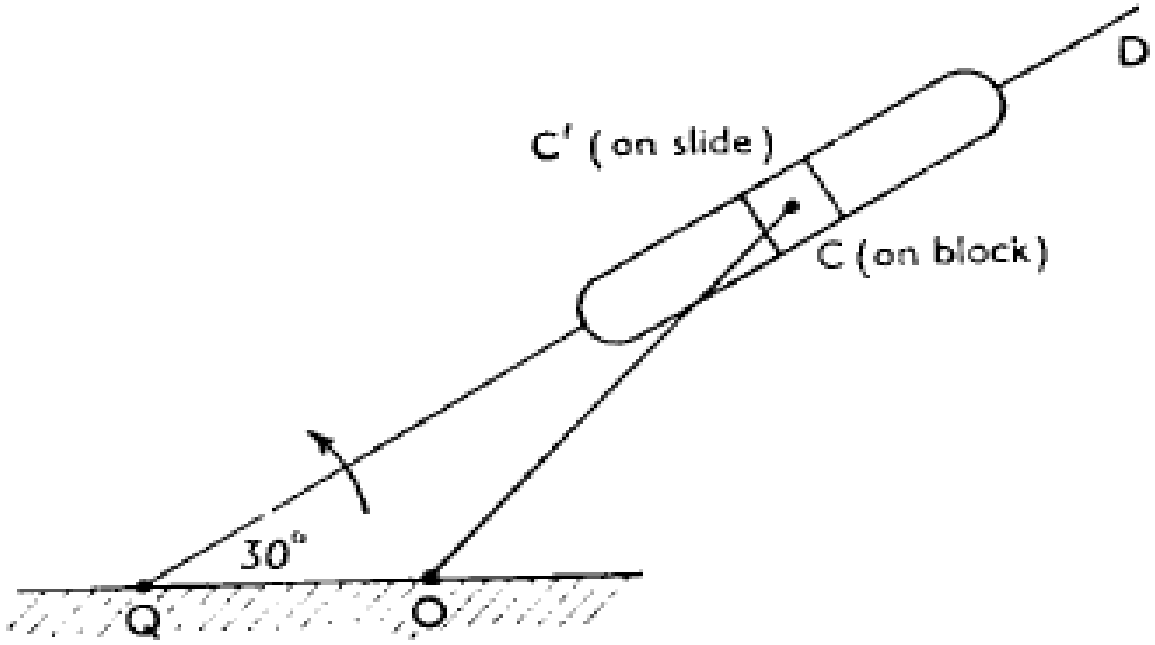
$${a_{AB}}^r = \frac{{v_{AB}}^2}{AB} = \frac{3.4^2}{0.6} = 19.3\ m/s^2$$

From the acceleration diagram

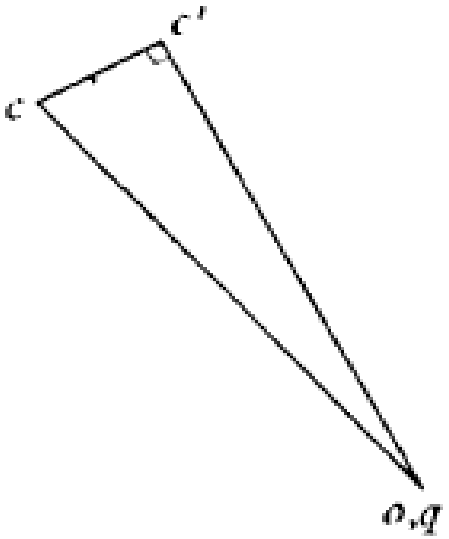
$$\alpha_{AB} = \frac{{a_{AB}}^t}{AB} = \frac{103^2}{0.6} = 171.67\ rad/s^2$$

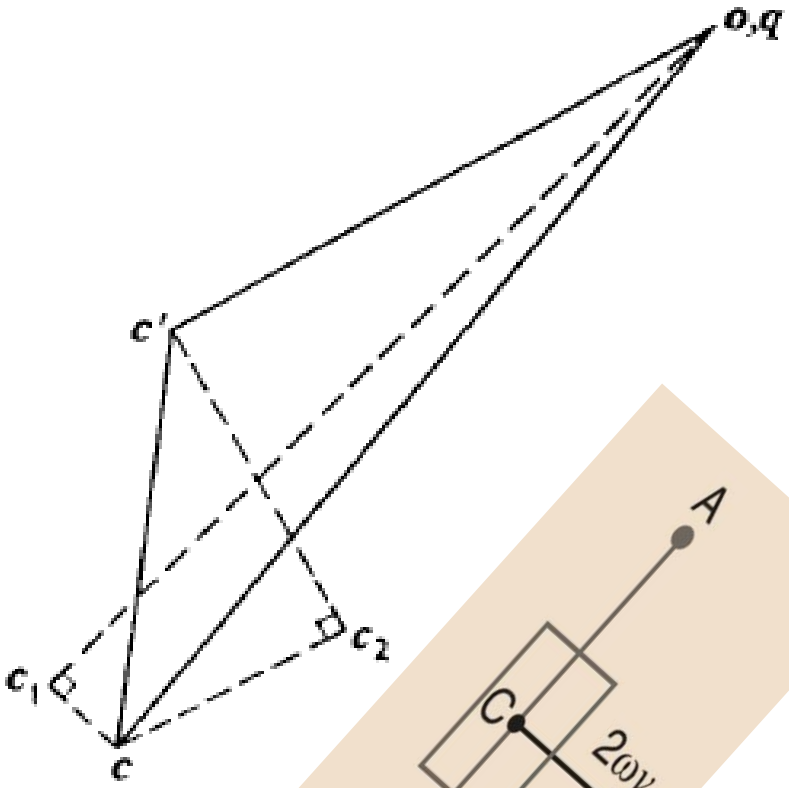
Example 2

In the mechanism shown, the slide QD is driven with a uniform angular velocity of 10 rad/s and the block C has a mass of 1.5 *kg. Draw the velocity and acceleration diagrams for the mechanism in the position shown. OQ* = 1 m; *OC* = 2 m.



solution





Velocity analysis

$$v_c = 28.8\ m/s$$

From the velocity diagram

$$v_{cc'} = 7.5\ m/s$$

Acceleration analysis

$$a_c{}^r = \frac{{v_c}^2}{OC} = 414\ m/s^2$$

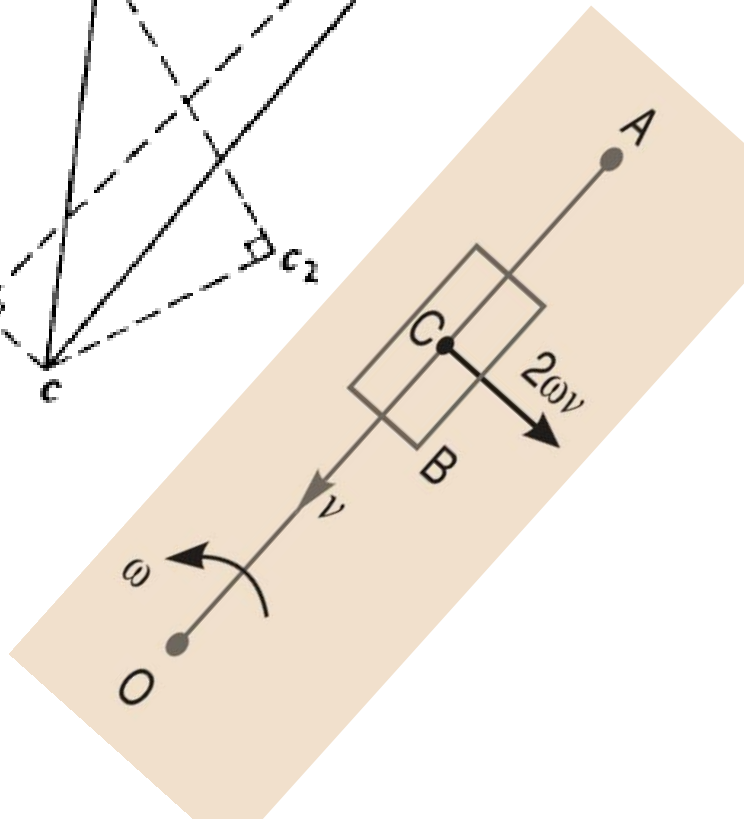
$$a_{C'Q}{}^r = 288\ m/s^2$$

$$a_{C'C}{}^C = 2v_{cc'}\omega_{C'Q}$$

$$a_{C'C}{}^C = 150\ \mathrm{m}/s^2$$

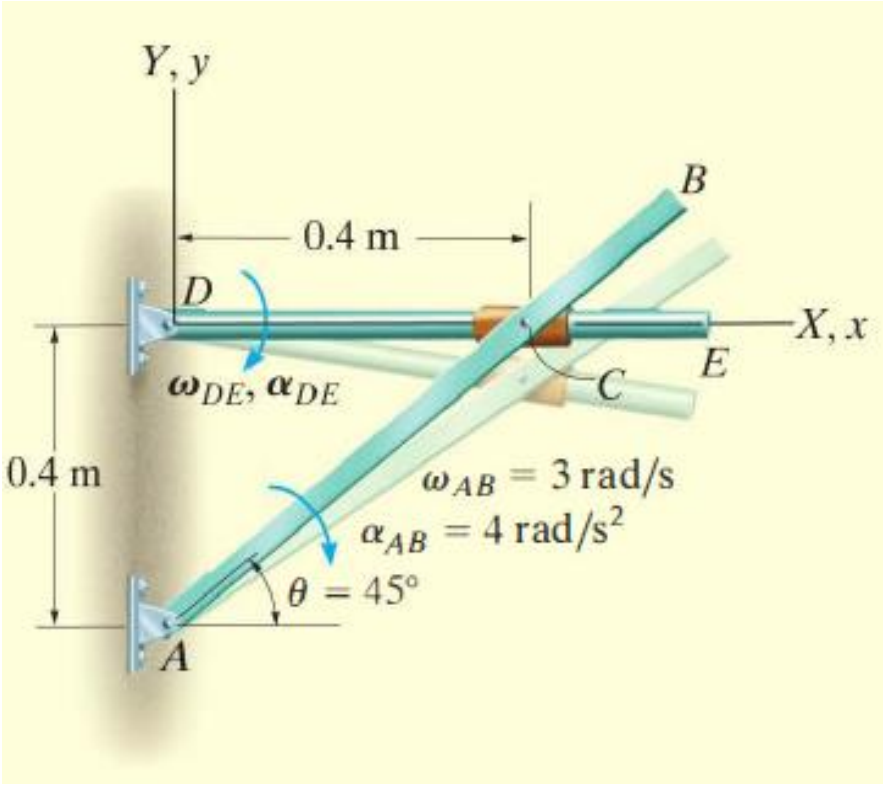
From the acceleration diagram

$$a_C = 417\ m/s^2$$



Assignment 1

Determine the angular velocity and angular acceleration of the rod DE at the instant shown. The collar at C is pin connected to AB and slides over rod DE.

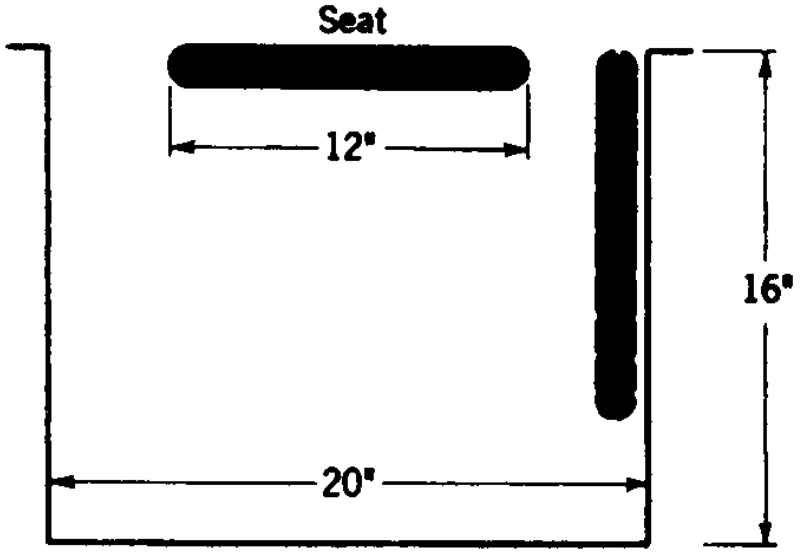


Assignment 2

The rocker of a crank-rocker linkage is to have a length of 500 mm and swing through a total angle of $45°$ with a time ratio of 1.25. Determine a suitable set of dimensions for $r_1, r_2 \; and \;\; r_3$.

Assignment 3

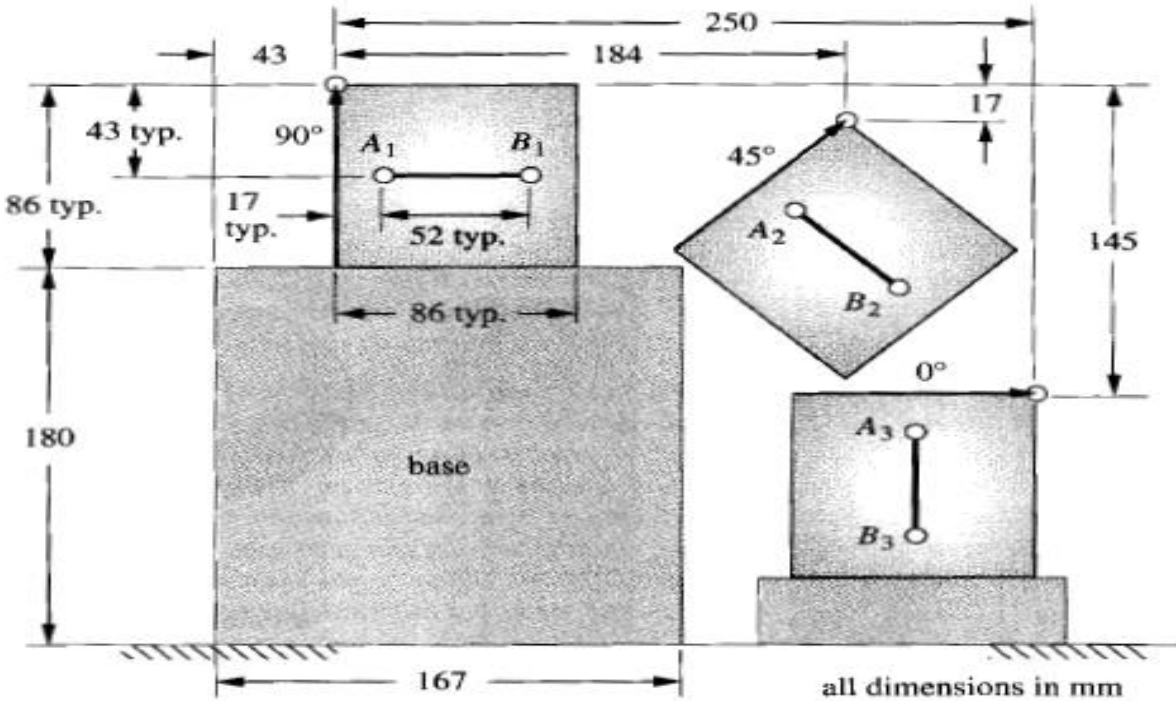
The figure shows two positions of a folding seat used in the aisles of buses to accommodate extra seated passengers. Design a four-bar linkage to support the seat so that it will lock in the open position and fold to a stable closing position along the side of the aisle.



3. Determine the link lengths of a slider-crank linkage to have a stroke of 600 mm and a time ratio of 1.2.

Assignment 4

Design a fourbar linkage to move the object below from position 1 to 2 using points A and B for attachment. Add a driver dyad to limit its motion to the range of positions shown making it a sixbar. All fixed pivots should be on the base.

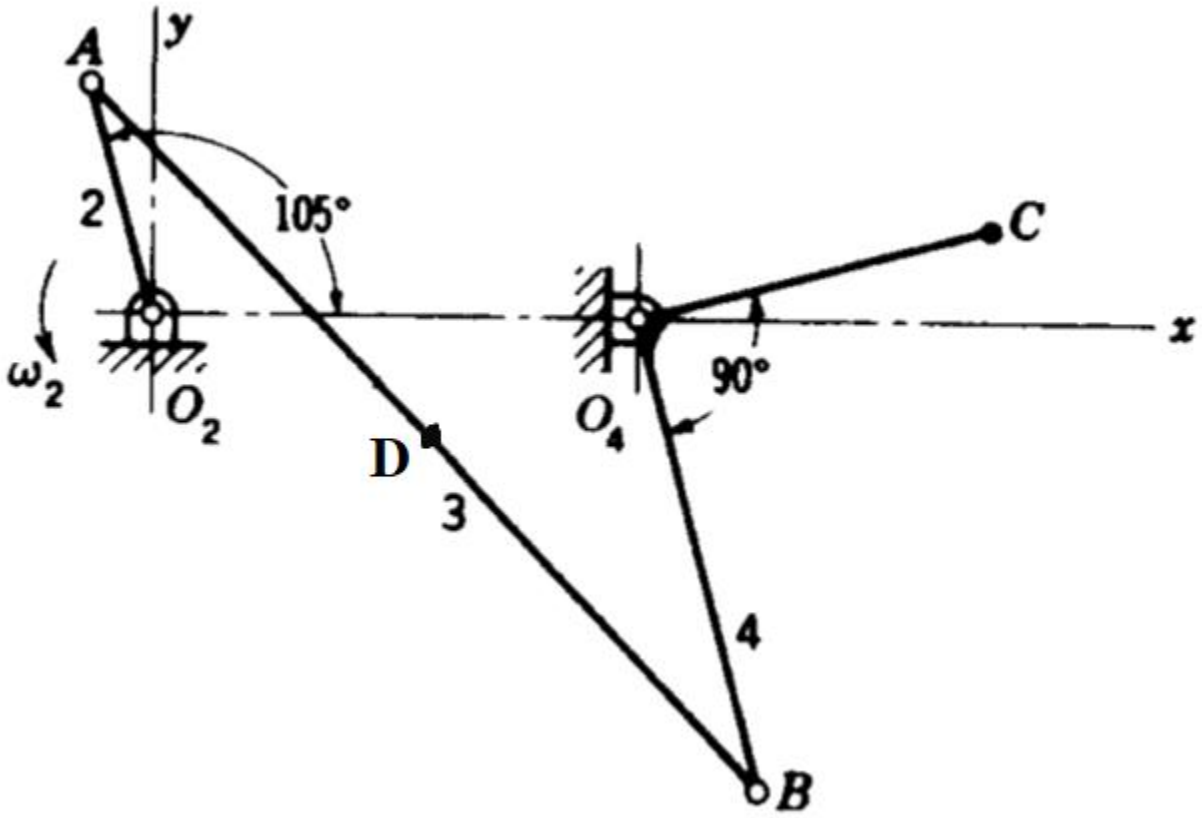


Assignment 5

Owusu House of Flapjacks wants to automate their flapjack production. They need a mechanism which will automatically flip the flapjacks “on the fly” as they travel through the griddle on a continuously moving conveyor. This mechanism must track the constant velocity of the conveyor, pick up a pancake, flip it over and place it back onto the conveyor.

Assignment 6

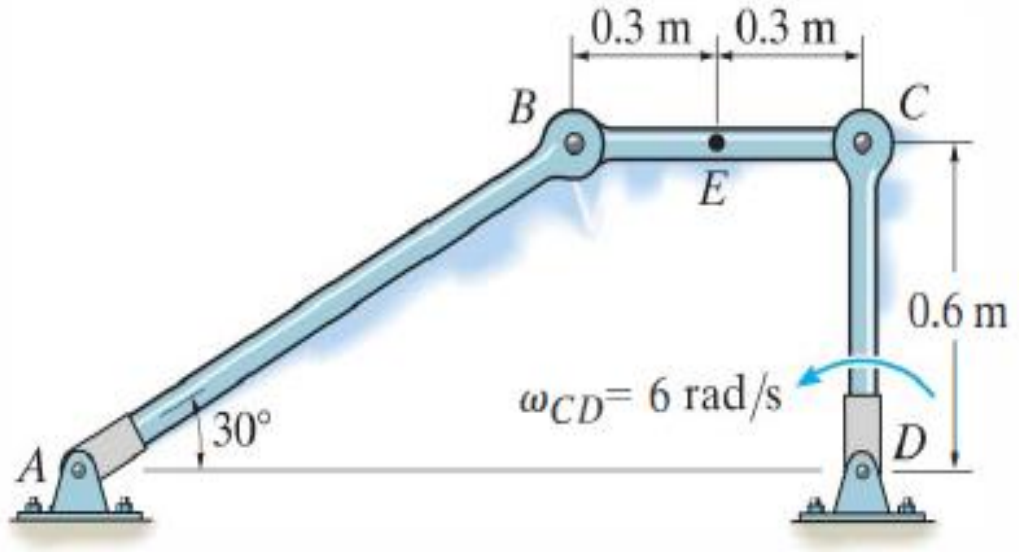
Draw the coupler curve of point D at the middle of the coupler AB for a complete revolution of link 2.



$O_2A = 8\ cm$
$AB = 32\ cm$
$O_4B = 16\ cm$
$O_2O_4 = 16\ cm$
$O_4C = 12\ cm$
$AD = 16\ cm$

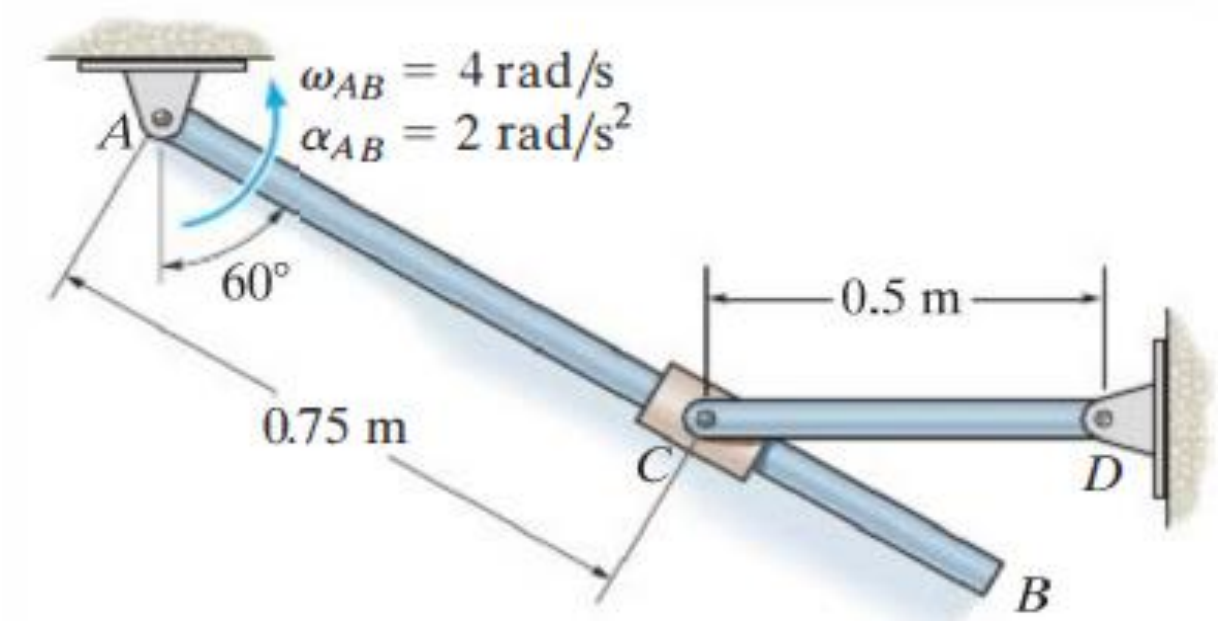
Assignment 7

If link CD has an angular velocity of = 6 rad/s, determine the velocity and acceleration of point E on link BC and the angular velocity and angular acceleration of link AB at the instant shown.



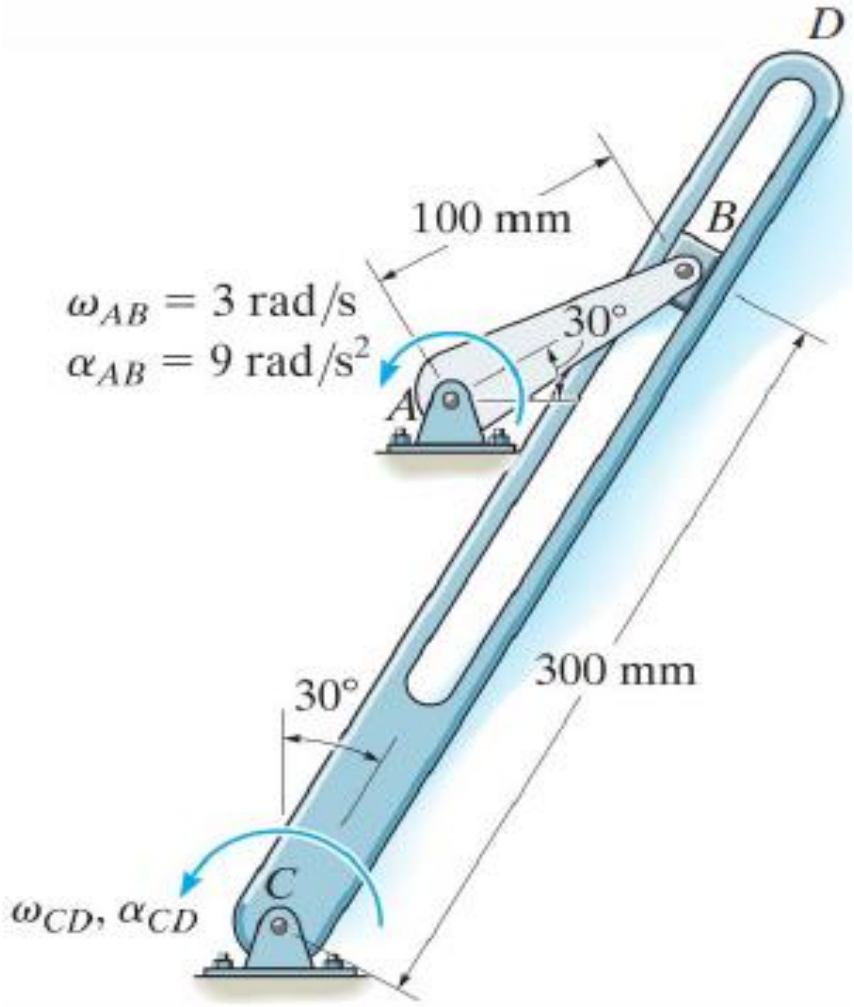
Assignment 8

At the instant shown rod AB has an angular velocity ω_{AB} = 4 rad/s and an angular acceleration $\omega_{CD} = 2\,rad/s^2$. Determine the angular velocity and angular acceleration of rod CD at this instant. The collar at C is pin connected to CD and slides freely along AB.



Assignment 9

The"quick-return" mechanism consists of a crank AB, slider block B, and slotted link CD. If the crank has the angular motion shown, determine the angular motion of the slotted link at this instant.



Assignment 10

Ball C moves with a speed of 3 m/s, which is increasing at a constant rate of 1.5 m/s^2, both measured relative to the circular plate and directed as shown. At the same instant the plate rotates with the angular velocity and angular acceleration shown. Determine the velocity and acceleration of the ball at this instant.

